M. Elena Vázquez-Cendón

Solving Hyperbolic Equations with Finite Volume Methods



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Solving Hyperbolic Equations with Finite Volume Methods

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Cover illustration: The cover figure represents the great *mandala of knowledge* in which each layer builds on the previous one. This is a special mandala, coloured by the author during one of her moments of relax.

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To Manuel Vázquez and Edita Cendón for giving me the initial condition, and especially for teaching me how to evolve being aware of my mistakes and with the example of their admirable vital paths

Foreword

Computational science has undergone explosive developments in recent years, thus generating tremendous potential for advances in many areas of scientific research. It is a highly multidisciplinary activity, in which empiricism, trial and error, and intuition are important parts of the game, all of which, however, coupled with the notorious lack of formal university training, as such, tends to undermine its power. It is here where sound training of the scientists who will be involved with mathematical modelling and computation enters the scene. This introductory book by Prof. M. Elena Vázquez-Cendón is a valuable contribution to this effort. The author fulfills what may be regarded as the essential requirements for writing a useful book on the subject. She has contributed to the research and development of improved numerical methods, has taught the subject at university level for many years to students from a wide variety of backgrounds, has been directly involved in the application of models and methods to solve problems of engineering relevance, and last but not least, can effectively communicate the subject to readers. Students will be fascinated by their first encounter with finite volume methods, a large class of numerical methods for solving partial differential equations. Lecturers will find useful material to prepare their courses with such essential ingredients as theory, practical implementation, computation of solution and critical assessment of the results.

Trento, Italy
February 2015

Eleuterio Toro

Preface

Finite volume methods are nowadays routinely used in numerous applications and by a broad multidisciplinary scientific community. The main aim of this book is to make this important tool available to students, researchers and academics involved in the training of students in different science and technology fields in which numerical methods for partial differential equations are used. This book, a revised version of the Spanish edition [35], introduces concepts and numerical methods, as well as exercises, examples and applications that will contribute to reinforce the theoretical concepts learned. In addition, some of the exercises and examples involve the execution of MATLAB/Octave codes, also included in the book. The choice of contents is based on the author's own experience in teaching Ph.D. and Master's courses in different universities. In particular, the Spanish version of this book [35] has been used for many years as a textbook for courses on finite volume methods in the Master's course in Mathematical Engineering and in the Master's course in Industrial Mathematics (http://www.m2i.es). The latter is an official degree jointly delivered by the University of Santiago de Compostela (USC), University of A Coruña (UDC), University of Vigo (UVigo), Carlos III University of Madrid (UC3M) and Technical University of Madrid (UPM). The structure of the book has been enriched by contributions from students who used [35] and its previous versions. This book can be used as a teaching aid and as a reference for lecturers. More than 40 exercises are included in the book, which can be used to design the continuous evaluation of the learning process, as required by the European Higher Education Area (EHEA). The book is meant to be a first contact with finite volume methods and aims at providing the necessary background to study more specific works, some of them introduced in the book, and at providing the skills to use commercial programs or open source software within the framework of Computational Fluid Dynamics (CFD). As the background of graduate students, in areas relevant to this book, tends to be rather varied, the author aims to develop the understanding of common terminology, with a balance between mathematical rigour and the physical intuition that characterizes the very origin of finite volume methods.

Since 1989 the author has been engaged in the application and development of finite volume methods to solve hyperbolic conservation laws, first as a mathematics Ph.D. student, then as a researcher and lecturer. The contents of this book are not her own contributions to this methodology but draw on her experience in learning these methods as a graduate student, as a researcher and as a lecturer. The first part of the book is an introduction to systems of hyperbolic conservation laws. The objective of this part is to study and illustrate with examples, the type of possible solutions of conservation laws. A good understanding of what is expected from the equations will enable the reader to develop sound criteria for assessing the correctness of approximate numerical solutions of such problems. The second part of the book is devoted to applications of finite volume methods. Concepts are presented via the simplest problem, namely the numerical solution of the scalar transport equation. The degree of difficulty increases gradually by introducing different ways of solving the Riemann problem. We study the exact solution of the Riemann problem and the Godunov method. We also study different approximate Riemann solvers and associated numerical methods for solving nonlinear one-dimensional problems, such as for example, the first-order upwind scheme and flux splitting methods. Finally, the third part contains MATLAB/Octave codes developed by the author to solve all the exercises in the book and for the implementation of all the numerical methods presented. The codes also allow the reader to assess the practical performance of the various numerical methods presented, for solving the transport and Burgers equations with different initial and boundary conditions.

Regarding the bibliography at the end of the book, I would like to share with the readers my *gratitude* to the authors of all the works referenced, especially monographs, which helped to consolidate the finite volume methodology in the temporal scale of Mathematics. As Toro remarks in [30], *in 1917 Richardson made the first reported attempt to forecast the weather by solving partial differential equations numerically, by hand!. It is generally accepted that this work marked the beginning of Computational Fluid Dynamics*. In representation of all the authors listed here, I have chosen three researchers who have made outstanding contributions to the topics of this book, and I have produced a brief biographical review for each of them: P. Lax, S. Godunov and E.F. Toro. Every time I read their works I learn something new. I had the privilege of direct personal contact with them, in lectures and discussions in Santiago de Compostela, Oxford and many other places in the world. In addition, I would like to share with readers my admiration for them, both as researchers as well as human beings. In the biographical review of each of them, I follow a precise timeline and give links to their publications and most important research on the subject, including new lines of research in the mathematical field. I note that the lives all three scientists I have mentioned above have been marked by turbulent historical events of their times, but nonetheless they remained enthusiastic about scientific research. I consider awareness of these personal experiences to be of educational value, especially for young researchers who begin their research careers.

I would like to thank two admired colleagues in the Department of Applied Mathematics at USC for reviewing a manuscript of the book that resulted in a much improved version. O. Lopez Pouso reviewed the Spanish version [35] and contributed with very relevant observations. A. Bermúdez was my teacher of numerical analysis, the supervisor of my Ph.D. and the director of the research group in Mathematical Engineering (http://www.usc.es/ingmat/?lang=en) I am part of; he gave me the opportunity to learn about this theme and is the co-author of my first and now, as I write this preface, my latest work on finite volume methods [1, 2]. The challenge of making this book suitable for engineers, physicists and mathematicians has facilitated discussions with co-authors of scientific publications from other disciplines, especially with collaborators such as J. Puertas and L. Cea from the University of A Coruña, and P. Garcia-Navarro and P. Brufau from the University of Zaragoza. My friend M. Gómez Marmol and collaborators from the University of Sevilla, who have followed the Spanish version [35] as teachers and students, respectively, have enriched it with valuable contributions, corrections and suggestions. In addition to all of those who helped me to improve [35], it has to be mentioned that I have been using it as teaching aid for more than eight courses by now. This fact allowed me to improve it by reducing errors and analysing which aspects required more detail to enhance the learning of the concepts and methods studied. I am deeply grateful to Eleuterio Toro for his work in reviewing formal and conceptual aspects of this book, which has helped significantly to improve the final version of the work. The convergence of [35] to this book is the result of a proposition by Francesca Bonadei (Springer) in 2011. I thank her for her encouragement and her ability to help during the entire process. The translation from Spanish to English has been done by two former students of the above-mentioned courses, L. García and M. Cobas. We considered this translation work as an opportunity to discuss again concepts and analyse the best way to present them. I would like to thank them both for their involvement in the project and the professionalism with which they carried out their work.

I would not conclude without a few words to challenge those readers who are also students. I hope the book will serve as a tool to get closer to the methodology of finite volume methods, helping to understand the basic concepts and some of the important contributions from people who once faced problems to which they did not know the solution. You are probably now the same age as the three scientists I mentioned above, when they first produced new contributions to the field. The admiration that these pioneers inspire should act as a stimulus to you to solve the problems you will face, thus contributing to the growth of the great *mandala of knowledge* in which each layer builds on the previous one. We may think that our contribution to the thickness of this layer might be negligible, but with contributions from all of us, its area will continue to increase.

Loyal to my parents' teachings, I declare myself directly responsible for all the mistakes made in this book. Although it has been my goal to minimize them, in every iterative process it is also necessary to introduce a stopping test to provide an

approximate solution to the reader, and this is what you have in your hands. From these lines I propose a critical reading that will help in learning and contributing to increase the accuracy of subsequent editions. I will especially appreciate your remarks, suggestions and corrections. These will be a relevant indicator of this open project, and I promise to address them all and, as I have done with the Spanish version, will incorporate them in the future editions.

Santiago de Compostela, Spain M. Elena Vázquez-Cendón
January 2015

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Part I
Basic Concepts and Examples of Environmental and Industrial Interest

Chapter 1
Motivation

Abstract This chapter constitutes an introduction to the problems that will be studied in the first part of the book as well as a summary of the state of the art regarding this particular theme. The main objective of this part consist of illustrating the difficulties and characteristics of the hyperbolic systems of conservation laws. All of them should be taken into account when designing the numerical methods to solve this kind of problems. In addition, different models of hyperbolic conservation laws will be presented in order to give a rough outline on the broad range of applications in which these equations can be used.

1.1 Introduction

In my opinion, the brevity and precision of the definitions established by Karni in [36] is not only adequate to properly transmit the meaning of the non-linear systems of conservation laws but also to retain the interest of the reader in the study of such systems:

> Conservation is a fundamental principle of the physical world. Matter may move around and redistribute but it does not appear or disappear. Hyperbolicity means that news that happen at a given point A, take time before they may affect affairs at another point B. Nonlinear means that the manner in which news propagate depends on what type of news it is. This makes the subject of nonlinear hyperbolic conservation laws fascinating, rich and challenging to study.

The combination of these terms leads to the objective of our theoretical study and, later on, to their numerical resolution by means of the finite volume methods:

A system of conservation laws is that in which the rate of change of a certain quantity contained in a region equals the total flux through the boundaries of that region.

For a space variable, the equations of such system have the form:

$$w_t(x,t) + f_x(w(x,t)) = 0, \tag{1.1}$$

where w is the vector of *conservative variables* or state variables, which depends on the position x and the time t, and the function f represents the *convective flux*.

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Equation (1.1) is to be completed with initial conditions and, in real cases, with boundary conditions. The simplest is the *initial value problem*, or *Cauchy problem*, for which (1.1) is verified and only initial conditions are specified:

$$w(x, 0) = w_0(x), \quad -\infty < x < \infty. \tag{1.2}$$

Hyperbolic systems of conservation laws are characterized by the fact that, for any value of w, the eigenvalues of the flux Jacobian matrix f are all real numbers, and the eigenvectors are linearly independent.

One of the best studied hyperbolic systems of conservation laws are the *Euler equations of gas dynamics*. Most of the general theory of conservation laws has been developed with these equations in mind, and many numerical methods were specifically designed to solve them. Indeed, it has been the focus of many monography on the topic of conservation laws (Godlewski and Raviart [8], LeVeque [17, 18] and Toro [30]).

Another model of interest is made by the *shallow water equations*, also called *Saint-Venant equations*. These equations model the behaviour of water in rivers, coastal areas, channels and in shallow regions in general.

The so-called *Riemann problem* is a fairly simple example that illustrates the behaviour of the solutions of hyperbolic conservation laws. It consists of a Cauchy problem in which the initial conditions are given in the following form:

$$w_0(x) = \begin{cases} w_l \text{ if } x < 0, \\ w_r \text{ if } x > 0, \end{cases} \tag{1.3}$$

with w_l and w_r being two constant states.

In the framework of shallow water equations, this problem models a *dam break*, and is equivalent to the *shock tube problem* in the Euler equations. This case presents a very interesting solution, in which the structure of the fluid is split into several regions separated by different kinds of waves, where even discontinuous solutions are admitted. The solution to this problem can be found in Stoker [26] and Toro [29].

Solutions of this kind clearly show that the *partial differential equations are not verified at every point in a classical sense*, since the derivatives are not defined at the discontinuities. Therefore, we need to define what we mean by solution of the conservation law in this particular case. In order to find the correct definition, we need to understand the meaning of the differential form of the equations from a physical point of view. This will lead us to the integral form of these equations, which is equivalent to the differential form only if additional regularity hypotheses are imposed.

The advantage of the integral form is that it remains valid even for discontinuous solutions. However, it is more difficult to work with this form of the equations, especially from the point of view of the discretization. Since the partial differential equations are valid everywhere but at the discontinuities, the next step is to add *jump conditions* that must be verified along the discontinuities. These conditions can be obtained from the integral form of the equations. In the context of hyperbolic

conservation laws there is one important issue at stake, namely the *non-uniqueness of weak solutions*. In general, there is more than one weak solution for a conservation law with the same data. Obviously, if our law models a real problem, only one solution would be meaningful.

Spurious solutions exist because the equations do not account for all the relevant physical effects. They are only a model of reality. Hyperbolic conservation laws in particular do not include the effects of diffusion nor viscosity. Although these effects are negligible almost everywhere in the fluid, they become relevant near the discontinuities. We may say that *what we are modelling with hyperbolic conservation laws is the limit of smooth solutions with a viscosity parameter in the right hand side of the equations, when this parameter tends to zero. This is, in fact, a weak solution of the system of conservation laws.*

Other weak solutions might be obtained with this methodology. Hence, we need a clever idea to choose the correct weak solution. One technique to define a unique weak solution of the systems of conservation laws consists in *introducing a diffusive term in the equations so that the resulting system presents only one smooth solution, and then make this term tend to zero*. This method is called *vanishing viscosity method*, and has very relevant implications for the analysis of conservation laws. However, this method is clearly suboptimal, since it involves solving more complex equations, which is what we initially wanted to avoid with the formulation of hyperbolic systems.

Therefore, we will try to find other conditions that, if imposed directly onto the conservation laws, will allow us to select the correct physical solution. This leads to the *mathematical notion of entropy and entropy conditions* [5], which are generalizations of the thermodynamic concept of entropy. In the case of gases, it means that gas molecules must increase their entropy if they move across a shock. In this way, we found a sufficient condition to tell apart a physically sound discontinuity, and therefore provide a unique solution. It is possible to calculate entropy functions for other systems and obtain the corresponding conditions. The so-called *entropy solution* of a hyperbolic system is defined by means of the concept of entropy. The existence and uniqueness of the entropy solution can be proven on the basis of Kruzcov's results [22], if certain assumptions on the initial conditions are satisfied.

Most of the problems that arise in technological or real-world applications are formulated in terms of *generalized conservation laws* or *balance laws*, in which the system of conservation laws has a non-zero right-hand side that represents a source term. The shallow water equations in variable depth domains (see Roe [23], Bermúdez and Vázquez-Cendón [2], Vázquez-Cendón [32], Toro [29] and Greenberg and LeRoux [13]) will be analyzed in this text as an example of this kind of generalized conservation laws, and some solutions will be examined.

The purpose of Part I is to expose the characteristics and inherent difficulties of hyperbolic systems of conservation laws. All of them should be taken into account in the design of the numerical methods to solve this kind of problems. In addition, different models of hyperbolic conservation laws will be presented in order to give a rough outline on the broad range of applications in which this equations can be used.

1.2 A Basic Model of Scalar Conservation Laws

We will start by introducing a basic model of scalar conservation laws that can be found in the text by Logan [19].

Let us consider a quantity $w(x,t)$ that might represent, for instance, the density of a certain magnitude. For simplicity we will assume that it is uniformly distributed in each section of a tube of constant section A. The dimensions of w are:

$$[w] = \frac{magnitude}{volume}. \tag{1.4}$$

If we consider an arbitrary segment of the tube V_I, with $I = [a,b]$, the total amount of w in V_I is

$$\int\int\int_{V_I} w(x,t)\,dxdydz = \int_a^b w(x,t)Adx = A\int_a^b w(x,t)dx.$$

We assume now that there exists a movement of particles inside the tube and call $f = f(x,t)$ the flux of w in x at time t. This is, f represents the amount of w that goes through section x at time t by unit volume and time. Therefore, the dimensions of f are:

$$[f] = \frac{magnitude \times longitude}{volume \times time} = [density] \times [velocity]. \tag{1.5}$$

Hereinafter, we will consider as positive a flux from left to right and as negative, a flux in the opposite direction. Then, at time t, the total amount of w that enters V_I, is the total amount that enters in $x = a$, minus the quantity that goes out through $x = b$. This is,

$$total\ flux\ of\ quantity\ w\ in\ V_I\ at\ time\ t = A\,f(a,t) - A\,f(b,t). \tag{1.6}$$

Let us finally assume that w may be created or destroyed in section x at time t, and that this creation or destruction is given by a function $g = g(x,t)$. The dimensions of the source or sink g are:

$$[g] = \frac{magnitude}{volume \times time}. \tag{1.7}$$

Once g is given we can calculate the creation/decrease of w in V_I by integrating:

$$rate\ of\ creation/decrease\ of\ w\ in\ V_I = \int_a^b g(x,t)A\,dx. \tag{1.8}$$

Taking all the considerations above into account, the conservation law for w can be formulated in any spatial interval I as:

$$\begin{aligned} \textit{rate of change of } w \textit{ in } I \; &= \textit{total flux of quantity } w \textit{ in } I \\ &\quad + \textit{rate of creation/decrease of } w \textit{ in } I. \end{aligned}$$

Therefore:

$$\frac{d}{dt}\int_a^b w(x,t)Adx = A\,f(a,t) - A\,f(b,t) + \int_a^b g(x,t)Adx. \tag{1.9}$$

Since A is constant the previous equation can be simplified as:

$$\frac{d}{dt}\int_a^b w(x,t)dx = f(a,t) - f(b,t) + \int_a^b g(x,t)dx. \tag{1.10}$$

This equation is an *integral conservation law*, which is also valid for low regularity functions. If functions w, f and g are smoother, for instance if they verify:

$$(i) \quad \int_a^b f_x(x,t)dx = f(b,t) - f(a,t). \tag{1.11}$$

$$(ii) \quad \frac{d}{dt}\int_a^b w(x,t)dx = \int_a^b w_t(x,t)dx. \tag{1.12}$$

Then the conservation law can be written as:

$$\int_a^b \left[w_t(x,t) + f_x(x,t) - g(x,t)\right] dx = 0, \quad \forall I = [a,b]. \tag{1.13}$$

Since this is satisfied for all intervals $[a,b]$, the integrand must vanish, hence

$$w_t(x,t) + f_x(x,t) - g(x,t) = 0, \quad \forall x \in \mathbf{R}, \quad t > 0, \tag{1.14}$$

which is a *differential conservation law.*

Observations

- We have assumed that f and g are functions of x and t, but we could have also considered that they depend explicitly on w, the *conservative variable*. This hypothesis leads to linear and non linear models that will be analized later.
- We have two unknowns w and f and only one equation. Therefore, we need another equation that relates w and f, which is known as *equation of state*.
- If we move g to the right hand side of the equation, we have

$$w_t(x,t) + f_x(w(x,t)) = g(x,t), \quad \forall x \in \mathbf{R}, \quad t > 0. \tag{1.15}$$

The term conservation law is normally used when $g \equiv 0$. For this reason, it is commonly specified when this term is not zero: *generalized conservation law* or *conservation law with source terms* or *conservation law with right hand side* g.

1.3 The Transport Equation

This classic example of a conservation law can be found in the text by Toro [30], among others.

The linear transport equation is a model in which *the flux is proportional to the density*:

$$f(w) = \lambda w, \tag{1.16}$$

with λ a constant that can be interpreted as a propagation velocity. By carrying out a non-dimensional analysis:

$$[f] = \frac{magnitude \times longitude}{volume \times time}, \tag{1.17}$$

$$[\lambda\, w] = [\lambda][w] = [\lambda]\frac{magnitude}{volume}, \tag{1.18}$$

hence

$$[\lambda] = \frac{longitude}{time}. \tag{1.19}$$

The corresponding conservation law is:

$$w_t(x,t) + \lambda w_x(x,t) = 0, \tag{1.20}$$

and given the initial condition

$$w(x,0) = w_0(x), \quad -\infty < x < \infty, \tag{1.21}$$

the solution of the equation (1.20) in terms of the initial condition is given by

$$w(x,t) = w_0(x - \lambda t), \quad -\infty < x < \infty, \quad t \in [0, +\infty). \tag{1.22}$$

This solution can be easily obtained by means of the characteristic curves since the solution is constant along them. Therefore, in order to know the value of the solution in a point x at time t, it suffices to trace the characteristic curve in the plane xt and obtain the intersection with the axis $t = 0$. The solution at (x, t) represents the value of the initial condition at the intersection calculated.

The formal procedure to obtain the solution is as follows:

(a) We introduce the characteristic curves:

$$t \longrightarrow X(t), \quad \frac{dX}{dt}(t) = \lambda. \tag{1.23}$$

(b) We check that the solution is constant along the characteristics:

$$\begin{aligned} \frac{dw}{dt}(X(t), t) &= w_x(X(t), t)\frac{dX}{dt}(t) + w_t(X(t), t) \\ &= w_x(X(t), t)\lambda + w_t(X(t), t) = 0, \end{aligned} \tag{1.24}$$

where the second equality holds as a consequence of the equation that verify the characteristic curves (1.23) and the third one derives from the fact that w is a solution of the transport equation (1.20).

(c) We calculate the solution at an arbitrary point $(x^\star, t^\star)$ of the xt plane and call $X^\star(t)$ the characteristic curve which passes through such point and that satisfies the initial value problem:

$$\frac{dX^\star}{dt}(t) = \lambda, \quad X^\star(t^\star) = x^\star, \tag{1.25}$$

being its expression:

$$X^\star(t) = X^\star(t^\star) + \lambda(t - t^\star), \tag{1.26}$$

and hence the point of intersection with the plane $t = 0$ (see Fig. 1.1) is

$$X^\star(0) = x^\star - \lambda t^\star. \tag{1.27}$$

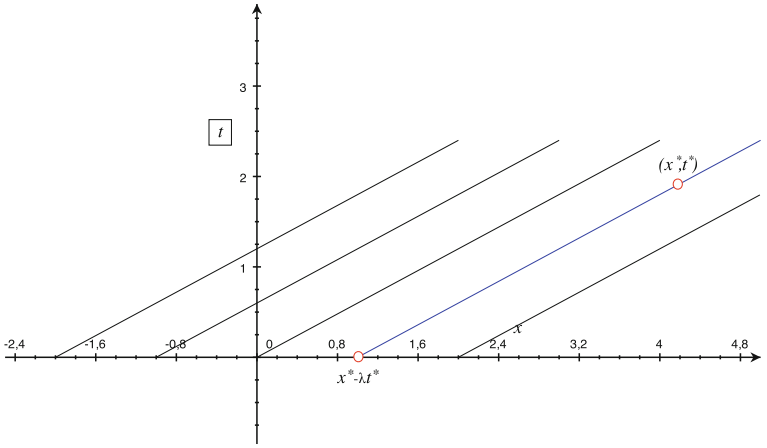


Fig. 1.1 Characteristics of the transport equation

(d) We calculate the solution taking into account (1.24) and the value of the initial condition (1.27)

$$w(x^\star, t^\star) = w(X^\star(0), 0) = w_0(X^\star(0)) = w_0(x^\star - \lambda t^\star). \tag{1.28}$$

Therefore, *the function w_0 moves along time at velocity λ with no deformation*.

If we consider now a discontinuous initial condition w_0

$$w_0(x) = \begin{cases} w_l \text{ if } x < 0, \\ w_r \text{ if } x > 0, \end{cases} \tag{1.29}$$

the solution of the conservation law is a solution in the sense of distributions. The initial discontinuity in $x = 0$ propagates a distance $d = \lambda t$ during time t. The characteristic curve $X(t) = \lambda t$ splits the rest of the characteristic curves in two groups: those lying to the left of $X(t)$ for which the solution is w_l, and those lying to its right, for which the solution is w_r.

As we have previously seen in the Introduction, the initial value problem associated with this initial conditions is known as *Riemann Problem* and, in this case, the solution is simply:

$$w(x, t) = w_0(x - \lambda t) = \begin{cases} w_l \text{ if } x - \lambda t < 0, \\ w_r \text{ if } x - \lambda t > 0. \end{cases} \tag{1.30}$$

1.3.1 Dispersion Relation and Amplification Coefficient

The conditions we will focus on in the following are verified by the harmonic waves that are solution of the transport equation. These waves constitute a relevant example when studying the properties of the numerical methods that are used to solve them.

Let us consider a harmonic wave of the form

$$w(x, t) = \mathrm{e}^{i(\omega t - kx)}, \tag{1.31}$$

where $\omega t - kx$ is the phase of the wave at point x and time t, ω represents the angular frequency and k the wave number.

It is immediate to check that the harmonic wave verifies the transport equation if the following condition holds

$$\omega - k\lambda = 0, \tag{1.32}$$

which is known as *dispersion relation*. This condition implies that if the phase remains constant ($\omega t - kx = cte$), the *phase velocity* V_p is precisely the velocity of propagation λ:

$$V_p := \frac{dx}{dt} = \frac{\omega}{k} = \lambda, \tag{1.33}$$

and it is, hence, independent of the *wave number* k. Moreover, if we still assume that the phase remains constant, the group velocity V_G also coincides with λ

$$V_G := \frac{d\,\omega}{dk} = \frac{x}{t} = \lambda, \tag{1.34}$$

and it is again independent of the wave number k.

The analysis of the phase in the numerical scheme will be related to the *dispersion* of the scheme.

In order to introduce the concept of amplification coefficient and to analyze these properties of the transport equation from a different point of view, we will consider the initial condition:

$$w_0(x) = \mathrm{e}^{ikx}. \tag{1.35}$$

The solution $w(x, t)$ given by (1.22) for this choice of initial condition equals to

$$\begin{aligned} w(x,t) = w_0(x - \lambda t) = \mathrm{e}^{ik(x-\lambda t)} = \mathrm{e}^{ikx}\,\mathrm{e}^{-ik\lambda t} \\ = w_0(x)\,\mathrm{e}^{-ik\lambda t} = w_0(x)\,G(k,t), \end{aligned} \tag{1.36}$$

where $G(k, t) = \mathrm{e}^{-ik\lambda t}$ is defined as the *amplification factor or amplification coefficient*, whose exact module is 1. This is, a phase difference equal to $k\lambda t$ occurs, but it does not produce any damping or amplification.

The analysis of the amplification factor of the numerical scheme will be related to a characteristic of the scheme which is known as *dissipation*. This phenomenon represents a decrease of the high frequency oscillations and implies a loss of accuracy in the approximation of discontinuities. The errors in the amplitude or in the amplification factor are related to the diffusion of the numerical scheme.

1.4 Acoustic Wave Propagation Model

We have previously seen that the characteristic curves play an important role in the definition of the solutions of the transport equation. In this section we will consider a more complex example stated and developed by Godunov in [9], for a system of equations

$$\frac{\partial w_1}{\partial t}(x,t) + \lambda_1 \frac{\partial w_1}{\partial x}(x,t) = 0, \quad \lambda_1 = constant, \tag{1.37}$$

$$\frac{\partial w_2}{\partial t}(x,t) + \lambda_2 \frac{\partial w_2}{\partial x}(x,t) = 0, \quad \lambda_2 = constant, \tag{1.38}$$

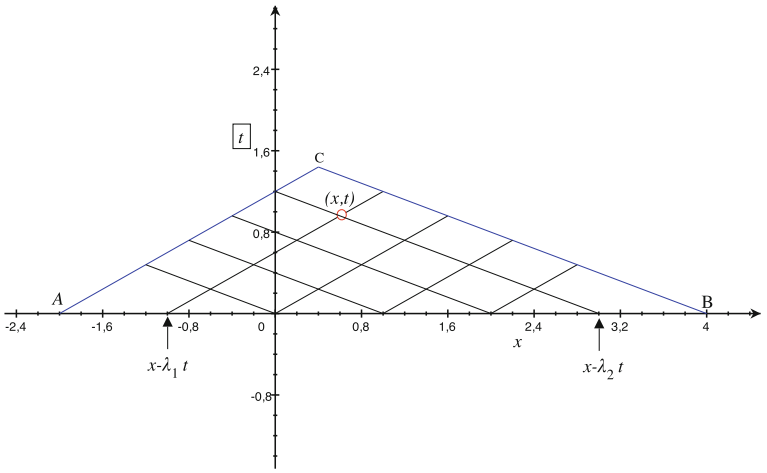


Fig. 1.2 Domain of definition of the solution of the linear system

composed of two independent solutions, $w_1(x,t)$ and $w_2(x,t)$, whose analytical expressions, in terms of their respective initial conditions $w_1^0(x)$ and $w_2^0(x)$, have the form:

$$w_1(x,t) = w_1^0(x - \lambda_1 t), \quad w_2(x,t) = w_2^0(x - \lambda_2 t). \tag{1.39}$$

If both initial conditions $w_1^0(x)$ and $w_2^0(x)$ are defined in an interval $[a,b]$, we can find a solution within the triangle ABC (see Fig. 1.2), with C the intersection point in the xt plane of the characteristics: $x - \lambda_1 t = constant$, which passes through B, and $x - \lambda_2 t = constant$ that passes through A. Only within this triangle the solution is unique.

This is an academic example, with a simple solution after having studied the transport equation. Next we will show how we can reduce to model (1.37) and (1.38) a system that at first sight may look more complex and that corresponds to a physical problem:

$$\frac{\partial u}{\partial t}(x,t) + \frac{1}{\rho_0}\frac{\partial p}{\partial x}(x,t) = 0, \tag{1.40}$$

$$\frac{\partial p}{\partial t}(x,t) + \rho_0 {c_0}^2 \frac{\partial u}{\partial x}(x,t) = 0. \tag{1.41}$$

This system describes the propagation of acoustic planar waves (of small perturbations) in a medium at rest:

- u speed of the perturbed medium.
- p pressure of the perturbed medium.
- ρ_0 density of the medium at rest.
- c_0 *compressibility* of the medium at rest.

This is a system of conservation laws in which the conservative variable is $\mathbf{w} = (u,\ p)^T$ and the physical flux is $\mathbf{f}(\mathbf{w}) = \left(\frac{p}{\rho_o},\ \rho_0 c_0^2 u\right)^T$.

We will show that these equations can be transformed into a simple model like the one introduced previously. To that end, we multiply the second equation by $\frac{1}{\rho_0 c_0}$:

$$\frac{\partial}{\partial t}\left(\frac{p}{\rho_0 c_0}\right) + c_0 \frac{\partial u}{\partial x} = 0. \tag{1.42}$$

Adding (1.40) to this relation, we have:

$$\frac{\partial}{\partial t}\left(u + \frac{p}{\rho_0 c_0}\right) + c_0 \frac{\partial}{\partial x}\left(u + \frac{p}{\rho_0 c_0}\right) = 0. \tag{1.43}$$

Subtracting both equations, we arrive at a similar relation:

$$\frac{\partial}{\partial t}\left(u - \frac{p}{\rho_0 c_0}\right) - c_0 \frac{\partial}{\partial x}\left(u - \frac{p}{\rho_0 c_0}\right) = 0. \tag{1.44}$$

Now we define two new variables v_1 and v_2 as:

$$v_1 := u - \frac{p}{\rho_0 c_0}, \quad v_2 := u + \frac{p}{\rho_0 c_0}. \tag{1.45}$$

Thus, v_1 and v_2 verify a linear system like the one presented before (1.37) and (1.38), where $\lambda_1 = -c_0$ and $\lambda_2 = c_0$. The general solution has the form

$$v_1(x,t) = v_1^0(x - \lambda_1 t) = v_1^0(x + c_0 t),\ v_2(x,t) = v_2^0(x - \lambda_2 t) = v_2^0(x - c_0 t). \tag{1.46}$$

Therefore, rewriting the original variables in terms of these equations, we obtain the following expressions:

$$u = \frac{v_1(x,t) + v_2(x,t)}{2} = \frac{v_1^0(x + c_0 t) + v_2^0(x - c_0 t)}{2}, \tag{1.47}$$

$$p = \rho_0 c_0 \frac{v_2(x,t) - v_1(x,t)}{2} = \rho_0 c_0 \frac{v_2^0(x + c_0 t) - v_1^0(x + c_0 t)}{2}, \tag{1.48}$$

that constitute the general solution of the equations of propagation of sound.

Let us suppose the distributions of pressure p and velocity u are known at the initial time in a spatial interval $[x_1, x_2]$. These distributions determine in a unique way the solution within the characteristic triangle of base $[x_1, x_2]$, defined by the inequalities

$$t > 0, \quad x - c_0 t > x_1, \quad x + c_0 t > x_2.$$

The quantities $u \pm \dfrac{p}{\rho_0 c_0}$ are called *Riemann invariants*. These quantities remain constant along the characteristic curves. Moreover, the formula

$$u + \frac{p}{\rho_0 c_0} = v_2^0(x - c_0 t),$$

shows that the distribution of this Riemann invariant moves to the right with speed $\lambda_2 = c_0$, preserving its shape. For this reason c_0 is known as *speed of propagation of acoustic waves* or *speed of sound*. Likewise, the formula

$$u - \frac{p}{\rho_0 c_0} = v_1^0(x + c_0 t),$$

shows that the distribution of the other Riemann invariant moves to the left preserving its shape, with speed $\lambda_1 = -c_0$.

1.5 An Equation of Transport with Source Terms

In this section we present an example of conservation laws with source terms, developed in [31]. We propose an expression of the solution w in which we consider a factor that represents the convective transport, denoted by $c(x - \lambda t)$, and two other factors that account for amplifications or dumpings in space and time, that will be called $b(x)$ and $g(t)$, respectively. This is, we will find solutions of the form

$$w(x, t) = c(x - \lambda t)\, b(x)\, g(t), \tag{1.49}$$

that verify the generalized conservation law or *balance law*

$$w_t(x, t) + \lambda w_x(x, t) = s(x, t, w), \tag{1.50}$$

where the initial condition is given by

$$w(x, 0) = c(x)\, b(x)\, g(0). \tag{1.51}$$

The solution proposed must verify (1.50) for an appropriate source term, obtained by simply taking the derivative of the solution (1.49) and replacing its expression in the conservation law:

$$s(x, t, w) = w(x, t) \left[\frac{g'(t)}{g(t)} + \lambda \frac{b'(x)}{b(x)} \right]. \tag{1.52}$$

1.6 The Traffic Flow Model

The equations of the traffic flow will help us develop a physical intuition that can be applied to more complex equations of flow dynamics. This example can be found in LeVeque [17].

Let us consider the traffic flow along a highway. We denote by ρ the density of cars, expressed in vehicles per kilometer, and by u the speed. For this application the domain of ρ corresponds to the interval $[0, \rho_m]$, with ρ_m the starting value after which the traffic flow collapses and a traffic jam is formed.

Since the number of cars must be constant, the density and velocity must hold the following relationship:

$$\rho_t + (\rho u)_x = 0. \tag{1.53}$$

In order to obtain a scalar conservation law only for ρ, we first assume in term that u is a function of ρ. This makes sense on the basis of the following interpretation: in a highway we can travel at the fastest allowed speed u_m, but in situations of dense traffic we must slow down, with decreasing speed as consequence of increasing flow density. The simplest model is the linear relationship:

$$u(\rho) = u_m \left(1 - \frac{\rho}{\rho_m}\right). \tag{1.54}$$

In the case of zero density, this is, with an empty highway, the speed can reach u_m, but it falls down to zero when ρ approaches ρ_m. By replacing the value of u in (1.53), we have:

$$\rho_t + f(\rho)_x = 0, \tag{1.55}$$

where the flow function is given by

$$f(\rho) = \rho u_m \left(1 - \frac{\rho}{\rho_m}\right). \tag{1.56}$$

Whitham states in [38] that a good fitting of the data for the Lincoln tunnel was found by Greenberg in 1959, and is given by the flow function

$$f(\rho) = \rho u_m \ln\left(\frac{\rho_m}{\rho}\right). \tag{1.57}$$

1.7 One-Dimensional Shallow Water Equations

The one-dimensional shallow water equations, or Saint-Venant equations [5], model the flow in channels of constant cross-section. Their development can be found in Stoker [26] and Toro [29]. It is assumed that the pressure is hydrostatic, and that the

horizontal velocity profile is nearly uniform in the vertical dimension. Likewise, it is assumed that the flow is incompressible, so that the density remains constant. Hence, we arrive, on the one hand, at the mass conservation equation:

$$\frac{\partial h}{\partial t} + \frac{\partial q}{\partial x} = 0, \tag{1.58}$$

and, in the case of constant depth, to the momentum conservation equation given by:

$$\frac{\partial q}{\partial t} + \frac{\partial}{\partial x}\left(\frac{q^2}{h} + \frac{1}{2}gh^2\right) = 0, \tag{1.59}$$

where h is the total height of the water column (from bottom to surface) in each point x and time t, u is the vertically averaged velocity and the conservative variable $q = hu$, is the flow by unit length. The conservative variable is formed by the two components of the vector $\mathbf{w} = (h, q)^T$ and the physical flow is defined by $\mathbf{f}(\mathbf{w}) = \left(q, \frac{q^2}{h} + \frac{1}{2}gh^2\right)^T$.

If the depth defined by $H(x)$, or equivalently, the bottom function $Z(x)$ ($Z(x) + H(x) = A = constant$) is not constant (as it is shown in Fig. 1.3), we should take into account a non-zero right hand side in equation (1.59) to account for this bottom variation:

$$\frac{\partial q}{\partial t} + \frac{\partial}{\partial x}\left(\frac{q^2}{h} + \frac{1}{2}gh^2\right) = -ghZ'(x). \tag{1.60}$$

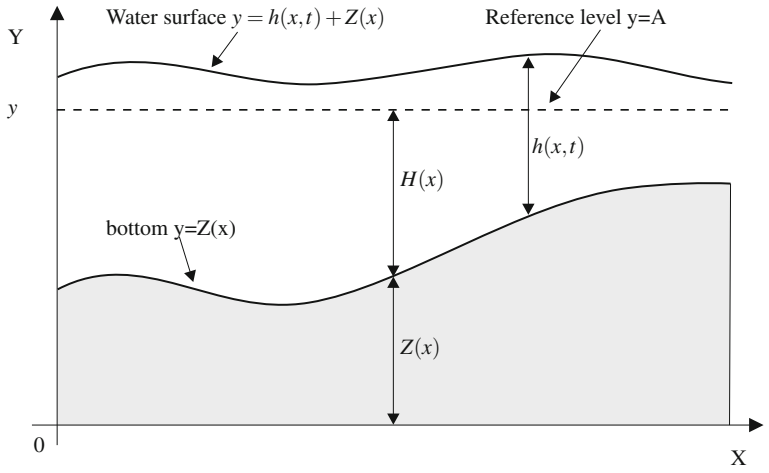


Fig. 1.3 Notation used in the one-dimensional shallow water equations

1.8 One-Dimensional Euler Equations

The Euler equations for gas dynamics in one scalar variable in conservative form are:

$$\mathbf{w}_t + \mathbf{f}(\mathbf{w})_x = \mathbf{0}, \tag{1.61}$$

where the vector of conservative variables $\mathbf{w}$ and the vector of the physical flux $\mathbf{f}(\mathbf{w})$ are defined by

$$\mathbf{w} = \begin{pmatrix} \rho \\ \rho u \\ E \end{pmatrix}, \quad \mathbf{f}(\mathbf{w}) = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ u(E+p) \end{pmatrix}, \tag{1.62}$$

where ρ represents the density, u is the velocity, p is the pressure and E is the total energy, which is given by

$$E = \rho(\frac{1}{2}u^2 + e), \tag{1.63}$$

with $\frac{1}{2}u^2$ the kinetic energy and e the specific internal energy. For an ideal gas, the following equation of state is considered:

$$e = e(\rho, p) = \frac{p}{\rho(\gamma - 1)}, \tag{1.64}$$

where γ represents the specific heats ratio. A thorough study of the properties of the Euler equations can be found in Chap. 3 by Toro [30].

1.9 The Linear Shallow Water Equations in Channels with Variable Bed and Width

The model for the shallow water equations that was introduced in Sect. 1.7 is nonlinear. We can introduce a simplification in the spatial derivatives of the conservative variables and obtain a linear version of the problem. In addition, we will include a source term that will allow us to take into account domains of variable geometry. We will also assume that water flows along a channel of variable bed, which is defined by $Z(x)$, with a constant regular section whose wet area is given by $h(x,t)B(x)$, with $B(x)$ the function that defines the width of the channel (see Fig. 1.4). In the same way as in Sect. 1.7 $h(x,t)$ is the water height and $u = u(x,t)$ the velocity:

$$h_t + \tilde{u}h_x + \tilde{h}u_x = -\tilde{h}\,\tilde{u}\,\frac{B'(x)}{B(x)}, \tag{1.65}$$

$$u_t + gh_x + \tilde{u}u_x = 2\,\tilde{u}^2\,\frac{B'(x)}{B(x)} - g\,Z'(x), \tag{1.66}$$

where $\tilde{h},\ \tilde{u},\ g = 9.8\,\mathrm{ms}^{-2}$ are constant values.

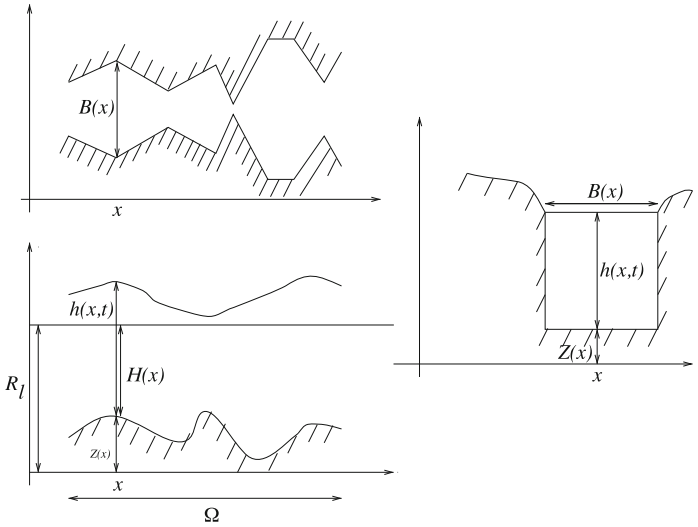


Fig. 1.4 Notation for the shallow water equations in a channel of constant section

This model is studied in [34] and the solutions corresponding to a Riemann problem are also presented.

In this first chapter we have introduced some models that can be solved by applying the finite volume method to the hyperbolic conservation laws. In Chap. 2 these models will be analyzed in a basic concept framework, studying the kind of solutions that can be obtained.

Chapter 2
Hyperbolic Conservation Laws. Basic Concepts and Examples

Abstract The purpose of this chapter is to present the basic concepts related to hyperbolic conservation laws. We will focus first on one-dimensional problems, and, at the end of the chapter, we will extend these concepts to multidimensional problems.

2.1 Conservation Laws

The models presented as a motivation in Chap. 1 are examples of scalar or vectorial conservation laws or systems of conservation laws, with p the number of components of the vector of conservative variables. The general form of one spatial dimension systems of conservation laws was introduced by the authors Godlewski and Raviart [18].

Definition 2.1 Let Ω be a subset of $\mathbf{R}^p$ and let $\mathbf{f}$ be a smooth function of Ω in $\mathbf{R}^p$:

$$\mathbf{f} : \Omega \longrightarrow \mathbf{R}^p. \tag{2.1}$$

The general form of a *system of conservation laws* is

$$\mathbf{w}_t + (\mathbf{f}(\mathbf{w}))_x = \mathbf{0}, \quad x \in \mathbf{R}, \quad t > 0, \tag{2.2}$$

where $\mathbf{w} = (w_1, \ldots, w_p)$, the *vector of conservative variables*, is the vectorial function:

$$\begin{aligned} \mathbf{w} : \mathbf{R} \times [0, \infty) &\longrightarrow \Omega, \\ (x, t) &\longrightarrow (w_1, \ldots, w_p). \end{aligned} \tag{2.3}$$

The set Ω is called *set of states* and the function $\mathbf{f} = (f_1, \ldots, f_p)$ physical flux ($f_i : \Omega \longrightarrow \mathbf{R}$).

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Formally, the system (2.2) represents a conservation of the quantities $w_1, \ldots, w_p$. If we consider an arbitrary interval $I = [a, b] \subset \mathbf{R}$ and integrate (2.2) over I, we have:

$$\frac{d}{dt} \int_I \mathbf{w}(x,t)\, dx = \mathbf{f}(\mathbf{w}(a,t)) - \mathbf{f}(\mathbf{w}(b,t).$$

Therefore, the variation of $\displaystyle\int_I \mathbf{w}(x,t)\, dx$ with respect to time equals the total net flux through the boundaries $\mathbf{f}(\mathbf{w}(a)) - \mathbf{f}(\mathbf{w}(b))$.

We will study the following Cauchy Problem for the hyperbolic systems of conservation laws (2.2):

Definition 2.2 *Cauchy Problem*:

Find a function $\mathbf{w} : (x,t) \in \mathbf{R}^d \times [0,\infty) \longrightarrow \mathbf{w}(x,t) \in \Omega$, solution of (2.2) that verifies the initial condition:

$$\mathbf{w}(x,0) = \mathbf{w}_0(x), \quad x \in \mathbf{R}, \tag{2.4}$$

where $\mathbf{w}_0 : x \in \mathbf{R} \longrightarrow \mathbf{w}_0(x) \in \Omega$ is a known function.

2.2 Examples of One Dimensional Conservation Laws

2.2.1 Transport Equation

This equation was presented and studied in Sect. 1.3, and is a linear scalar model:

$$p = 1, \ f(w) = \lambda w.$$

2.2.2 Burgers Equation

The parabolic scalar equation:

$$\frac{\partial w}{\partial t} + w\frac{\partial w}{\partial x} - \nu \frac{\partial^2 w}{\partial x^2} = 0, \tag{2.5}$$

where ν is the viscosity coefficient, was introduced by Burgers as the simplest differential model for fluids. He mainly focused on the equation obtained as a limit when ν tends to zero, which, in conservative form, is as follows:

$$\frac{\partial w}{\partial t} + \frac{\partial}{\partial x}\left(\frac{w^2}{2}\right) = 0, \tag{2.6}$$

and is known as *Burgers equation*. This equation presents many of the peculiarities of a *nonlinear* hyperbolic system in one spatial variable. In particular, we will see that the Cauchy problem for the Burgers equation, depending on the initial conditions, can have more than one solution. This model illustrates how discontinuous solutions can be obtained even if the initial conditions are continuous.

2.2.3 Traffic Flow Model

The traffic flow model introduced in Sect. 1.6 is also scalar and nonlinear:

$$p = 1, \quad w = \rho, \quad f(\rho) = \rho u_m \left(1 - \frac{\rho}{\rho_m}\right).$$

2.2.4 Acoustic Equations

The acoustic planar wave propagation model in a medium at rest introduced in Sect. 1.4 is vectorial and linear:

$$p = 2, \quad \mathbf{w} = (u,\ p)^T, \quad \mathbf{f}(\mathbf{w}) = \left(\frac{p}{\rho_o},\ \rho_0 c_0^2 u\right)^T.$$

It is a linear system, since the flux function admits the following representation in terms of a matrix of constant coefficients:

$$\mathbf{f}(\mathbf{w}) = \begin{pmatrix} 0 & \dfrac{1}{\rho_0} \\ \rho_0 c_0^2 & 0 \end{pmatrix} \begin{pmatrix} u \\ p \end{pmatrix} = \begin{pmatrix} 0 & \dfrac{1}{\rho_0} \\ \rho_0 c_0^2 & 0 \end{pmatrix} \mathbf{w}.$$

2.2.5 One-Dimensional Shallow Water Equations

The one-dimensional shallow water equations with constant bottom, introduced in Sect. 1.7 constitute a nonlinear system of conservation laws:

$$p = 2, \quad \mathbf{w} = (h,\ q)^T, \quad \mathbf{f}(\mathbf{w}) = \left(q,\ \frac{q^2}{h} + \frac{1}{2} g h^2\right)^T,$$

where h is the water height in each x point and time t, u is the vertically averaged velocity, and the conservative variable q is the flow per unit length, $q = hu$. As we mentioned in the previous chapter, the first equation represents the mass conservation equation, and the second the momentum conservation equation.

2.2.6 One-Dimensional Euler Equations

The one-dimensional Euler equations, introduced in Sect. 1.8 also constitute a non-linear system of conservation laws:

$$p = 3, \quad \mathbf{w} = (\rho,\ \rho u,\ E)^T, \quad \mathbf{f}(\mathbf{w}) = \left(\rho u,\ \rho u^2 + p, u(E + p)\right)^T,$$

where ρ represents density, u is velocity and p is pressure. E is the total energy. All three equations of the system correspond with the mass conservation equation, momentum conservation equation and energy equation, respectively.

2.2.7 Cauchy-Riemann Equations

The Cauchy-Riemann equations form a linear system that, as we shall see, do not verify the definition of hyperbolic system:

$$p = 2, \quad \mathbf{w} = (u,\ v)^T, \quad \mathbf{f}(\mathbf{w}) = (-v,\ u)^T.$$

2.3 One-Dimensional Hyperbolic Systems of Conservation Laws

Definition 2.3 Let us consider a system of the form (2.2), and, for every flux function $\mathbf{f}$, let us define its Jacobian matrix $\mathbf{A}$ as:

$$\mathbf{A}(\mathbf{w}) = \left(\frac{\partial f_i}{\partial w_k}(\mathbf{w})\right)_{1\le i,k\le p}, \tag{2.7}$$

where f_i is the ith component of the flux function. We say that the *system* (2.2) *is hyperbolic* if the matrix $\mathbf{A}(\mathbf{w})$ has p real eigenvalues $\lambda_1(\mathbf{w}) \le \lambda_2(\mathbf{w}) \le \cdots \le \lambda_p(\mathbf{w})$, and the corresponding p eigenvectors are linearly independent. In addition, if all the eigenvalues are different, we say that the system is *strictly hyperbolic*.

2.3.1 Examples of One-Dimensional Hyperbolic System

Example 2.1 The Jacobian matrix of the acoustic planar wave propagation model in a medium at rest is

$$\mathbf{A} = \begin{pmatrix} 0 & \dfrac{1}{\rho_0} \\ \rho_0 c_0^2 & 0 \end{pmatrix},$$

and the eigenvalues and eigenvectors of this matrix are:

$$\lambda_1 = -c_0, \quad \lambda_2 = c_0, \quad \mathbf{e}_1 = (-1,\, c_0\rho_0)^T, \quad \mathbf{e}_2 = (1,\, c_0\rho_0)^T.$$

Hence, it is a strictly hyperbolic system of conservation laws, unless $c_0 = 0$.

Exercise 2.1 Check that the one-dimensional shallow water equations with constant bottom constitute a hyperbolic system of conservation laws. Find the eigenvalues and eigenvectors of the Jacobian matrix.

Solution 2.1 Left to the reader.

Exercise 2.2 Check that the one-dimensional Euler equations constitute a hyperbolic system of conservation laws. Find the eigenvalues and eigenvectors of the Jacobian matrix.

Solution 2.2 Left to the reader.

Exercise 2.3 Check that the Cauchy-Riemann equations do not constitute a hyperbolic system of conservation laws. Find the eigenvalues and eigenvectors of the Jacobian matrix.

Solution 2.3 Left to the reader.

2.4 One-Dimensional Linear Hyperbolic System of Conservation Laws

A one-dimensional *linear* hyperbolic system of conservation laws has the form:

$$\mathbf{w}_t + \mathbf{A}\mathbf{w}_x = 0, \quad \mathbf{f}(\mathbf{w}) = \mathbf{A}\mathbf{w}, \tag{2.8}$$

where $\mathbf{A}$ is a $p \times p$ matrix of constant coefficients. Since it is a hyperbolic system, the matrix $\mathbf{A}$ has p real eigenvalues λ_k and p linearly independent eigenvectors $\mathbf{e_k}$.

2.4.1 Solution of a Cauchy Problem Associated with a Linear Hyperbolic System of Conservation Laws

We will see how we can find the solution of a Cauchy problem associated with (2.8), with initial conditions $\mathbf{w}(x, 0) = \mathbf{w}^0(x)$, knowing the solution of the transport equation:

- We define the diagonal matrix Λ formed by the eigenvalues ($\Lambda_{kk} = \lambda_k$), $k = 1, \ldots, p$, and the change of basis matrix $\mathbf{P}$, of constant coefficients, whose columns are the corresponding eigenvectors of $\mathbf{A}$, such that $\mathbf{A}\mathbf{e_k} = \lambda_k \mathbf{e_k}$ (right eigenvectors).
- We define the characteristic variables $\mathbf{u} = \left(u_1, \ldots, u_p\right)^T$ in terms of the change of basis matrix $\mathbf{P}$, $\mathbf{w} = \mathbf{P}\mathbf{u}$. Hence, since $\mathbf{P}$ is a constant coefficient matrix, the spatial and vectorial derivatives of the conservative variables, in terms of the characteristic variables, are given by:

$$\mathbf{w}_t = \mathbf{P}\mathbf{u}_t, \quad \mathbf{w}_x = \mathbf{P}\mathbf{u}_x, \tag{2.9}$$

  and replacing the partial derivatives of $\mathbf{w}$ with respect to t and x in (2.8), we have

$$\mathbf{P}\mathbf{u}_t + \mathbf{A}\mathbf{P}\mathbf{u}_x = 0 \Longrightarrow \mathbf{P}^{-1}\mathbf{P}\mathbf{u}_t + \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{u}_x = 0. \tag{2.10}$$

  Moreover, taking into account that $\Lambda = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$, this law can be expressed in the form:

$$\mathbf{u}_t + \Lambda \mathbf{u}_x = 0. \tag{2.11}$$

- Since Λ is a diagonal matrix, the former expression is reduced to p transport equations:

$$(u_k)_t + \lambda_k (u_k)_x = 0 \quad k = 1, \ldots, p. \tag{2.12}$$

- As we have previously seen, the solution of each of these equations is know:

$$u_k(x, t) = u_k^0(x - \lambda_k t) \quad k = 1, \ldots, p, \tag{2.13}$$

  where the components of the initial conditions $u_k^0(x)$ of the characteristic variables are given by the expression:

$$\mathbf{u}^0(x) = \mathbf{P}^{-1}\mathbf{w}^0(x). \tag{2.14}$$

- Finally, undoing the change of variable, we obtain the solution $\mathbf{w}$ at every point and time.

$$\mathbf{w}(x, t) = \mathbf{P}\mathbf{u}(x, t). \tag{2.15}$$

Exercise 2.4 Express the acoustic equations as a system of hyperbolic conservation laws and find the solutions. Identify the conservative and characteristic variables.

Solution 2.4 Left to the reader.

2.4.2 Solution of a Riemann Problem Associated with a Hyperbolic Linear System

In this section we will examine the form taken by (2.15) if the Cauchy problem corresponds to a Riemann problem. We will use this expression both to develop numerical methods and to validate them.

Let us assume that the initial conditions are discontinuous and have the form:

$$\mathbf{w}(x,0) = \mathbf{w}_0(x) = \begin{cases} \mathbf{w}_L \text{ if } x < 0, \\ \mathbf{w}_R \text{ if } x > 0. \end{cases} \tag{2.16}$$

Following (2.15), the solution $\mathbf{w}$ can be expressed as:

$$\mathbf{w} = \sum_{k=1}^{p} \alpha_k \mathbf{e}_k, \tag{2.17}$$

where we denote by $\mathbf{e}_k$, $k = 1, \ldots, p$ the columns of $\mathbf{P}$, which are coincident with the eigenvectors of the matrix $\mathbf{A}$. Therefore, the coordinates of $\mathbf{w}$ in the basis of eigenvectors are given by α_k, $k = 1, \ldots, p$.

In order to build the solution of the linear Riemann problem, and following LeVeque [17], we must sort the eigenvalues of $\mathbf{A}$ from smallest to largest. Since there are p different eigenvalues, we can write:

$$\lambda_1 < \lambda_2 < \cdots < \lambda_p. \tag{2.18}$$

Next we use, on the one side, the form (2.17) to express the initial states on both sides of the discontinuity:

$$\mathbf{w}_L = \sum_{k=1}^{p} \alpha_k^L \mathbf{e}_k \quad , \quad \mathbf{w}_R = \sum_{k=1}^{p} \alpha_k^R \mathbf{e}_k. \tag{2.19}$$

On the other hand, note that, in the characteristic variables $\mathbf{u} = \left(u_1, \ldots, u_p\right)^T$, the Riemann problem is also decoupled into p scalar Riemann problems governed by the transport equation

$$\mathbf{u}_t + \Lambda \mathbf{u}_x = 0 \tag{2.20}$$

and the initial conditions

$$u_k(x,0) = u_k^0(x) = \begin{cases} \alpha_k^L \text{ if } x < 0, \\ \alpha_k^R \text{ if } x > 0. \end{cases}$$

The exact solution of these p Riemann problems is given by

$$u_k(x,t) = u_k^0(x - \lambda_k t) = \begin{cases} \alpha_k^L \text{ if } x - \lambda_k t < 0, \\ \alpha_k^R \text{ if } x - \lambda_k t > 0. \end{cases}$$

We can now use this information to reconstruct the exact solution as a function of the conserved variables $\mathbf{w}$ by means of the relation

$$\mathbf{w}(x,t) = \mathbf{P}\mathbf{u}(x,t). \tag{2.21}$$

Moreover, taking into account that the column vector of the exact solution in characteristic variables $\mathbf{u}(x,t)$ can be expressed in the following form

$$\mathbf{u}(x,t) = \begin{pmatrix} \alpha_1^R \\ \vdots \\ \alpha_I^R \\ \vdots \\ \alpha_p^L \end{pmatrix},$$

where I represents the index where the ordered feet of the characteristics change sign from positive to negative, as expressed by the following inequality, based on (2.18)

$$x - \lambda_1 t > x - \lambda_2 t > \cdots > x - \lambda_I t > 0 > x - \lambda_{I+1} t > \cdots > x - \lambda_p t. \tag{2.22}$$

Hence, $\mathbf{w}$ can be rewritten as

$$\mathbf{w}(x,t) = \left(\mathbf{e}_1 \,\middle|\, \cdots \,\middle|\, \mathbf{e}_p \right) \begin{pmatrix} \alpha_1^R \\ \vdots \\ \alpha_I^R \\ \vdots \\ \alpha_p^L \end{pmatrix} \begin{matrix} \leftarrow \text{ Coefficients corresponding to} \\ \text{the right state } \mathbf{w}_R \\ \hline \leftarrow \text{ Coefficients corresponding to} \\ \text{the left state } \mathbf{w}_L \end{matrix}.$$

Therefore, the first I coordinates of the solution in the eigenvector base, i.e., the α_k coefficients, correspond to the first I right states, and the remaining $p - I$ coordinates correspond to the left states

$$\mathbf{w}(x,t) = \sum_{k=1}^{I} \alpha_k^R \mathbf{e}_k + \sum_{k=I+1}^{p} \alpha_k^L \mathbf{e}_k. \tag{2.23}$$

2.4.2.1 A Particular Case of a System of Two One-Dimensional Equations with Two Eigenvalues of Opposite Sign

In order to analyze and better understand the structure of the solution of the linear Riemann problem (2.23) we consider a particular case of $p = 2$, that can be identified with the linear system of the acoustic Eq. (1.41) considering, for instance, initial conditions of the form (2.16).

Let us distinguish three kind of points (one in each of the regions defined by the characteristics $\frac{dX}{dt} = \lambda_1$ and $\frac{dX}{dt} = \lambda_2$, both of them intercepting the origin and dividing the xt plane). At a certain time $t^* > 0$ we depict in Fig. 2.1 the three points (x_i, t^*), $i = 1, 2, 3$. We assume, as stated in the title of this section, that $\lambda_1 < 0$ and $\lambda_2 > 0$ without loss of generality.

- Point 1 of coordinates (x_1, t^*) on the left of the characteristic curve ($\frac{dX}{dt} = \lambda_1$, $X(0) = 0$). The two interception points with the x axis of the characteristic curves that pass through this point are negative:

$$x_1 - \lambda_1 t^* < 0,\ x_1 - \lambda_2 t^* < 0,$$

  then $I = 0$ and hence the expression of the solution is

$$\mathbf{w}(x_1, t^*) = \sum_{k=1}^{2} \alpha_k^L \mathbf{e}_k = \mathbf{w}_L, \tag{2.24}$$

  that coincides with value $\mathbf{w}_L$.
- Point 2 of coordinates (x_2, t^*) is characterized by the fact that the characteristics that pass through this point intercept the x axis at both sides of the origin:

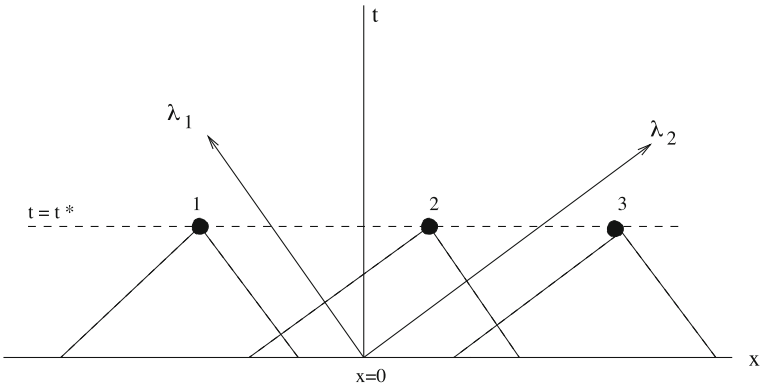


Fig. 2.1 Characteristic curves of the linear Riemann problem

$$x_2 - \lambda_2 t^* < 0,\ x_2 - \lambda_1 t^* > 0,$$

so that the solution at this point is written as

$$\mathbf{w}(x_2, t^*) = \alpha_1^R \mathbf{e}_1 + \alpha_2^L \mathbf{e}_2. \tag{2.25}$$

- Point 3 of coordinates (x_3, t^*), like point 1, has the feet of the two characteristic at the same side of the origin, positive in this case:

$$x_3 - \lambda_1 t^* > 0,\ x_3 - \lambda_2 t^* > 0,$$

meaning that $I = 2$, that leads to the following solution

$$\mathbf{w}(x_3, t^*) = \sum_{k=1}^{2} \alpha_k^R \mathbf{e}_k = \mathbf{w}_R. \tag{2.26}$$

Below, we see that it is possible to rewrite the solution for the generic point 2 (2.25) in the form:

$$\mathbf{w}(x_2, t^*) = \alpha_1^L \mathbf{e}_1 + \alpha_2^L \mathbf{e}_2 + (\alpha_1^R - \alpha_1^L)\mathbf{e}_1 = \mathbf{w}_L + (\alpha_1^R - \alpha_1^L)\mathbf{e}_1, \tag{2.27}$$

or

$$\mathbf{w}(x_2, t^*) = \alpha_1^R \mathbf{e}_1 + \alpha_2^R \mathbf{e}_2 - (\alpha_2^R - \alpha_2^L)\mathbf{e}_2 = \mathbf{w}_R - (\alpha_2^R - \alpha_2^L)\mathbf{e}_2. \tag{2.28}$$

Both expressions state that the solution at point 2 can be interpreted on the basis of the initial values that define the Riemann problem and the property that a "jump" occurs at those points for each of the characteristic curve that is crossed. These jumps behave as waves adding or substracting their contribution depending on the employed expression ((2.27) or (2.28), respectively).

In each of the regions between characteristic curves, the solution is constant and a jump of value

$$[\mathbf{w}] = (\alpha_k^R - \alpha_k^L)\mathbf{e}_k, \tag{2.29}$$

is produced when the kth characteristic is crossed. Note that, due to the properties of the conservative variables, these jumps verify:

$$\begin{aligned}[\mathbf{f}] = \mathbf{A}\,[\mathbf{w}] &= (\alpha_k^R - \alpha_k^L)\mathbf{A}\mathbf{e}_k = (\alpha_k^R - \alpha_k^L)\lambda_k \mathbf{e}_k \\ &= \lambda_k (\alpha_k^R - \alpha_k^L)\mathbf{e}_k = \lambda_k\,[\mathbf{w}]\,,\end{aligned}$$

this is, λ_k is the propagation velocity of each of these jumps.

Finally, we can take advantage of the double definition of the solution at a generic point (2.27) and (2.28) to express the difference between the initial left and right states as a sum of jumps

$$\mathbf{w}_R - \mathbf{w}_L = (\alpha_1^R - \alpha_1^L)\mathbf{e}_1 + (\alpha_2^R - \alpha_2^L)\mathbf{e}_2. \tag{2.30}$$

Exercise 2.5 Given the system of partial differential equations that represents the linearized shallow water equations problem introduced in Sect. 1.9 when the bottom is flat ($Z'(x) = 0$) and the channel width is constant ($B'(x) = 0$):

$$h_t + \tilde{u}h_x + \tilde{h}u_x = 0, \tag{2.31}$$
$$u_t + gh_x + \tilde{u}u_x = 0, \tag{2.32}$$

with $\tilde{h},\ \tilde{u},\ g = 9.8\,\mathrm{ms}^{-2}$ constants and $h = h(x,t),\ u = u(x,t)$ representing the water height and the velocity, respectively. It is requested:

1. To represent it as a system of linear conservation laws.
2. To determine the hypothesis under which the constants $\tilde{h},\ \tilde{u}$ lead to a hyperbolic system and a strictly hyperbolic system.
3. To provide the exact solution of a Riemann problem associated with this system of conservation laws and implement it in a computer code. Assume that data $\tilde{h},\ \tilde{u},\ g$ and the initial conditions of the Riemann problem $h_l,\ u_l,\ h_r,\ u_r$ are given.

Solution 2.5 1. These equations can be written as a system of linear conservation laws:

$$\mathbf{w}_t + (\mathbf{f}(\mathbf{w}))_x = \mathbf{0}, \tag{2.33}$$

where the conservative variables and the flux functions of this linear system are defined by:

$$\mathbf{w} = (h,u)^T, \quad \mathbf{f}(\mathbf{w}) = \left(\widetilde{u}h + \widetilde{h}u, gh + \widetilde{u}u\right)^T. \tag{2.34}$$

It is a linear system since the flux can be written as:

$$\mathbf{f}(\mathbf{w}) = \mathbf{A}\mathbf{w}, \quad \text{where} \quad \mathbf{A} = \begin{pmatrix} \widetilde{u} & \widetilde{h} \\ g & \widetilde{u} \end{pmatrix}. \tag{2.35}$$

2. The Jacobian matrix of the flux is $\frac{\partial \mathbf{f}(\mathbf{w})}{\partial \mathbf{w}} = \mathbf{A}$, since we are dealing with a linear system. The eigenvalues of $\mathbf{A}$ are:

$$\lambda_1 = \widetilde{u} + \sqrt{g\widetilde{h}}, \quad \lambda_2 = \widetilde{u} - \sqrt{g\widetilde{h}}. \tag{2.36}$$

A necessary condition for the system to be hyperbolic is that $\widetilde{h} \geq 0$, which holds immediately if $\widetilde{h}$ represents a mean water height. The system would be strictly hyperbolic if $\widetilde{h} > 0$. The eigenvectors of matrix $\mathbf{A}$ are:

$$\mathbf{e}_1 = \left(\frac{\sqrt{g\widetilde{h}}}{g}, 1\right)^T, \quad \mathbf{e}_2 = \left(-\frac{\sqrt{g\widetilde{h}}}{g}, 1\right)^T. \tag{2.37}$$

It is trivial to prove that they are linearly independent if $\widetilde{h} > 0$.

3. In order to solve the Riemann problem, let us assume that the initial condition is:

$$\mathbf{w}(x,0) = \mathbf{w}_0(x) = \begin{cases} \mathbf{w}_L & \text{if} \quad x < 0, \\ \mathbf{w}_R & \text{if} \quad x > 0, \end{cases} \tag{2.38}$$

where $\mathbf{w}_L = (h_L, u_L)^T$ and $\mathbf{w}_R = (h_R, u_R)^T$. Hence, the solution of the Riemann problem is

$$\mathbf{w}(x,t) = \begin{cases} \mathbf{w}_L & \text{if } x - \lambda_1 t < 0, \\ \mathbf{w}_L + \left(\alpha_1^R - \alpha_1^L\right)\mathbf{e}_1 & \text{if } (x - \lambda_1 t)(x - \lambda_2 t) < 0, \\ \mathbf{w}_R & \text{if } x - \lambda_2 t > 0, \end{cases} \tag{2.39}$$

where the constants α_1^L and α_1^R, are defined by

$$\alpha_1^L = u_L + \sqrt{\frac{g}{\widetilde{h}}} h_L, \quad \alpha_1^R = u_R + \sqrt{\frac{g}{\widetilde{h}}} h_R. \tag{2.40}$$

The solution (2.39) was implemented in MATLAB. Figure 2.2 represents the solution at $t = 0,1$ s for the following choices of parameters and initial conditions:

$$\begin{aligned} &\tilde{h} = 7\,\text{m}, \quad \tilde{u} = 0.5\,\text{ms}^{-1}, \\ &\mathbf{w}_L = (h_L, u_L)^T = (10,0)^T, \quad \mathbf{w}_R = (h_R, u_R)^T = (5,0)^T. \end{aligned} \tag{2.41}$$

The figure depicts a dam break in a channel 6 m long, with constant rectangular section and flat bottom. The initial conditions are two different water levels at rest of heights $h_L = 10$ and $h_R = 5$, being the dam located at $x = 0$.
In Fig. 2.2 we see that the values of the solution for the points in Zone 1 (defined by $x - \lambda_1 t < 0$) are the same as the left state, this is, the initial values upstream the dam $\mathbf{w}(x,t) = \mathbf{w}_L$. In Zone 3 (defined by $x - \lambda_2 t > 0$), the solution reproduces the right state $\mathbf{w}(x,t) = \mathbf{w}_R$. However, in Zone 2 (defined by $(x - \lambda_1 t)(x - \lambda_2 t) < 0$), where the feet of the characteristics are located on both sides of the initial discontinuity ($x = 0$ in this case), the solution contains information from both the left and right initial states (recall the theoretical case treated in Fig. 2.1). This can be observed graphically in Fig. 2.2, and in equation (2.39) the coordinates of the states are given in the basis of eigenvectors, this is, $\mathbf{w}(x,t) = \mathbf{w}_L + \left(\alpha_1^R - \alpha_1^L\right)\mathbf{e}_1$.

Exercise 2.6 Provide the exact solution of a Riemann problem associated with the planar wave acoustic propagation model in a medium at rest.

Solution 2.6 Left to the reader.

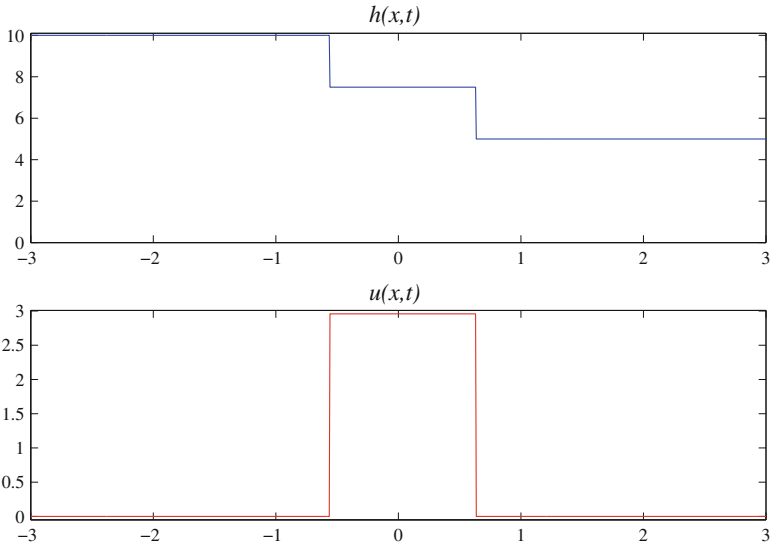


Fig. 2.2 Solution of the Riemann problem for the linear shallow water equations at $t = 0.08$ s

2.5 Multidimensional Systems of Hyperbolic Conservation Laws

The concepts presented in this chapter for one-dimensional problems can be extended to the multidimensional case. The spatial variable $\mathbf{x} = (x_1, \ldots, x_d)$ belongs to $\mathbf{R}^d$ in the multidimensional problems, the flux will have now d components, each of which will have its image in $\mathbf{R}^p$. The spatial derivative of the flux in the one-dimensional case transforms into its divergence $\sum_{j=1}^{d} \left(\mathbf{f}_j(w)\right)_{x_j}$ when extending the problem to the multidimensional case. This section will be devoted to the general definitions that are given in [8].

Definition 2.4 Let Ω be a subset of $\mathbf{R}^p$ and $\mathbf{f}_j$, $1 \leq j \leq d$, d smooth functions from Ω to $\mathbf{R}^p$:

$$\mathbf{f}_j : \Omega \longrightarrow \mathbf{R}^p,\ 1 \leq j \leq d. \tag{2.42}$$

The general form of a *system of conservation laws* is

$$\mathbf{w}_t + \sum_{j=1}^{d} \left(\mathbf{f}_j(\mathbf{w})\right)_{x_j} = \mathbf{0}, \quad \mathbf{x} = (x_1, \ldots, x_d) \in \mathbf{R}^d, \quad t > 0, \tag{2.43}$$

where $\mathbf{w} = (w_1, \ldots, w_p)$, the *vector of conservative variables*, is the vectorial function:

$$\mathbf{w} : \mathbf{R}^d \times [0, \infty) \longrightarrow \Omega, \tag{2.44}$$

$$(x_1, \ldots, x_d, t) \longrightarrow (w_1, \ldots, w_p). \tag{2.45}$$

The set Ω is called *set of states* and the functions $\mathbf{f}_j = (f_{1j}, \ldots, f_{pj})$, physical flux $\big(f_{ij} : \Omega \longrightarrow \mathbf{R}\big)$.

Definition 2.5 *Cauchy Problem*:

Find a function $\mathbf{w} : (\mathbf{x}, t) \in \mathbf{R}^d \times [0, \infty) \longrightarrow \mathbf{w}(\mathbf{x}, t) \in \Omega$, solution of (2.2) that verifies the initial condition:

$$\mathbf{w}(\mathbf{x}, 0) = \mathbf{w}_0(\mathbf{x}), \quad \mathbf{x} \in \mathbf{R}^d, \tag{2.46}$$

where $\mathbf{w}_0 : \mathbf{x} \in \mathbf{R}^d \longrightarrow \mathbf{w}_0(\mathbf{x}) \in \Omega$ is a given function.

2.5.1 Two-Dimensional Shallow Water Equations

The two-dimensional shallow water equations with a flat bottom (Stoker [26]) are:

$$d = 2, \quad p = 3, \quad \mathbf{w} = (h,\, q_1,\, q_2)^T,$$

$$\mathbf{f}_1(\mathbf{w}) = \left(q_1,\, \frac{q_1^2}{h} + \frac{1}{2}gh^2,\, \frac{q_1 q_2}{h}\right)^T, \quad \mathbf{f}_2(\mathbf{w}) = \left(q_2,\, \frac{q_1 q_2}{h},\, \frac{q_2^2}{h} + \frac{1}{2}gh^2\right)^T.$$

The notation is analogous to the previous one and, in this case, the vector $\mathbf{u} = (u_1, u_2)$ is the vertically averaged horizontal velocity and $\mathbf{q} = (q_1, q_2) = h\mathbf{u}$. Figure 2.3 depicts the water height h and the velocity $\mathbf{u}$ that are obtained as a solution of the Riemann problem that represents a dam break in a domain with constant bottom by using a finite volume method.

The definition of hyperbolic systems of conservation laws in the multidimensional case is as follows:

Definition 2.6 Let us consider a system of the form (2.43). For every flux function $\mathbf{f}_j$, $j = 1, \ldots, d$, we define its Jacobian matrix $\mathbf{A}_j$:

$$\mathbf{A_j}(\mathbf{w}) = \left(\frac{\partial f_{ij}}{\partial w_k}(\mathbf{w})\right)_{1 \le i,k \le p}.$$

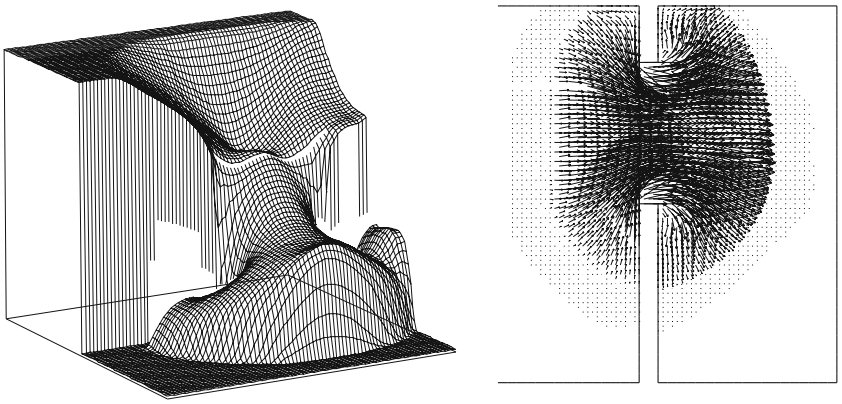


Fig. 2.3 Solution of the 2D shallow water equations for a dam break problem with constant bottom, water height (*left*) and velocity (*right*)

System (2.2) *is called hyperbolic* if, for any $\eta = (\eta_1, \ldots, \eta_d) \in \mathbf{R}^d$, the matrix

$$\mathbf{A}(\mathbf{w}, \eta) = \sum_{j=1}^{d} \eta_j \mathbf{A_j}(\mathbf{w}),$$

has p real eigenvalues $\lambda_1(\mathbf{w}, \eta) \leq \lambda_2(\mathbf{w}, \eta) \leq \cdots \leq \lambda_p(\mathbf{w}, \eta)$ and the p corresponding eigenvectors are linearly independent. Moreover, if all the eigenvalues are different, then the system is called *strictly hyperbolic*.

Exercise 2.7 Check that the two-dimensional shallow water equations with a flat bottom constitute a hyperbolic system of conservation laws. Find the eigenvalues and eigenvectors of the Jacobian matrix.

Solution 2.7 Left to the reader.

In this chapter we have dealt with the basic concepts that allow us to define the framework of application of the finite volume method in this book: the hyperbolic systems of conservation laws. In the next chapter we will review the different kind of solutions of these problems. Through these solutions, we will face the difficulties that the numerical methods considered should overcome in order to solve these problems. In addition, the solutions will be used to validate the numerical methods and, in some cases they will allow us to define them, as we will see when introducing the Godunov method.

Chapter 3
Types of Solutions to Hyperbolic Systems of Conservation Laws

Abstract This chapter is aimed at knowing the different types of solutions of hyperbolic conservation laws as well as some examples. The study of these solutions will allow us to identify the difficulties that the numerical methods should overcome in order to approximate them. Hence, we will better understand and implement different numerical methods, such as Godunov's method. The solutions of the hyperbolic conservation laws will also be used in the analysis and validation of the different numerical methods.

3.1 Classical Solution

Definition 3.1 Classical solution:

We say that a function $\mathbf{w} : \mathbf{R}^d \times [0, \infty) \longrightarrow \boldsymbol{\Omega}$ is a *classical solution* of the Cauchy problem (2.43)–(2.46) if $\mathbf{w}$ is a function of class C^1 that verifies (2.43)–(2.46) pointwise.

One essential characteristic of the Cauchy problem (2.43)–(2.46) is that, *in general, no solutions in the classical sense exist.* This is because, even if the initial condition is a smooth function, solutions with discontinuities may arise from a certain finite time interval on.

3.2 Construction of the Solution by the Method of Characteristics: Calculation of the Critical Time

The starting point is the scalar one-dimensional equation:

$$w_t + \lambda(w) w_x = 0, \quad \lambda(w) = f'(w) \quad \text{in} \quad \mathbf{R} \times [0, T]. \tag{3.1}$$

The Cauchy problem is defined for this equation in the same domain with the initial conditions:

$$w(x, 0) = w_0(x), \quad \text{in} \quad \mathbf{R} \times [0, T]. \tag{3.2}$$

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The corresponding characteristic curves, that are obtained by solving

$$\frac{dX}{dt}(t) = \lambda(w), \quad X(0) = \xi, \tag{3.3}$$

where ξ is the intersection of the corresponding characteristic with axis $t = 0$, are of the form:

$$X(t) - \xi = \lambda(w_0(\xi))t. \tag{3.4}$$

We will prove next that a sufficient condition for there to be singularities is that w_0 must be strictly decreasing in any open interval I:

- Let us assume that $w_0(x) > 0$ and that $w_0'(x) < 0$ in I.
- We consider the characteristic curves emanating from points ξ_1 and ξ_2 in the x axis with velocities $\lambda(w_0(\xi_1))$ and $\lambda(w_0(\xi_2))$, respectively.
- If we take into account the expression for λ in (3.4):

$$\lambda(w_0(\xi)) = \frac{X(t)}{t} - \frac{\xi}{t} \Longrightarrow \frac{d\lambda}{d\xi}(w_0(\xi)) = -\frac{1}{t} < 0, \tag{3.5}$$

  i.e. λ is decreasing as a function of ξ. Therefore, we have:

$$\xi_1 < \xi_2 \Longrightarrow \lambda(w_0(\xi_1)) > \lambda(w_0(\xi_2)). \tag{3.6}$$

  This is, the characteristic curve emanating from ξ_1 is faster than the one emanating from ξ_2.[1] This fact implies that the characteristic curves intersect, which contradicts the definition of the solution since:

$$w(X(t), t) = w_0(\xi) = \overline{w}, \tag{3.7}$$

  with $\overline{w}$ a constant which is different for every characteristic curve.
- Consequently, it is not possible that a smooth solution exists for every $t > 0$. In order to determine the time when a gradient *chatastrophe* occurs, or when the gradient is not defined anymore, we calculate w_x along the characteristic curve (3.4).
- We define $g(t) := w_x(X(t), t)$ as the gradient along the characteristic curves. Then, considering (3.4):

$$\frac{dg}{dt} = w_{tx} + w_{xx}\lambda(w).$$

- Moreover, if we take the derivative of the conservation law (3.1) with respect to x we have:

[1] The slopes are the inverse of λ, we represent them in the xt plane and calculate $\frac{dx}{dt}$.

$$w_{tx} + w_{xx}\lambda(w) = -(w_x)^2\lambda'(w) = -(g(t))^2\lambda'(w).$$

- Therefore, taking into account both conditions above, every characteristic curve verifies:

$$\frac{dg}{dt} = -\lambda'(w)g^2(t). \tag{3.8}$$

- If we integrate the ordinary differential equation (3.8) along the characteristics and consider (3.7):

$$\int_0^t \frac{1}{g^2(s)}\frac{dg}{ds}\,ds = -\int_0^t \lambda'(\overline{w})ds, \tag{3.9}$$

we obtain the following expression for the gradient $g(t) = w_x$:

$$g(t) = \frac{g(0)}{1 + \lambda'(\overline{w})tg(0)} = \frac{w_0'(\xi)}{1 + \lambda'(w_0(\xi))tw_0'(\xi)}. \tag{3.10}$$

- On the basis of the previous hypothesis, w_0' and λ' have opposite signs. Hence, the denominator of (3.10) may vanish. Then, if we analyze w_x along all the characteristics parameterized by ξ, the *critical breaking time* will occur at the characteristic curve for which the denominator first vanishes.
- If we call $F(\xi) := \lambda(w_0(\xi))$, we can conclude that wave breaking occurs at the characteristic curve $\xi = \xi_b$ for which $|F'(\xi)| = |\lambda'(w_0(\xi))w_0'(\xi)|$ is a maximum. Thus, the breaking time (see Fig. 3.1) is:

$$t_b = -\frac{1}{F'(\xi_b)}. \tag{3.11}$$

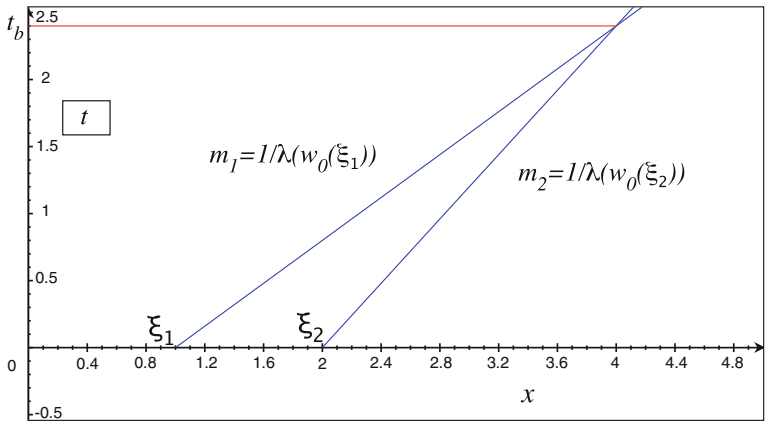


Fig. 3.1 Calculation of the critical time t_b

- If the initial condition w_0 is not monotone, breaking will occur for the first time at the characteristic $\xi = \xi_b$ for which $F'(\xi) < 0$ and $|F'(\xi)|$ is a maximum:

$$t_b = -\min_{\xi} \frac{1}{F'(\xi)}. \tag{3.12}$$

Exercise 3.1 Calculate the breaking time for the Burgers equation with initial condition $w_0(x) = \mathrm{e}^{-x^2}$.

Solution 3.1 $t_b = \frac{\sqrt{2}}{2}\mathrm{e}^{1/2}$.

Exercise 3.2 Solve the Cauchy problem that corresponds to the Burgers equation with the initial condition:

$$w_0(x) = \begin{cases} 1 & \text{if } x \le 0, \\ 1-x & \text{if } 0 < x \le 1, \\ 0 & \text{if } x > 1, \end{cases} \tag{3.13}$$

for times smaller than the breaking time $t = 1$.

Solution 3.2 We need to calculate the breaking time, the characteristic curves and evaluate the initial condition at the crossing points of the corresponding characteristics.

In the Burgers equation $\lambda(w) = f'(w) = w$, if we solve the characteristic equation for the time before the breaking time, $t \le 1$, we have:

$$x(\xi, t) = \xi + t w_0(\xi) = \begin{cases} \xi + t & \text{if } \xi \le 0, \\ \xi + t(1-\xi) = \xi(1-t) + t & \text{if } 0 < \xi \le 1, \\ \xi & \text{if } \xi > 1. \end{cases}$$

Figure 3.2 shows the characteristic curves of Burgers equation with initial condition (3.13).

We obtain the intersection point of the characteristics with axis $t = 0$ as a function of x and t:

$$\xi(x, t) = \begin{cases} x - t & \text{if } x \le t \le 1, \\ \dfrac{x-t}{1-t} & \text{if } t < x \le 1, \\ x & \text{if } x > 1. \end{cases} \tag{3.14}$$

Finally, we calculate the expression of the solution at (x, t) taking into account that it is constant along the characteristics, i.e. $w(x, t) = w_0(\xi)$:

$$W(x, t) = W_0(\xi) = \begin{cases} 1 & \text{if } x \le t < 1, \\ 1 - \dfrac{x-t}{1-t} = \dfrac{1-x}{1-t} & \text{if } t < x \le 1, \\ 0 & \text{if } x > 1,\ t \le 1. \end{cases} \tag{3.15}$$

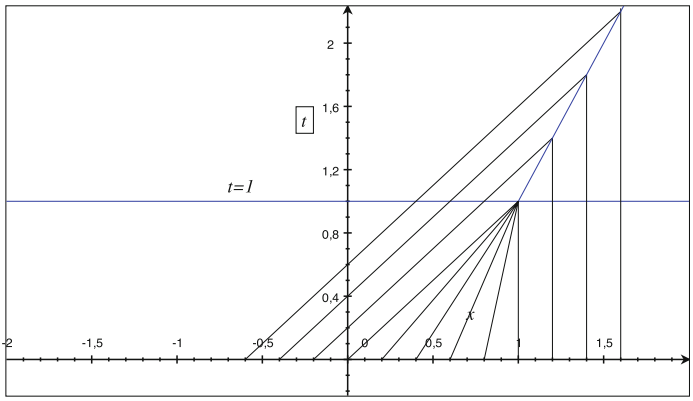


Fig. 3.2 Characteristic curves of Burgers equation with initial condition (3.13)

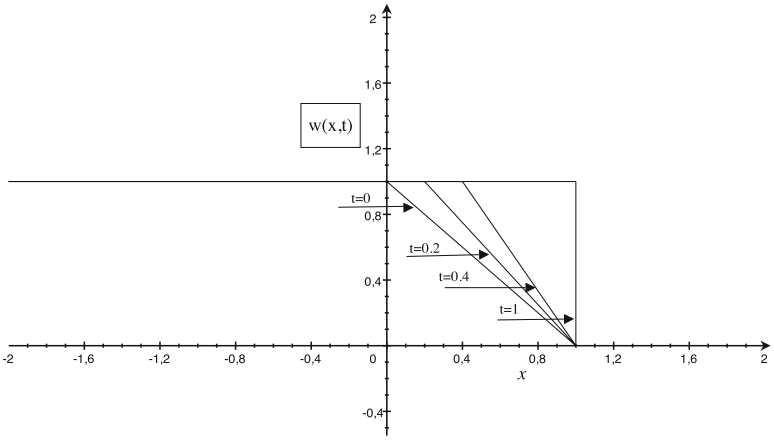


Fig. 3.3 Solution of the Burgers equation with initial condition (3.13)

The solution for different values of t is depicted in Fig. 3.3.

Example 3.1 Representation of the process of formation of a discontinuity in the shallow water equations.

If we assume an initial wave as shown in Fig. 3.4, at time $t = t_0$, the velocities of points a, b, c would not be the same.

After a certain time, point a would reach point b and this, in turn, would reach point c. In particular, at time $t = t_1$, this process would have already occurred and the map of the trajectories in the (x, t) plane shows that this situation is reflected by the intersection of two characteristic curves. In hydraulics, these discontinuities are associated with hydraulic jumps, either stationary or movile when the profile of the surface shows significant differences at both sides. In gas dynamics, we would be dealing with the so-called shock wave [30].

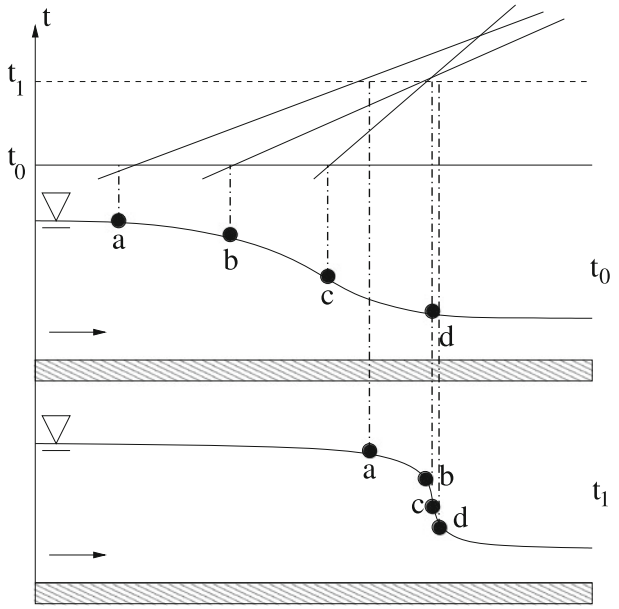


Fig. 3.4 Representation of the process of formation of a discontinuity in the shallow water equations

3.3 Weak Solution

The existence of discontinuous solutions presented in the previous section leads us to the introduction of the *weak solutions of the Cauchy problem*.

3.3.1 Integral Form of the Hyperbolic Systems of Conservation Laws

Let $C_0^1(\mathbf{R}^d \times [0, +\infty))$ be the space of functions C^1 with compact support in $\mathbf{R}^d \times [0, +\infty)$.

If $\mathbf{w}$ is a classical solution and $\varphi \in C_0^1(\mathbf{R}^d \times [0, +\infty))^p$, by using Green's formula we have:

$$
\begin{aligned}
0 &= \int_0^\infty \int_{\mathbf{R}^d} \left\{ \frac{\partial \mathbf{w}}{\partial t} + \sum_{j=1}^{d} \frac{\partial}{\partial x_j} \left(\mathbf{f}_j(\mathbf{w}) \right) \right\} \cdot \varphi \, dx dt \\
&= - \int_0^\infty \int_{\mathbf{R}^d} \left\{ \mathbf{w} \frac{\partial \varphi}{\partial t} + \sum_{j=1}^{d} \mathbf{f}_j(\mathbf{w}) \frac{\partial \varphi}{\partial x_j} \right\} dx dt - \int_{\mathbf{R}^d} \mathbf{w}(\mathbf{x}, 0) \cdot \varphi(\mathbf{x}, 0) \, dx.
\end{aligned}
$$

Therefore, any classical solution of the Cauchy problem ($\mathbf{w}(\mathbf{x}, 0) = \mathbf{w_0}(\mathbf{x})$) verifies:

$$\int_0^\infty \int_{\mathbf{R}^d} \left\{ \mathbf{w} \frac{\partial \varphi}{\partial t} + \sum_{j=1}^{d} \mathbf{f}_j(\mathbf{w}) \frac{\partial \varphi}{\partial x_j} \right\} dxdt + \int_{\mathbf{R}^d} \mathbf{w}_0(\mathbf{x}) \cdot \varphi(\mathbf{x}, 0)\, dx = 0,$$
$$\forall \varphi \in C_0^1(\mathbf{R}^d \times [0, +\infty))^p. \tag{3.16}$$

It is interesting to highlight that (3.16) makes sense if $\mathbf{w} \in L^\infty_{loc}(\mathbf{R}^d \times [0, +\infty))^p$, where L^∞_{loc} is the space of locally bounded measurable functions.

Definition 3.2 Weak solution:
Let us assume that $\mathbf{w}_0 \in L^\infty_{loc}(\mathbf{R}^d)^p$. A function $\mathbf{w} \in L^\infty_{loc}(\mathbf{R}^d \times [0, +\infty))^p$ is called a weak solution of the Cauchy problem (2.2)–(2.4) if it verifies (3.16) for any function $\varphi \in C_0^1(\mathbf{R}^d \times [0, +\infty))^p$.

Hence, any classical solution of (2.2)–(2.4) is also a weak solution. Conversely, if we consider $\varphi \in C_0^\infty(\mathbf{R}^d \times (0, +\infty))^p$, we have that any weak solution $\mathbf{w}$ verifies (2.2) in the sense of distributions on $\mathbf{R}^d \times (0, +\infty)$. Moreover, if $\mathbf{w}$ is a function of class C^1, then it is a classical solution. Indeed, if $\varphi \in C_0^1(\mathbf{R}^d \times (0, +\infty))^p$, by integrating (3.16) by parts,

$$\int_0^\infty \int_{\mathbf{R}^d} \left\{ \frac{\partial \mathbf{w}}{\partial t} + \sum_{j=1}^{d} \frac{\partial}{\partial x_j} \left(\mathbf{f}_j(\mathbf{w}) \right) \right\} \cdot \varphi \, dxdt = 0,$$

and hence (2.2) is verified pointwise. If we multiply (2.2) by a test function $\varphi \in C_0^1(\mathbf{R}^d \times [0, +\infty))^p$, integrate by parts and compare with (3.16), we obtain:

$$\int_{\mathbf{R}^d} \{\mathbf{w}(\mathbf{x}, 0) - \mathbf{w}_0(\mathbf{x})\} \cdot \varphi \, dx = 0,$$

which yields (2.4) pointwise.

Let us focus now on solutions of (2.2) in the sense of distributions that are only piecewise continuous and hence, admit discontinuities.

On the basis of the previous results, it is proven that any solution in the sense of distributions is a classical solution of (2.2) in any domain where $\mathbf{w}$ is C^1. We will prove next that not all the discontinuities are admissible.

Definition 3.3 We say that *a function* $\mathbf{w}$ *is piecewise* C^1 if there exists a finite number of smooth orientable surfaces Σ in the $(\mathbf{x}, t)$-space, such that $\mathbf{w}$ is C^1 out of these surfaces and presents a jump discontinuity across them.

Given a surface of discontinuity Σ, we denote by

$$\mathbf{n} = (n_{x_1}, \ldots, n_{x_d}, n_t) \neq 0,$$

the outward pointing normal vector and by $\mathbf{w}_+$ and $\mathbf{w}_-$ the limits of $\mathbf{w}$ at both sides of Σ:

$$\mathbf{w}_\pm(\mathbf{x}, t) = \lim_{\varepsilon \to 0} \mathbf{w}((\mathbf{x}, t) \pm \varepsilon \mathbf{n}).$$

Theorem 3.1 *Let* $\mathbf{w} : \mathbf{R}^d \times [0, \infty) \longrightarrow \Omega$ *be a piecewise* C^1 *function. Then,* $\mathbf{w}$ *is a solution of (2.2) in the sense of distributions in* $\mathbf{R}^d \times (0, +\infty)$ *if and only if the following two conditions hold:*

(i) $\mathbf{w}$ *is a classical solution of (2.2) in the domain where* $\mathbf{w}$ *is* C^1.
(ii) $\mathbf{w}$ *verifies the jump condition:*

$$(\mathbf{w}_+ - \mathbf{w}_-)\, n_t + \sum_{j=1}^{d} \left(\mathbf{f}_j(\mathbf{w}_+) - \mathbf{f}_j(\mathbf{w}_-)\right) n_{x_j} = 0, \tag{3.17}$$

along the surfaces of discontinuity.

Proof It can be found in Godlewski and Raviart [8].

The jump expression (3.17) is known as *Rankine-Hugoniot jump condition*. If we call:

$$[\mathbf{w}] = \mathbf{w}_+ - \mathbf{w}_-,$$

the jump of $\mathbf{w}$ across Σ, and analogously

$$[\mathbf{f}_j(\mathbf{w})] = \mathbf{f}_j(\mathbf{w}_+) - \mathbf{f}_j(\mathbf{w}_-),$$

the jump of $\mathbf{f}_j(\mathbf{w})$, $1 \leq j \leq d$, Eq. (3.17) can be written as:

$$n_t[\mathbf{w}] + \sum_{j=1}^{d} n_{x_j}[\mathbf{f}_j(\mathbf{w})] = 0. \tag{3.18}$$

If $(n_{x_1}, \ldots, n_{x_d}) \neq 0$ we can take $\mathbf{n} = (\nu, -s)$, where $s \in \mathbf{R}$ and $\nu = (\nu_1, \ldots, \nu_d)$ is a unit vector in $\mathbf{R}^d$. Then (3.18) is equivalent to

$$s[\mathbf{w}] = \sum_{j=1}^{d} \nu_j [\mathbf{f}_j(\mathbf{w})], \tag{3.19}$$

where s is known as *shock velocity* or velocity of propagation of the discontinuity.

Remark 3.1 In the *one dimensional case* ($d = 1$), if we assume that Σ is a smooth curve, it can be parameterized as $(\sigma(t), t)$ and we have

$$\mathbf{n} = (1, -s), \quad s = \frac{d\sigma}{dt}.$$

Hence, the Rankine-Hugoniot condition is

$$s[w] = [f(w)]. \tag{3.20}$$

Figure 3.5 illustrates the meaning of the parameterization of the surface of discontinuity. For a more detailed view, we zoom in a rectangle of area $\Delta x \times \Delta t$ sufficiently small to assume that the solution is constant on both sides of the discontinuity, represented by w_- and w_+ in this figure.

We find very didactic and graphical the interpretation of the Rankine-Hugoniot jump condition by Karni in [36], characterizing the admissible discontinuities as moving internal boundaries. The rule that governs the propagation of the admissible discontinuities can be obtained in a non-formal way if we appeal to the principle of conservation, that will be described on the rectangle depicted in Fig. 3.5. In this rectangle we will assume that the velocity of the discontinuity can be approximated by $s = \frac{\Delta x}{\Delta t}$.

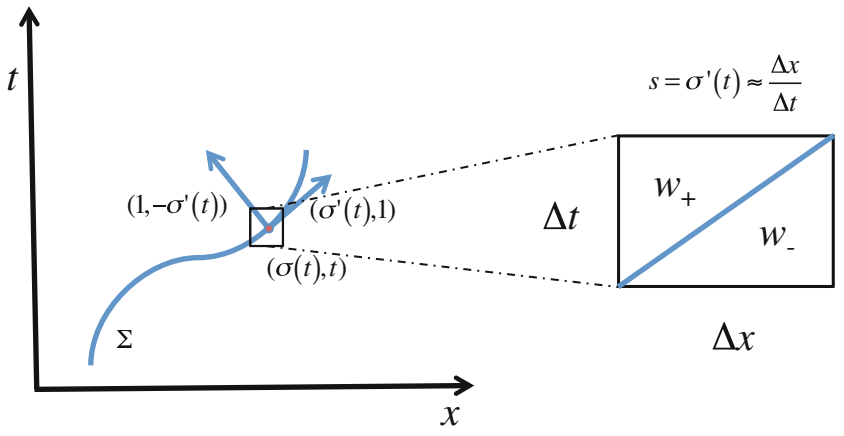


Fig. 3.5 Interpretation of the Rankine-Hugoniot jump condition

If we assume that the total amount of w after a certain time interval Δt is $\Delta x w_+$, which is coincident with the initial amount $\Delta x w_-$ plus the total net flux that crosses the boundary, $\Delta t f(w_-) - \Delta t f(w_+)$:

$$\Delta x\, w_+ \approx \Delta x\, w_- + \Delta t f(w_+) - \Delta t f(w_-) \Longrightarrow \frac{\Delta x}{\Delta t}(w_+ - w_-) \approx f(w_+) - f(w_-).$$

If $\Delta x,\ \Delta t \to 0$ with $s = \frac{\Delta x}{\Delta t}$, we have the Rankine-Hugoniot jump condition, $s[w] = [f(w)]$.

Remark 3.2 A simple way of confirming that $\mathbf{w}$ is a solution in the sense of distributions, if $\mathbf{w}$ is a piecewise C^1 function, consists of checking that it verifies properties (i) and (ii) of the previous theorem.

We will see next how to use the Rankine-Hugoniot jump condition to construct weak solutions of the Cauchy problem:

Example 3.2 Let us consider the Burgers equation (2.6). In this case, condition (3.20) becomes

$$s(w_+ - w_-) = \frac{1}{2}(w_+^2 - w_-^2) \Longrightarrow s = \frac{w_+ + w_-}{2}. \tag{3.21}$$

If the initial conditions are given by (3.13), then we obtain a continuous solution up to the critical time $t = 1$. From this time on, the velocity of propagation of the discontinuity s can be calculated from the jump condition, being $s = \frac{1}{2}$ in this case.

Example 3.3 Two problems with the same classical solutions and different weak solutions:

If we consider the Burgers equation (2.6) and multiply it by $2w$, we obtain:

$$2ww_t + 2w\left(\frac{w^2}{2}\right)_x = 0 \Longrightarrow \left(w^2\right)_t + \left(\frac{2w^3}{3}\right)_x = 0. \tag{3.22}$$

(2.6) and (3.22) have the same classical solutions. However, the jump conditions are different:

- Propagation velocity of (2.6):

$$s = \frac{w_+ + w_-}{2}. \tag{3.23}$$

- Propagation velocity of (3.22):

$$s = \frac{2}{3}\frac{w_+^3 - w_-^3}{w_+^2 - w_-^2}. \tag{3.24}$$

Exercise 3.3 Calculate the jump propagation velocity for the traffic flow model

$$f(\rho) = \rho u_m \left(1 - \frac{\rho}{\rho_m}\right), \tag{3.25}$$

if we consider the following initial conditions

$$\rho(x, 0) = \begin{cases} \rho_l & \text{if } x < 0, \\ \rho_r & \text{if } x > 0. \end{cases} \tag{3.26}$$

Solution 3.3 Left to the reader.

3.4 Solution of the Riemann Problem for the Burgers Equation

The Cauchy problem corresponding to a conservation law with initial conditions given by a constant function with a discontinuity, as was mentioned earlier for the linear systems, is know as *Riemann problem*:

$$w_0(x) = \begin{cases} w_l & \text{if } x < 0, \\ w_r & \text{if } x > 0. \end{cases} \tag{3.27}$$

Considering the *Burgers equation* (2.6) with the mentioned initial conditions, the form of this solution depends on the relation between w_l and w_r:

- If $w_l > w_r$, the jump condition shows that a weak solution of the corresponding Riemann problem, in which the discontinuity propagates with velocity $s = \frac{w_l + w_r}{2}$, is given by (Fig. 3.6):

$$w(x, t) = \begin{cases} w_l & \text{if } \dfrac{x}{t} \leq s, \\ \\ w_r & \text{if } \dfrac{x}{t} > s. \end{cases} \tag{3.28}$$

- If $w_l \leq w_r$, we will see that there are infinitely many weak solutions. Since the characteristics *emanate* from the shock, i.e., from the straight line

$$x = st,$$

where s is given by (3.23), the solution (3.28) is not stable (Fig. 3.7).
Among all the solutions, there is also one continuous solution:

  – By means of the characteristics method we can determine the solution everywhere except in the region $w_l \leq \frac{x}{t} \leq w_r$.

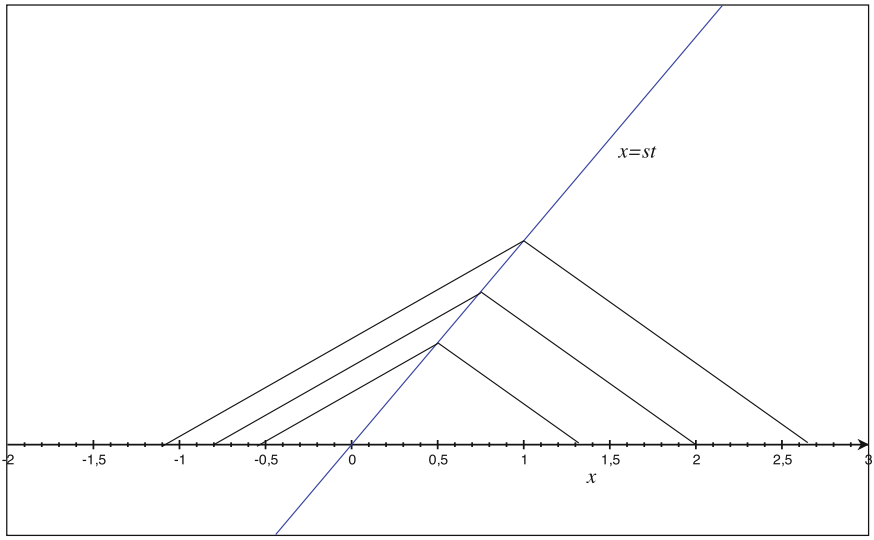


Fig. 3.6 Characteristic lines of the shock wave $w_l > w_r$

– It is immediate to verify that the function $v(x,t) = \dfrac{x}{t}$ is a classic solution of the Burgers equation.
– Hence, the continuous solution:

$$w(x,t) = \begin{cases} w_l & \text{if } x \le w_l t, \\ \dfrac{x}{t} & \text{if } w_l \le \dfrac{x}{t} \le w_r, \\ w_r & \text{if } x \ge w_r t, \end{cases} \tag{3.29}$$

is also a weak solution of the problem, known as *expansion wave* or *rarefaction wave* (Fig. 3.8).
– This solution is stable, and can be obtained using the *vanishing viscosity method* (Godlewski and Raviart [8]).

3.4.1 Non-uniqueness of the Weak Solutions. Examples

In addition to the solutions explained previously, it is also possible to obtain a *family* of weak solutions for the Riemann problem associated with the Burgers equation:

$$w(x,t) = \begin{cases} w_l & \text{if } x < s_1 t, \\ -a & \text{if } s_1 t < x < 0, \\ a & \text{if } 0 < x < s_2 t, \\ w_r & \text{if } x > s_2 t, \end{cases} \tag{3.30}$$

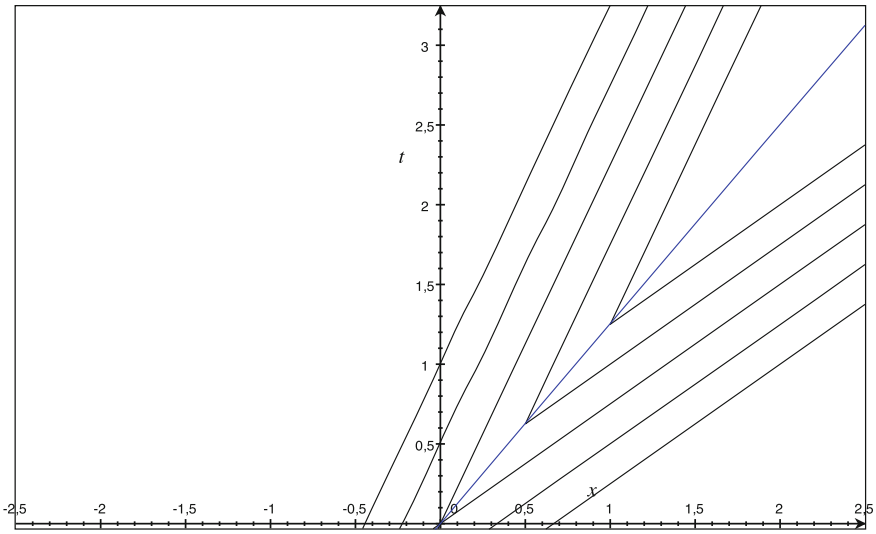


Fig. 3.7 Characteristic lines of the shock-rarefaction wave $w_l \leq w_r$

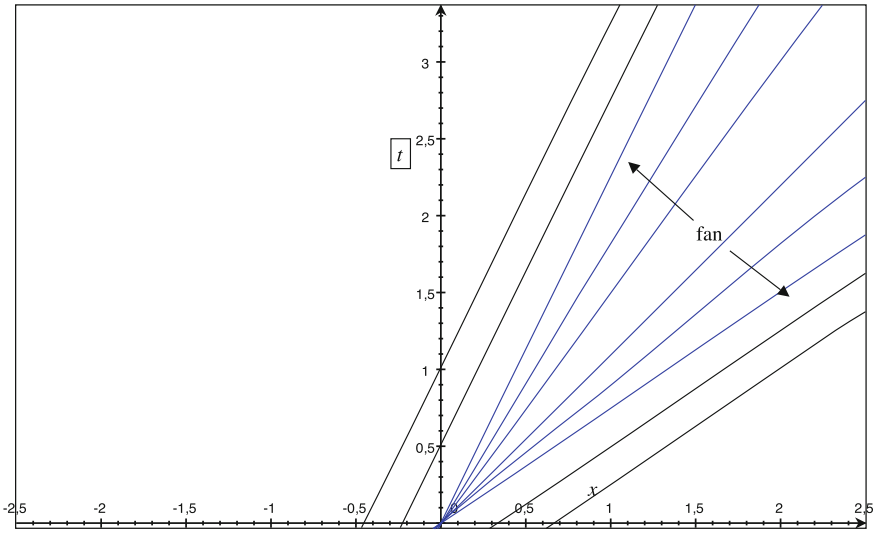


Fig. 3.8 Characteristic lines of a rarefaction wave $w_l \leq w_r$

with

$$s_1 = \frac{w_l - a}{2}, \quad s_2 = \frac{w_r + a}{2}, \quad \text{and} \quad a \geq \max\{w_l, -w_r\}.$$

Exercise 3.4 Show that the previous solution is a weak solution of the Burgers equation with the initial condition (3.27). *Suggestion:* Check that the jump conditions are satisfied.

Solution 3.4 Left to the reader.

3.5 Mathematical Notion of Entropy. Entropy Solutions

Given any smooth solution $\mathbf{w}$ of (2.2), let us examine whether $\mathbf{w}$ *verifies an additional conservation law of the form*:

$$(U(\mathbf{w}))_t + \sum_{j=1}^{d} (F_j(\mathbf{w}))_{x_j} = 0,\ \mathbf{x} = (x_1, \ldots, x_d) \in \mathbf{R}^d,\ t > 0, \tag{3.31}$$

where U and $F_j,\ j = 1, \ldots, d$ are sufficiently smooth functions from Ω to $\mathbf{R}$. It is immediate to prove that this holds if

$$\mathbf{U}'(\mathbf{w})\,\mathbf{f}_\mathbf{j}'(\mathbf{w}) = \mathbf{F}_\mathbf{j}'(\mathbf{w}), \quad j = 1, \ldots, d, \tag{3.32}$$

with $\mathbf{U}'(\mathbf{w}) = \nabla U^T$, $\mathbf{F}_\mathbf{j}'(w) = \nabla F_j^T$ y $\mathbf{f}_\mathbf{j}' = \mathbf{A}_\mathbf{j} = \left(\dfrac{\partial f_{ij}}{\partial w_k}(w)\right)_{1 \le i,k \le p}$.

Multiplying (2.2) by $\mathbf{U}'$ we get

$$\mathbf{U}'(\mathbf{w})(\mathbf{w})_t + \mathbf{U}'(\mathbf{w}) \sum_{j=1}^{d} (F_j(\mathbf{w}))_{x_j} = 0$$

$$\Longrightarrow (U(\mathbf{w}))_t + \sum_{j=1}^{d} \mathbf{U}'(\mathbf{w}) \mathbf{f}_j'(\mathbf{w}) \mathbf{w}_{x_j} = 0. \tag{3.33}$$

Therefore, any classical solution of (2.2) verifies the additional conservation law (3.31) as long as (3.32) is satisfied.

On the other hand, this is not true, in general, for every weak solution. In particular, it is not true for a piecewise C^1 weak solution. This solution must verify the Rankine-Hugoniot jump condition along the discontinuity surfaces (3.19), whereas the solution of the additional conservation law must verify the following jump condition:

$$n_t [U(\mathbf{w})] + \sum_{j=1}^{d} n_{x_j} [F_j(\mathbf{w})] = 0, \tag{3.34}$$

which is, in general, incompatible with Rankine-Hugoniot (3.19).

Definition 3.4 Entropy function and entropy flux:

Let us assume that Ω is convex. A *convex function* $U : \Omega \longrightarrow \mathbf{R}$ is called an *entropy* for the system of conservation laws (2.2) if there exist d functions $F_j : \Omega \longrightarrow \mathbf{R}$, $1 \leq j \leq d$, called *entropy flux*, that verify the equation

$$\mathbf{U}'(\mathbf{w})\,\mathbf{f}_\mathbf{j}'(\mathbf{w}) = \mathbf{F}_\mathbf{j}'(\mathbf{w}), \quad j = 1, \ldots, d. \tag{3.35}$$

The problem lies in finding entropy functions and, if possible, all the entropy functions associated with a nonlinear system of conservation laws.

For a scalar conservation law it is easily found, since any convex function U is an entropy function. A particular choice for a uniparametric family of entropy functions is given by

$$U(w) = |w - k|,\ k \in \mathbf{R}, \quad F_j(w) = \operatorname{sgn}(w - k)\left(f_j(w) - f_j(k)\right),$$

with *sgn* the sign function.

Theorem 3.2 *Let us assume that (2.43) admits an entropy function U with F_j, $1 \leq j \leq d$ the entropy fluxes. Let $\{\mathbf{w}_\varepsilon\}_\varepsilon$ be a sequence of solutions of*

$$(\mathbf{w}_\epsilon)_t + \sum_{j=1}^{d} \left(\mathbf{f}_j(\mathbf{w}_\varepsilon)\right)_{x_j} = \varepsilon \triangle \mathbf{w}_\varepsilon,$$

sufficiently smooth, such that

$$||\mathbf{w}_\varepsilon||_{L^\infty(\mathbf{R}^d \times [0,\infty))} \leq C, \quad \mathbf{w}_\varepsilon \longrightarrow \mathbf{w} \text{ when } \varepsilon \longrightarrow 0 \text{ a.e. } \mathbf{R}^d \times [0, \infty), \tag{3.36}$$

where $C > 0$ is a constant independent of ε. Then, $\mathbf{w}$ is a solution of (2.43) and verifies the entropy condition

$$n_t[U(\mathbf{w})] + \sum_{j=1}^{d} n_{x_j}[F_j(\mathbf{w})] \leq 0 \tag{3.37}$$

in the sense of distributions on $\mathbf{R}^d \times (0, \infty)$, i.e.,

$$\int_0^\infty \int_{\mathbf{R}^d} \left\{ U(\mathbf{w})\,(\varphi)_t + \sum_{j=1}^{d} F_j(\mathbf{w})\,(\varphi)_{x_j} \right\} dxdt \geq 0,$$
$$\forall \varphi \in C_0^\infty(\mathbf{R}^d \times (0, +\infty)),\ \varphi \geq 0. \tag{3.38}$$

Proof It can be found in Godlewski and Raviart [8].

Definition 3.5 Entropy solution

A *weak solution* of (2.43)–(2.46) is said to be an entropy solution if it verifies the entropy condition

$$n_t[U(\mathbf{w})] + \sum_{j=1}^{d} n_{x_j}[F_j(\mathbf{w})] \leq 0, \tag{3.39}$$

for all entropy function U and for all test function $\varphi \in C_0^1(\mathbf{R}^d \times [0, +\infty))$, $\varphi \geq 0$ in the sense of distributions in $\mathbf{R}^d \times [0, \infty)$, i.e.,

$$\int_0^\infty \int_{\mathbf{R}^d} \left\{ U(\mathbf{w})\,(\varphi)_t + \sum_{j=1}^{d} F_j(\mathbf{w})\,(\varphi)_{x_j} \right\} dxdt$$
$$+ \int_{\mathbf{R}^d} U(\mathbf{w}_0(\mathbf{x}))\varphi(\mathbf{x}, 0)dx \geq 0. \tag{3.40}$$

Example 3.4 Entropy function for the Burgers equation.

An entropy function for the Burgers equation is given by

$$U(w) = w^2,$$

and the corresponding entropy flux is

$$F(w) = \frac{2}{3}w^3.$$

(See Example 3.3).

Remark 3.3 By using similar arguments to those employed in the proof of the theorem that introduces the Rankine-Hugoniot jump conditions, it is easy to show that a *piecewise* C^1 *function* $\mathbf{w}$ that is a solution of the conservation law (2.2) in the sense of distributions, verifies the entropy condition if and only if:

(i) $\mathbf{w}$ is a classic solution of (2.2) in the domain where $\mathbf{w}$ is C^1.
(ii) $\mathbf{w}$ verifies the jump condition:

$$n_t[U(\mathbf{w})] + \sum_{j=1}^{d} n_{x_j}[F_j(\mathbf{w})] \leq 0, \tag{3.41}$$

along the surfaces of discontinuity.

In the one-dimensional case the previous inequality becomes

$$s[U(w)] \geq [F(w)], \tag{3.42}$$

where s is given by the Rankine-Hugoniot jump condition.

Exercise 3.5 Prove that a rarefaction-shock wave, defined by (3.28) is not a an entropy solution for the Burgers equation if $w_l \leq w_r$.

Solution 3.5 Left to the reader.

Exercise 3.6 Prove that a shock wave, defined by (3.28) is an entropy solution for the Burgers equation if $w_l > w_r$.

Solution 3.6 Left to the reader.

Note 3.1 The code for the entropy solution for the Riemann problem of the Burgers equation can be found in Sect. 9.1.

The texts by LeVeque (Sect. 3.8 [17] and Sect. 11.13 [18]) for the scalar conservation laws present different versions of the entropy condition that do not explicitly involve the concept of entropy function:

Theorem 3.3 *Entropy condition (version I, by Peter Lax):*
For a scalar conservation law in which the flux function f is convex, a discontinuity that propagates with velocity s satisfying the Rankine-Hugoniot jump condition, verifies the entropy condition if

$$f'(w_l) > s > f'(w_r). \tag{3.43}$$

Remark 3.4 Recall that $f'(w)$ denotes the characteristic velocities. For a given function f, either concave or convex, by definition, the Rankine-Hugoniot jump velocity s must lie between $f'(w_l)$ and $f'(w_r)$. Hence, (3.43) reduces simply to $f'(w_l) > f'(w_r)$.

Theorem 3.4 *Entropy condition (version II):*
$w(x, t)$ is said to be an entropy solution if all the discontinuities have the following property:

$$\frac{f(w) - f(w_l)}{w - w_l} \geq s \geq \frac{f(w) - f(w_r)}{w - w_r}, \tag{3.44}$$

for all values of w between w_l and w_r.

Example 3.5 For the solutions of the Burgers equation, the entropy condition reduces to

$$w_l \geq s \geq w_r, \tag{3.45}$$

with s the jump propagation velocity given by the Rankine-Hugoniot condition.

Exercise 3.7 Verify that the entropy condition for the traffic flow equation reduces to $\rho_l < \rho_r$.

Solution 3.7 Left to the reader.

Exercise 3.8 Verify that for the traffic flow equation a stationary shock wave is obtained, this is, the wave propagation velocity is null, and therefore the solution coincides with the initial condition, in any of the following cases:

(i) $\rho_l = \rho_m$ and $\rho_r = 0$, which represents a situation in which a traffic jam exists before the discontinuity and no car flow after the discontinuity.
(ii) $\rho_l = 0$ and $\rho_r = \rho_m$, which represents a situation in which there are no cars before the discontinuity and there is a traffic jam after it.

Analyse which of the two solutions is an entropy solution.

Solution 3.8 Left to the reader.

3.5.1 Some Remarks on the Entropy Inequalities

The entropy inequalities are related to the second law of thermodynamics, which states that in a closed physical system, the entropy cannot decrease. Intuitively, the physical concept of entropy is related to the disorder of the system or the loss of information.

Regarding the admissible shock waves, note how the characteristics *converge* in the shock (3.43). This can be interpreted as a loss of information, i.e., *an increase of the entropy of the system*. To sum up, we could say that a discontinuous weak solution is admissible if the characteristics converge in the shock (see Fig. 3.6).

These ideas can be formalized by using energy estimators, starting with weak solutions of the viscous problem and analysing the limit of these solutions when the viscous parameter ε tends to zero.

3.6 Conclusions

- The physically admissible solutions will be chosen taking into account certain *entropy inequalities*.
- The *weak solutions of the viscous Burgers equation* verify the entropy inequalities, whereas the solutions of the non-viscous equation do not always verify these inequalities.
- For this equation, from a physical point of view, only the *expansion waves* are admissibles in the case $w_l < w_r$.
- This matter will also be taken into account in the *design of the numerical methods* that solve non-viscous problems.
- A minimum amount of *artificial viscosity* will have to be implemented in the numerical schemes to avoid convergence to *non-entropy* solutions.

We would like to finish the first part of the book with a biographical summary of Professor Peter Lax.

Chapter 4
Biographical Summary of Professor Peter Lax

Abstract Professor Peter Lax is one of the most relevant experts in hyperbolic conservation laws and the finite volume method. His work is devoted to this and other branches of Mathematics, for he is considered one of the most versatile mathematicians of his generation.

Professor Peter Lax was born on the 1st of May 1926, into a Jewish family in Budapest, and, at the age of 15 years, he moved to the United States where he met John von Neumann, through his mentor Dénes König.

During his visit to Santiago de Compostela in November 2007 (Fig. 4.1), in the framework of the "Programa ConCiencia"[1] he mentioned in an interview that his inspiration to devote his life to Mathematics came through "*a woman and a book*".

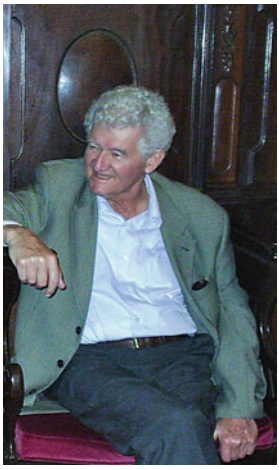


Fig. 4.1 Peter Lax in the University of Santiago de Compostela (November 2007)

[1] His conference in the Galician Center of Contemporary Art, "Life and time of John von Neumann", can be fully reviewed in http://www.usc.es/en/cursos/conciencia/lax07.html

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Later on in that interview he revealed both unknowns: the woman is Rózsa Péter, his teacher and mentor, and "Playing with Infinity", a book written by her [20]. We, who have had the chance to learn from him, share the same respect and admiration for him as he shows for his mentors.

His availability to cooperate and to know what is being done currently in the context of conservation laws can only be expressed in terms of positive energy.

In the present book, the scheme he developed with his supervisor, Kurt Friedrichs, is studied, along with the theorems that characterize the entropy solutions, which are also named after him. We list in the Bibliography section the work he published in collaboration with Harten et al. [15], in which they state a good number of the concepts introduced in this book.

The large record of his contributions, mentions and honors, include the outstanding Abel Prize he was awarded in 2005 "for his groundbreaking contributions to the theory and application of partial differential equations and to the computation of their solutions". Details on his work and life can be reviewed in the following links:

- Peter Lax in MathSciNet:
  http://www.ams.org/mathscinet/search/author.html?mrauthid=111160
- Peter Lax in MacTutor History of Mathematics:
  http://www-history.mcs.st-and.ac.uk/Biographies/Lax_Peter.html
- Peter Lax in Wikipedia:
  http://en.wikipedia.org/wiki/Peter_Lax
- Peter Lax in his personal web site:
  http://math.nyu.edu/faculty/lax/

Part II
Finite Volume Methods Applied to the Hyperbolic Conservation Laws

Chapter 5
Numerical Resolution of One-Dimensional Hyperbolic Linear Systems

Abstract In this chapter we will begin the study of finite volume numerical methods to solve hyperbolic systems of conservation laws. We will introduce the basic concepts, with special attention to the *first-order methods for linear systems*. More precisely, we will focus on the upwind schemes, since they will be the fundamental starting point for the study of the methods used in the resolution of the non linear systems of conservation laws that we will study in the next chapter.

Godunov's method [17, 18, 30] will allow us to link this chapter with the results studied in the previous one. We will take into account the solution of the *Riemann problem for linear systems* to present the aforementioned method, and lay the foundations for the generalization of the schemes to nonlinear systems.

Finite volume methods are closely related to finite difference methods, and a finite volume method is often directly interpreted as a finite differences approximation of the differential equation. Nevertheless, as LeVeque states in [18], *finite volume methods are derived on the basis of the integral form of the conservation law, a starting point that turns out to have many advantages.*

5.1 Basic Concepts

Finite volume methods in one spatial dimension are based on the division of the spatial domain into several intervals, called *finite volumes*, or cells, and the approximation of the integral of the conservative variable w[1] in each of them. In each time step we will update these values by using the approximations of the flux through the boundaries of the finite volumes.

The spatial domain $[a, b]$ is discretized with an arbitrary grid $\mathscr{C}_{\Delta x}$, where $\{x_i : i = 1, \ldots, M\}$ is the set of nodes of $\mathscr{C}_{\Delta x}$, with Δx the norm of the grid $\mathscr{C}_{\Delta x}$:

$$\Delta x = \sup_{x_i \in \mathscr{C}_{\Delta x}} |x_i - x_{i-1}| . \tag{5.1}$$

[1] For simplicity, in this chapter, except in the section devoted to linear systems, we will assume that the model is scalar, and therefore denote the conservative variable as w.

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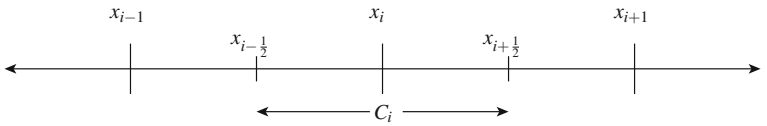


Fig. 5.1 Finite volume grid scheme

The cell or finite volume C_i is defined as follows:

$$C_i = \left(x_{i-\frac{1}{2}}\,,\, x_{i+\frac{1}{2}}\right) = \left(x_i - \frac{x_i - x_{i-1}}{2}\,,\, x_i + \frac{x_{i+1} - x_i}{2}\right), \tag{5.2}$$

and therefore its length in the 1D case is $A_i = x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}}$, see Fig. 5.1.

For simplicity, we will consider in most cases a constant distance between nodes, and therefore $A_i = x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}} = \Delta x = x_i - x_{i-1},\ i = 1, \ldots, M$.

The integral of the conservation law $w_t + f(w)_x = 0$, in each cell C_i, is given by

$$\frac{d}{dt}\int_{C_i} w(x,t)\,dx = f(w(x_{i-\frac{1}{2}},t)) - f(w(x_{i+\frac{1}{2}},t)). \tag{5.3}$$

Integrating in time (5.3) from $t = t^n$ to $t = t^{n+1}$ we obtain

$$\begin{aligned}&\int_{C_i} w(x,t^{n+1})\,dx - \int_{C_i} w(x,t^n)\,dx\\ &= \int_{t^n}^{t^{n+1}} f(w(x_{i-\frac{1}{2}},t))\,dt - \int_{t^n}^{t^{n+1}} f(w(x_{i+\frac{1}{2}},t))\,dt.\end{aligned} \tag{5.4}$$

Rearranging the terms and dividing by Δx we get

$$\begin{aligned}&\frac{1}{\Delta x}\int_{C_i} w(x,t^{n+1})\,dx = \frac{1}{\Delta x}\int_{C_i} w(x,t^n)\,dx\\ &-\frac{1}{\Delta x}\left[\int_{t^n}^{t^{n+1}} f(w(x_{i+\frac{1}{2}},t))\,dt - \int_{t^n}^{t_{n+1}} f(w(x_{i-\frac{1}{2}},t))dt\right].\end{aligned} \tag{5.5}$$

The previous expression states exactly how the average value of w over the cell must be updated within one time step. Therefore, defining the value w_i^n as an approximation of the average value of w over each cell C_i at time t_n

$$w_i^n \approx \frac{1}{\Delta x}\int_{x_{i-1/2}}^{x_{i+1/2}} w(x,t^n)\,dx = \frac{1}{\Delta x}\int_{C_i} w(x,t^n)\,dx, \tag{5.6}$$

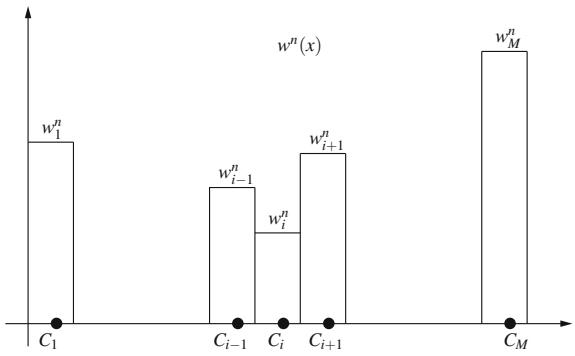


Fig. 5.2 Representation of a piecewise constant function $w^n(x)$ at time t^n

in such a way that the approximate solution is given by the piecewise constant function

$$w^n(x) = w_i^n \quad \text{for} \quad x \in C_i, \tag{5.7}$$

represented in Fig. 5.2. We can employ (5.5) to develop an explicit algorithm for temporal evolution: *Given w_i^n, the average approximations over each of the cells at time t^n, we want to approximate w_i^{n+1}, the average values over each of the cells at the next time step t^{n+1}, after a time step of length $\Delta t = t^{n+1} - t^n$.* If w is a smooth function, then the integral of (5.6) coincides with the value of w at the node of the cell with order $O(\Delta x^2)$. Nevertheless, by working with the averages over the cells instead of the values at the nodes, it is easier to use important properties of the conservation laws in the development of the numerical methods.

In particular, we must make sure that the numerical method is *conservative* in the sense that *it mimics* the exact solution, which is specially important when trying to *correctly calculate shock waves*. This is due to the fact that $\sum_{i=1}^{M} w_i^n \Delta x$ approximates the integral of w over the spatial domain $[a, b]$, and if we employ a numerical method in the *conservative form*, only the values at the boundaries $x = a$ and $x = b$ will make the mentioned discrete sum change. The total amount of the conservative variable within the computational domain will be conserved, or, at most, it will change depending on the boundary conditions imposed.

However, in general we cannot exactly evaluate the time integrals of the right-hand-side of (5.5), since $w(x_{i\pm\frac{1}{2}}, t)$ will change with time along each side of the cell, and we do not have the exact solution to work with. Therefore, we must study numerical methods of the kind

$$w_i^{n+1} = w_i^n - \frac{\Delta t}{\Delta x}\left(f_{i+\frac{1}{2}}^n - f_{i-\frac{1}{2}}^n\right), \tag{5.8}$$

where $f_{i-\frac{1}{2}}^n$ and $f_{i+\frac{1}{2}}^n$ are approximations of the mean value of the flux in the xt-plane along the straight lines $x = x_{i-\frac{1}{2}}$ and $x = x_{i+\frac{1}{2}}$, respectively, with t varying between t^n and t^{n+1}:

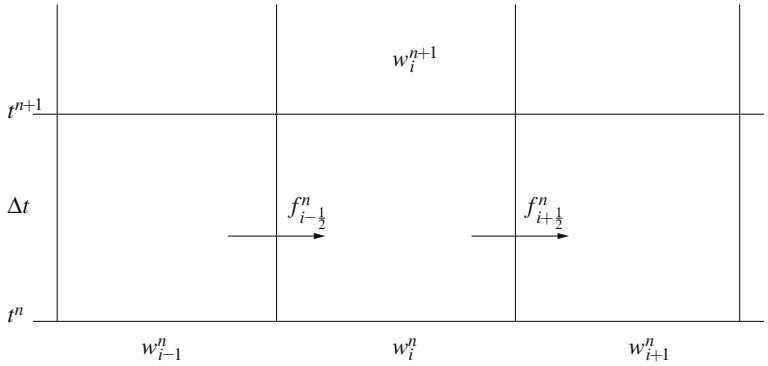


Fig. 5.3 Detail of the finite volume method. Updating process of the value of w_i^n to w_i^{n+1} using an explicit method in time

$$f^n_{i\pm\frac{1}{2}} \approx \frac{1}{\Delta t}\int_{t^n}^{t_{n+1}} f(w(x_{i\pm\frac{1}{2}}, t))\, dt. \tag{5.9}$$

If we can approximate the mean values of the flux at the boundaries of the cells using the values of $w^n(x)$ and the conservation law, then we have a complete discretization of the Cauchy problem. A graphical interpretation of the finite volume method in conservative form is depicted in Fig. 5.3.

In hyperbolic problems the information propagates at a finite velocity, and therefore, it seems reasonable to assume initially that we can obtain $f^n_{i-\frac{1}{2}}$ based only on the values of w^n_{i-1} and w^n_i, the average values on both sides of the boundary $x_{i-\frac{1}{2}}$, and likewise for $f^n_{i+\frac{1}{2}}$, using the values w^n_i and w^n_{i+1}. Hence, we can use an expression of the form

$$f^n_{i-\frac{1}{2}} = \phi(w^n_{i-1}, w^n_i), \quad f^n_{i+\frac{1}{2}} = \phi(w^n_i, w^n_{i+1}), \tag{5.10}$$

where ϕ is a certain *numerical flux function*. Using this notation the numerical scheme (5.8) becomes

$$w_i^{n+1} = w_i^n - \frac{\Delta t}{\Delta x}\left[\phi(w_i^n, w_{i+1}^n) - \phi(w_{i-1}^n, w_i^n)\right]. \tag{5.11}$$

The numerical method obtained depends on the choice of the expression of ϕ, but in general, any such method is a *three-point-stencil explicit method*, which means that the value of w_i^{n+1} will depend on the three values w^n_{i-1}, w^n_i and w^n_{i+1} from the previous time step.

Moreover, it is said to be in *conservative form*, since it mimics the property (5.5) of the exact solution: if we sum up w_i^{n+1} defined by (5.8) over any set of cells $(i = I, \ldots, J)$ and multiply by Δx, we obtain

$$\Delta x \sum_{i=I}^{J} w_i^{n+1} = \Delta x \sum_{i=I}^{J} w_i^n - \Delta t \left(f^n_{J+\frac{1}{2}} - f^n_{I-\frac{1}{2}} \right), \tag{5.12}$$

since the sum of the fluxes cancels out everywhere but in the extreme values $x = x_{I-\frac{1}{2}}$ and $x = x_{J+\frac{1}{2}}$.

5.1.1 Finite Differences Approximation

The method (5.11) can also be regarded as a direct *finite differences approximation* of the conservation law $w_t + f(w)_x = 0$, since it can be rewritten as

$$\frac{w_i^{n+1} - w_i^n}{\Delta t} + \frac{f^n_{i+\frac{1}{2}} - f^n_{i-\frac{1}{2}}}{\Delta x} = 0. \tag{5.13}$$

Likewise, many methods can be considered both as finite differences approximations of the transport equation or as finite volume methods, but the interpretation of w_i^n is different.

5.1.2 Godunov's Method

As we mentioned in the beginning of this Section, the approximate solution $w^n(x)$ of w at t^n, defined in (5.7), is considered as a piecewise constant function with discontinuities in each of the boundaries of every cell C_i, with $i = 1, \ldots, M$. The basis of Godunov's method is that a Riemann problem can be defined in each interval $[x_i, x_{i+1}]$ taking t^n as the initial time and $w^n(x)$ as initial condition in $[x_i, x_{i+1}]$ (see Fig. 5.4).

Godunov suggests that we approximate the value of $w(x,t)$ in C_i at time $t^{n+1} = t^n + \Delta t$ by calculating first the values of the solution of the two Riemann problems $w(x, t^{n+1})$ (denoted in Fig. 5.4 as RP $[x_{i-1}, x_i]$ and RP $[x_i, x_{i+1}]$, respectively). Subsequently, the approximate solution at time t^{n+1} can be obtained as the piecewise constant function $w^{n+1}(x) = w_i^{n+1}$ if $x \in C_i$, with w_i^{n+1} the integral mean value of the two solutions corresponding to the cell C_i (see Fig. 5.5):

$$w_i^{n+1} = \frac{1}{\Delta x} \int_{x_{i-1/2}}^{x_{i+1/2}} w(x, t^{n+1})\, dx. \tag{5.14}$$

In this chapter we will detail the expression of Godunov's method for linear equations and systems.

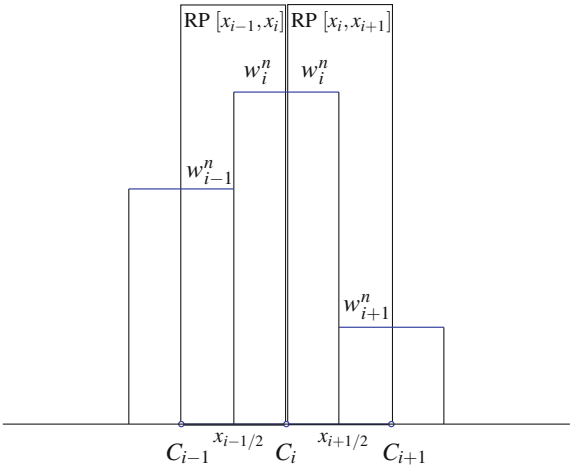


Fig. 5.4 Initial conditions of the two Riemann problems (RP $[x_{i-1}, x_i]$ and RP $[x_i, x_{i+1}]$) considered to define Godunov's method

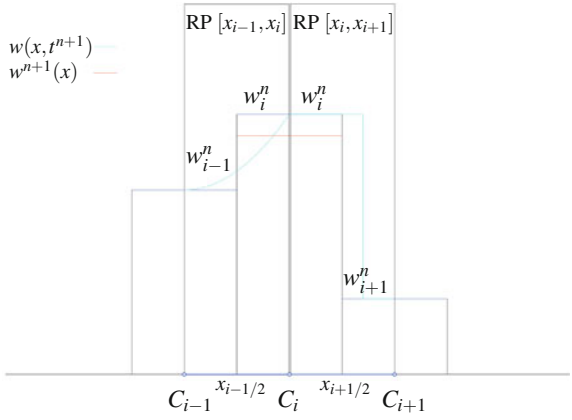


Fig. 5.5 Representation of $w(x, t^{n+1})$, the solution of the two Riemann problems (RP $[x_{i-1}, x_i]$ and RP $[x_i, x_{i+1}]$), and $w^{n+1}(x)$, approximate solution by Godunov's method in the cell C_i

5.1.3 Convergence

Recall that, in order to assess if a particular scheme provides good approximations of the conservation law, an essential requirement is that the resulting numerical method must be *convergent*. This is, the numerical solution obtained must converge to the exact solution of the differential equation when $\Delta x,\ \Delta t \to 0$. The convergence of the method usually requires two conditions:

- The method must be *consistent* with the differential equation, which means that the differential equation is locally well approximated.
- The method must be *stable* in a particular sense, i.e., the errors introduced in each time step must remain bounded.

5.1.4 Consistency

The numerical flux must approximate the integral of (5.9). In particular, if the function $w(x,t) \equiv \overline{w}$ is constant in x, then w will not change with time, and the integral in (5.9) reduces simply to $f(\overline{w})$. As a consequence, if $w_{i-1}^n = w_i^n = \overline{w}$, the numerical flux function ϕ becomes the physical flux in that state $f(\overline{w})$. Therefore, it is required that

$$\phi(\overline{w}, \overline{w}) = f(\overline{w}), \tag{5.15}$$

for any value $\overline{w}$. This is part of the basic consistency condition. In general, continuity will also be a requirement, depending on the variation of w_{i-1}^n and w_i^n. This is $\phi(w_{i-1}^n, w_i^n) \to f(\overline{w})$ when $w_{i-1}^n,\ w_i^n \to \overline{w}$. Additionally, the "Lipchitz continuity condition" is usually needed, i.e., that there exists a constant L such that

$$|\phi(w_{i-1}^n, w_i^n) - f(\overline{w})| \leq L \max\left[|w_i^n - \overline{w}|, |w_{i-1}^n - \overline{w}|\right]. \tag{5.16}$$

5.1.5 von Neumann Stability Analysis

Stability analysis of linear models is easily done in the 2-norm, since the Fourier analysis can be used to simplify the problem. This is the basis of von Neumann stability analysis, which is described in detail in many texts on the subject of finite difference methods for partial differential equations (see, for example, Strikwerda [27]). We summarize next the ideas and concepts in which this method is based.

Let us consider an explicit finite difference scheme that provides an approximate[2] solution of the Cauchy problem at time t^n, given by w_j^n

$$w_j^{n+1} = \sum_{l=-1}^{l=1} c_l w_{j+l}^n, \tag{5.17}$$

where the expression of the coefficients c_l is obtained, in the case of three-point conservative schemes, from the numerical flux, like the one proposed in (5.11). Let us assume that $\left\{w_j^n\right\}$ has finite norm

$$\sum_{j=-\infty}^{\infty} |w_j^n|^2 < +\infty. \tag{5.18}$$

[2] In this section we will change the index of nodes from i to j, so that we do not confuse them with $i = \sqrt{-1}$.

The Fourier transform of the corresponding piecewise constant function $w^n(x)$ defined by (5.7) is:

$$\widehat{w}^n(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} w^n(x) \mathrm{e}^{-ikx}\, dx.$$

As we mentioned before, using the 2-norm, the Parseval's identity of the Plancherel Theorem [27] states that

$$\left\| w^n \right\|_2 = \left\| \widehat{w}^n \right\|_2,$$

where

$$\left\| w^n \right\|_2 = \left(\Delta x \sum_{j=-\infty}^{\infty} |w_j^n|^2 \right)^{1/2} \quad \text{and} \quad \left\| \widehat{w}^n \right\|_2 = \left(\int_{-\infty}^{\infty} |\widehat{w}^n(k)|^2 dk \right)^{1/2}.$$

We can deduce from the expression of the three-point finite difference scheme that the two following identities are satisfied for the functions $w^{n+1}(x)$ and $\widehat{w}^{n+1}(k)$ as a function of $w^n(x)$ and $\widehat{w}^n(k)$, respectively:

$$w^{n+1}(x) = \sum_{l=-1}^{l=1} c_l w^n(x + l\Delta x), \quad \forall x \in \mathbf{R}, \tag{5.19}$$

$$\widehat{w}^{n+1}(k) = \left(\sum_{l=-1}^{l=1} c_l \mathrm{e}^{ikl\Delta x} \right) \widehat{w}^n(k), \quad \forall k \in \mathbf{R}. \tag{5.20}$$

Identity (5.19) follows from the application of the definition of the Fourier transform. Next, we detail the steps needed to verify identity (5.20):

$$\begin{aligned}
\widehat{w}^{n+1}(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} w^{n+1}(x) \mathrm{e}^{-ikx}\, dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sum_{l=-1}^{l=1} c_l w^n(x + l\Delta x) \mathrm{e}^{-ikx}\, dx \\
&= \frac{1}{\sqrt{2\pi}} \sum_{l=-1}^{l=1} c_l \int_{-\infty}^{\infty} w^n(x + l\Delta x) \mathrm{e}^{ikl\Delta x} \mathrm{e}^{-ik(x+l\Delta x)}\, dx \\
&= \frac{1}{\sqrt{2\pi}} \sum_{l=-1}^{l=1} c_l \mathrm{e}^{ikl\Delta x} \int_{-\infty}^{\infty} w^n(x + l\Delta x) \mathrm{e}^{-ik(x+l\Delta x)}\, dx
\end{aligned}$$

$$= \sum_{l=-1}^{l=1} c_l \mathrm{e}^{ikl\Delta x} \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} w^n(y) \mathrm{e}^{-iky}\, dy \right)$$
$$= \left(\sum_{l=-1}^{l=1} c_l \mathrm{e}^{ikl\Delta x} \right) \widehat{w}^n(k),$$

where we have considered the change of variable $y = x + l\Delta x$. Finally, we have

$$\widehat{w}^{n+1}(k) = g(k, \Delta x, \Delta t)\widehat{w}^n(k),$$

where $g(k, \Delta x, \Delta t)$ is the amplification factor for each wave number k, given by:

$$g(k, \Delta x, \Delta t) = \sum_{l=-1}^{l=1} c_l \mathrm{e}^{ikl\Delta x}. \tag{5.21}$$

In order to prove that the 2-norm of w^n remains bounded, it suffices to show that the 2-norm of $\widehat{w}^n$ does. Note that, while the components of w^n, as j changes, are coupled by the difference equations, every element of $\widehat{w}^n$, as k changes, verifies the relation that defines the amplification factor, which is uncoupled from the other wave numbers. The Fourier transform diagonalizes the linear differential operator. Hence, it suffices to consider a single wave number k and a function of the form

$$w_j^n = \widehat{w}^n(k) \mathrm{e}^{ikx_j}. \tag{5.22}$$

From here we can obtain the amplification factor. A sufficient condition for stability is that the module of the amplification factor remains smaller or equal to one for all wave number k. Applying the recurrence relation

$$\widehat{w}^n(k) = [g(k, \Delta x, \Delta t)]^n\, \widehat{w}^0(k), \tag{5.23}$$

therefore:

$$\left\| \widehat{w}^n \right\|_2 \leq |g(k, \Delta x, \Delta t)|_\infty^n \left\| \widehat{w}^0 \right\|_2, \tag{5.24}$$

and, owing to Parseval's identity:

$$\left\| w^n \right\|_2 \leq |g(k, \Delta x, \Delta t)|_\infty^n \left\| w^0 \right\|_2.$$

Theorem 5.1 *A difference scheme is L-2 stable if and only if its amplification factors* $g(k, \Delta x, \Delta t)$ *verify*

$$|g(k, \Delta x, \Delta t)| \leq 1, \quad \textit{for all wave number } k. \tag{5.25}$$

Applying Theorem 5.1 to the different finite volume schemes that will be presented, we will be able to conclude whether the scheme is unconditionally stable, unconditionally unstable or, if it results conditionally stable, we will be able to obtain, from (5.25), the corresponding stability conditions. Next, we will analyse the well-known *CFL condition*.

As we mentioned before in this section, the amplification factor can be obtained from the relation (5.21) or replacing the expression (5.22) in the finite volume method.

5.1.5.1 CFL Condition

The *CFL condition* (Courant, Friedrichs and Lewy, 1928)[3]

$$\frac{|\lambda|\Delta t}{\Delta x} \leq 1, \tag{5.26}$$

is a necessary condition that any *explicit* finite volume or finite difference method must verify if we want it to be stable and to converge to the solution of the linear difference equation when the grid is refined.

This condition states that *a method must be applied in such a way that the information can propagate at the physically correct velocities, determined by the eigenvalues of the Jacobian matrix* of the flux.

In the explicit scheme presented in (5.11) the value w_i^{n+1} depends only on three values, as we mentioned before. Let us assume that we apply this method to the transport equation $w_t + \lambda w_x = 0$ with $\lambda > 0$. We saw in Chap. 2 that the exact solution propagates a distance $\lambda \Delta t$ every time step

$$w(x, t^n + \Delta t) = w(x - \lambda \Delta t, t^n).$$

Figure 5.6a shows a situation in which $\lambda \Delta t < \Delta x$, and therefore the information travels a distance smaller than the length of the cell in each time step. In this case it makes sense to define the flux at $x_{i-\frac{1}{2}}$ depending only on w_{i-1}^n and w_i^n.

Figure 5.6b illustrates a case in which $\lambda \Delta t > \Delta x$. We see how the exact solution at the cell C_i and time t_{n+1} depends also on w_{i-2}^n, and therefore the exact flux at $x_{i-\frac{1}{2}}$ also depends on this value. The method suggested in (5.11) will become unstable if the numeric flux function depends only on w_{i-1}^n and w_i^n.

This is a consequence of the CFL condition, in which

$$\mu = \frac{\lambda \Delta t}{\Delta x}, \tag{5.27}$$

[3] The condition is named after Richard Courant, Kurt Friedrichs, and Hans Lewy who described it in their 1928 paper, republished in the IBM Journal in 1967 (English translation), [21].

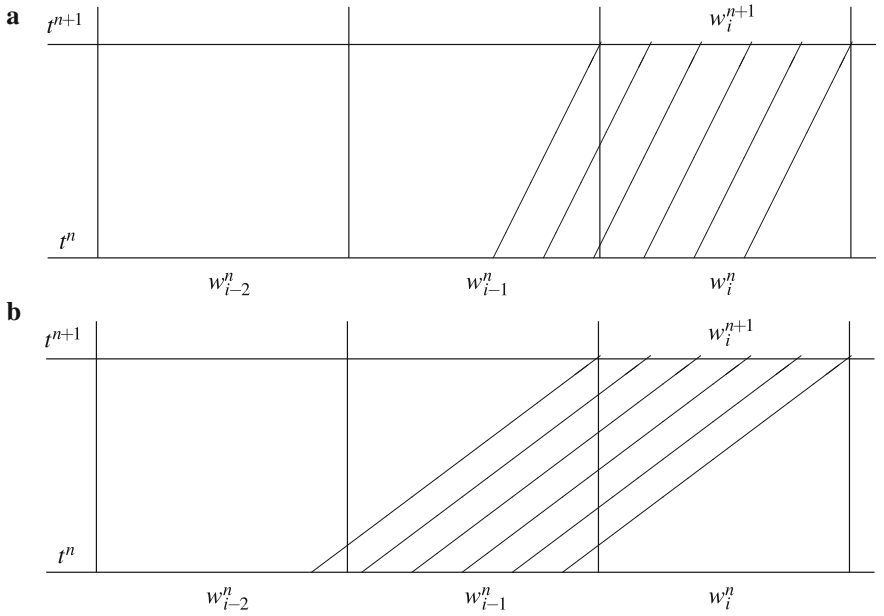


Fig. 5.6 CFL condition applied to the transport equation. **a** Values employed in the parameters of the figure: $\Delta x = 4$, $\Delta t = 2.4$, $\lambda = 1.2 \Longrightarrow \lambda \Delta t = 2.88 < 4 = \Delta x \Longrightarrow \frac{\lambda \Delta t}{\Delta x} \leq 1$. The CFL condition is satisfied; **b** Values employed in the parameters of the figure: $\Delta x = 4$, $\Delta t = 2.4$, $\lambda = 2 \Longrightarrow \lambda \Delta t = 4.8 > 4 = \Delta x \Longrightarrow \frac{\lambda \Delta t}{\Delta x} > 1$. The CFL condition is not satisfied

is known as *CFL number* or *Courant number*, and can be interpreted in two ways:

- Length ratio:
  $\lambda \Delta t =$ distance travelled by a signal that propagates at velocity λ for a time Δt.
  $\Delta x =$ length of the spatial discretization step.
  In this case the interpretation of the Courant condition ($\lambda \Delta t \leq \Delta x$) is the one explained before: *the information of the discrete problem must travel a distance smaller than the length of a cell in each time step*.
- Velocity ratio:

$$\mu = \frac{\lambda}{\lambda_p}, \tag{5.28}$$

  where:
  $\lambda_p = \dfrac{\Delta x}{\Delta t} =$ velocity of propagation of the discretization (or grid).
  $\lambda =$ velocity of propagation of the continuous problem.
  In this case the interpretation of the Courant condition ($|\lambda| \leq \lambda_p$) is as follows: *the absolute value of the velocity of propagation of the continuous problem* $|\lambda|$ *must be smaller than the velocity at which the information propagates in the grid* λ_p.

The definition of the Courant number considered in this book (5.27) coincides with that used by other authors, like Toro [30]. Nevertheless, LeVeque [18] defines it as the absolute value of the Courant number employed in the present text. In our opinion, the definition used in this book simplifies the expressions in terms of the CFL number of several schemes considered, including explicitly the sign of λ for the transport equation.

5.1.6 Approximation of Derivatives

Although our aim is the study of finite volume methods, in order to explain the basic concepts, we will present some finite difference approximations.

In the following we will introduce several approximation of the derivatives by finite difference schemes.

Given a sufficiently smooth function g, by means of the Taylor series expansion, we can obtain the value of $g(x)$ at $x_0 + \Delta x$:

$$g(x_0 + \Delta x) = g(x_0) + g'(x_0)\Delta x + \frac{g''(x_0)}{2!}(\Delta x)^2 + O(\Delta x)^3.$$

Likewise, at $x_0 - \Delta x$,

$$g(x_0 - \Delta x) = g(x_0) - g'(x_0)\Delta x + \frac{g''(x_0)}{2!}(\Delta x)^2 + O(\Delta x)^3.$$

This allows us to make the following approximations:

- Forward finite difference approximation:

$$D^+ g(x_0) = \frac{g(x_0 + \Delta x) - g(x_0)}{\Delta x}, \tag{5.29}$$

$$g'(x_0) = D^+ g(x_0) + O(\Delta x). \tag{5.30}$$

- Backward finite difference approximation:

$$D^- g(x_0) = \frac{g(x_0) - g(x_0 - \Delta x)}{\Delta x}, \tag{5.31}$$

$$g'(x_0) = D^- g(x_0) + O(\Delta x). \tag{5.32}$$

- Central approximation:

$$D^0 g(x_0) = \frac{g(x_0 + \Delta x) - g(x_0 - \Delta x)}{2\Delta x}, \tag{5.33}$$

$$g'(x_0) = D^0 g(x_0) + O(\Delta x)^2. \tag{5.34}$$

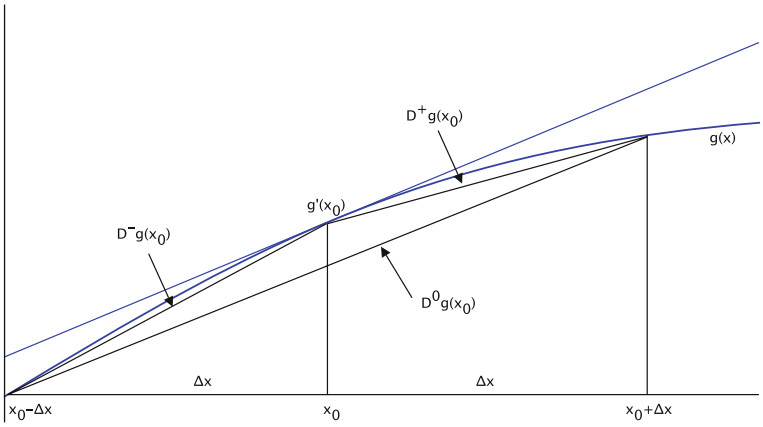


Fig. 5.7 Backward finite difference approximation $D^- g(x_0)$, forward finite difference approximation $D^+ g(x_0)$ and centered finite difference approximation $D^0 g(x_0)$

Figure 5.7 shows the finite difference approximations considered. After having introduced these three approximations to the first derivative, we will study possible numerical schemes to discretize the transport equation.

5.2 A First Scheme for the Transport Equation: Unconditionally Unstable

Let us consider the following difference quotients as approximations of the partial differential equations:

$$w_t \simeq \frac{w_i^{n+1} - w_i^n}{\Delta t}, \quad w_x \simeq \frac{w_{i+1}^n - w_{i-1}^n}{2\Delta x},$$

this is, a *forward discretization in time* and a *centered discretization in space*, respectively.

If we recall the transport equation, $w_t(x,t) + \lambda w_x(x,t) = 0$, and replace the corresponding discretizations with the previous expressions, we get the approximation:

$$\frac{w_i^{n+1} - w_i^n}{\Delta t} + \lambda \frac{w_{i+1}^n - w_{i-1}^n}{2\Delta x} = 0. \tag{5.35}$$

Isolating w_i^{n+1}, we have

$$w_i^{n+1} = w_i^n - \frac{\lambda \Delta t}{2\Delta x}\left(w_{i+1}^n - w_{i-1}^n\right), \tag{5.36}$$

and replacing the Courant number $\mu = \frac{\lambda \Delta t}{\Delta x}$, the corresponding finite volume method has an expression of the form

$$w_i^{n+1} = w_i^n - \frac{\mu}{2}\left(w_{i+1}^n - w_{i-1}^n\right). \tag{5.37}$$

This scheme can be written in conservative form by defining the numerical flux function:

$$f_{i-\frac{1}{2}}^n = \phi(w_{i-1}^n, w_i^n) = \frac{1}{2}\left[f(w_{i-1}^n) + f(w_i^n)\right] = \frac{\lambda}{2}\left[w_{i-1}^n + w_i^n\right], \tag{5.38}$$

$$f_{i+\frac{1}{2}}^n = \phi(w_i^n, w_{i+1}^n) = \frac{1}{2}\left[f(w_i^n) + f(w_{i+1}^n)\right] = \frac{\lambda}{2}\left[w_i^n + w_{i+1}^n\right], \tag{5.39}$$

which satisfies the consistency condition given by (5.15).

Example 5.1 In order to analyze the stability of the scheme in terms of the CFL condition and verify that the propagation velocity of the grid is given by $\lambda_p = \frac{\Delta x}{\Delta t}$ we consider the solution $\left(w_i^n\right)$ given by the centered scheme with the initial condition

$$w_i^0 = 0, \quad \text{si} \quad i \neq 0, \quad w_0^0 = 1. \tag{5.40}$$

The solution provided by this scheme presents the following properties:

- It is null in the outside of the cone defined by the lines:

$$D_{0,\mu}^- = \left\{(x,t) \,/\, \frac{x}{t} = -\frac{\Delta x}{\Delta t}\right\}, \quad D_{0,\mu}^+ = \left\{(x,t) \,/\, \frac{x}{t} = +\frac{\Delta x}{\Delta t}\right\}.$$

- It is non-null over $D_{0,\mu}^-$ and $D_{0,\mu}^+$.
- It is easy to prove that

$$w_{-n}^n = \left(-\frac{\mu}{2}\right)^n, \quad (x_{-n}, t^n) \in D_{0,\mu}^-,$$
$$w_n^n = \left(\frac{\mu}{2}\right)^n, \quad (x_n, t^n) \in D_{0,\mu}^+.$$

Therefore, in addition to the propagation towards the positive values of x, another propagation towards the negative values of x takes place, contrarily to what occurs in the continuous equation. The propagation velocity in both directions is equal to λ_p (see Fig. 5.8).

Exercise 5.1 Programme the previous case and check the incorrect propagation of information.

Solution 5.1 Code included in Sect. 8.1.

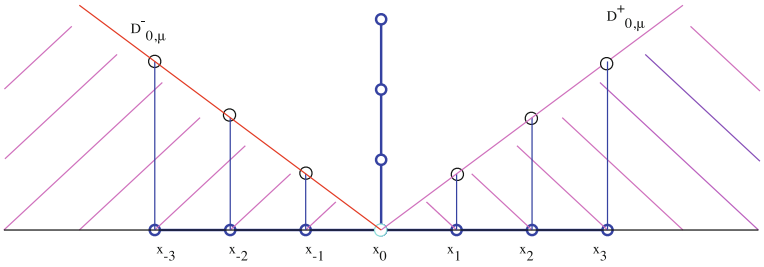


Fig. 5.8 Centered scheme defined in (5.37): regions of incorrect propagation of the information. Initial condition: pulse

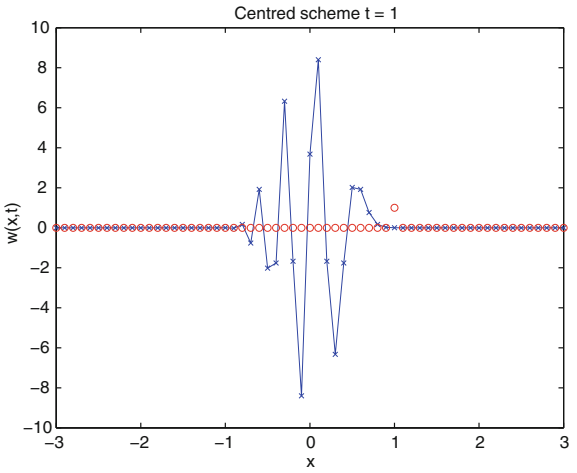


Fig. 5.9 Centered scheme defined in (5.37): exact solution (*red* ∘) and approximate solution (*blue* ×). Initial condition: pulse, $\mu = 1, t = 1$

Figure 5.9 contrasts the numerical solution obtained by the centered scheme with the exact solution. An incorrect propagation of the information is clearly seen: for negative values of x, non-null values appear, which points out an incorrect propagation of the information. Moreover, the approximate solution in the positive part of the OX-axis is likewise incorrect. The parameters employed are $\Delta x = 0.1$, $\Delta t = 0.1$ and $\mu = 1$.

As we will prove in the next section, although it is unconditionally unstable, it is a conservative scheme. It is easy to prove that the following equation holds:

$$\sum_{i=-M}^{M} w_i^n = \sum_{i=-M}^{M} w_i^0 = 1. \tag{5.41}$$

5.2.1 *von Neumann Analysis of the Centered Scheme Defined in (5.37)*

Regarding the expression of the general linear scheme defined by (5.17), the coefficients c_{-1}, c_0 and c_1 of the centered scheme introduced in (5.37) are

$$c_{-1} = \frac{1}{2}\mu, \quad c_o = 1, \quad c_1 = -\frac{1}{2}\mu. \tag{5.42}$$

Therefore, applying Theorem 5.1, the amplification factor is:

$$g(k, \Delta x, \Delta t) = \frac{1}{2}\mu e^{-ik\Delta x} + 1 - \frac{1}{2}\mu e^{ik\Delta x} = 1 - i\mu \sin(k\Delta x). \tag{5.43}$$

We can obtain the amplification factor in a different way, by considering an initial condition like the one introduced in (5.22)

$$w_j^0 = \widehat{w}^0 e^{ikx_j}, \quad \forall j \in \mathbf{Z},$$

where $\widehat{w}^0$ denotes a complex amplitude and k a real wave number.[4]

In order to calculate the amplification coefficient of the scheme, we assume that the approximate solution at time t_n is known, and is given by $w_j^n = \widehat{w}^n(k)e^{ikx_j}$, $\forall j \in \mathbf{Z}$. Then, if we replace this expression in the centered scheme:

$$w_j^{n+1} = w_j^n - \frac{\mu}{2}\left(w_{j+1}^n - w_{j-1}^n\right), \tag{5.44}$$

obtaining:

$$\widehat{w}^{n+1}e^{ikx_j} = \widehat{w}^n e^{ikx_j} - \frac{\mu}{2}\left(\widehat{w}^n e^{ikx_{j+1}} - \widehat{w}^n e^{ikx_{j-1}}\right). \tag{5.45}$$

Dividing the previous expression by e^{ikx_j} and taking into account that $x_j = j\Delta x$, we have

$$\widehat{w}^{n+1} = \widehat{w}^n\left[1 - \frac{\mu}{2}\left(e^{ik\Delta x} - e^{-ik\Delta x}\right)\right]. \tag{5.46}$$

Expanding the complex exponentials,

$$\widehat{w}^{n+1} = \widehat{w}^n\left[1 - \frac{\mu}{2}\left(\cos(k\Delta x) + i\sin(k\Delta x) - \cos(-k\Delta x) - i\sin(-k\Delta x)\right)\right],$$

and simplifying:

$$\widehat{w}^{n+1} = \widehat{w}^n\left[1 - \frac{\mu}{2}\left(2i\sin(k\Delta x)\right)\right] = \widehat{w}^n\left[1 - i\mu\sin(k\Delta x)\right]. \tag{5.47}$$

[4] In this section, the index of the nodes was changed from i to j so we do not confuse it with $i = \sqrt{-1}$.

Hence, we get that the amplification factor $g(k, \Delta x, \Delta t)$ is given by

$$g(k, \Delta x, \Delta t) = \frac{\widehat{w}^{n+1}}{\widehat{w}^n} = 1 - i\mu \sin(k\Delta x), \tag{5.48}$$

as is stated in the following Lemma:

Lemma 5.1 *A solution* $\left\{w_j^n\right\}$ *obtained with the explicit centered scheme defined by (5.36) with the initial condition* $w_j^0 = \widehat{w}^0 \mathrm{e}^{ikx_j}$, $\forall j \in \mathbf{Z}$ *is*

$$w_j^n = \widehat{w}^n e^{ikx_j}, \quad \forall j \in \mathbf{Z}, \quad \forall n \in \mathbf{N},$$

where the amplitude $\widehat{w}$ *is given by the recurrence relation*

$$\widehat{w}^{n+1} = g(k, \Delta x, \Delta t)\widehat{w}^n,$$

which is the discrete Fourier transform of w^n*, where*

$$g(k, \Delta x, \Delta t) = 1 - i\mu \sin(k\Delta x), \tag{5.49}$$

is the amplification coefficient of the scheme.

Corollary 5.1 *The following is also true for the centered scheme defined by (5.36):*

(i) $\widehat{w}^n = g(k, \Delta x, \Delta t)^n \widehat{w}^0$.
(ii) $|\widehat{w}^n| = |g(k, \Delta x, \Delta t)|^n |\widehat{w}^0|$.
(iii) $|g(k, \Delta x, \Delta t)| = \left(1 + \mu^2 \sin^2(k\Delta x)\right)^{\frac{1}{2}} > 1$.

Hence, the centered scheme defined by (5.36) is *unconditionally unstable*.

Exercise 5.2 Work with the programmed method defined by (5.36) and verify it is unconditionally unstable for any choice of Δx and Δt.

Solution 5.2 Left to the reader.

5.3 Lax-Friedrichs Method

The classical Lax-Friedrichs scheme for the equation of transport is given by

$$w_j^{n+1} = \frac{1}{2}\left(w_{j-1}^n + w_{j+1}^n\right) - \frac{\Delta t}{2\Delta x}\left(\lambda w_{j+1}^n - \lambda w_{j-1}^n\right). \tag{5.50}$$

This method is *centered in space* and is also stable. It is obtained by replacing w_j^n in the centered scheme defined by (5.36) by the average $\frac{1}{2}\left(w_{j-1}^n + w_{j+1}^n\right)$.

Although apparently it does not seem to be a conservative scheme, after some algebra we get the expression of the corresponding numerical flux:

$$\begin{aligned}\phi(w_j^n, w_{j+1}^n) &= \frac{\lambda w_j^n + \lambda w_{j+1}^n}{2} - \frac{\Delta x}{2\Delta t}(w_{j+1}^n - w_j^n) \\ &= \frac{f(w_j^n) + f(w_{j+1}^n)}{2} - \frac{\Delta x}{2\Delta t}(w_{j+1}^n - w_j^n). \end{aligned} \tag{5.51}$$

It is immediate to check the consistency of the numerical flux function verifying (5.15).

This scheme is very similar to the one studied in the previous section but, as we propose in Exercise 5.3, the replacement of w_i^n by $\frac{1}{2}\left(w_{j-1}^n + w_{j+1}^n\right)$ leads to a conditionally stable scheme, and therefore convergent, once the consistency is proven.

Exercise 5.3 Prove that the Lax-Friedrichs scheme is conditionally stable for the linear case, and obtain the stability condition.

Solution 5.3 Lax-Friedrichs scheme for the transport equation is given by (5.50). One of the procedures to obtain the amplification factor studied in this book consists of assuming that the approximate solution at time t^n is known, given by t^n, $w_j^n = \hat{w}^n e^{ikx_j}$, $\forall j \in \mathbf{Z}$, and replacing this expression in (5.50):

$$\hat{w}^{n+1} e^{ikx_j} = \frac{1}{2}(\hat{w}^n e^{ikx_{j-1}} + \hat{w}^n e^{ikx_{j+1}}) - \frac{\lambda \Delta t}{2\Delta x}(\hat{w}^n e^{ikx_{j+1}} - \hat{w}^n e^{ikx_{j-1}}). \tag{5.52}$$

Now we isolate $\hat{w}^{n+1}$ and rearrange terms

$$\begin{aligned}\widehat{w}^{n+1} &= \widehat{w}^n \frac{1}{2}\left[e^{-ik\Delta x} + e^{ik\Delta x} - \mu(e^{ik\Delta x} - e^{-ik\Delta x})\right] \\ &= \widehat{w}^n \left[\cos(k\Delta x) - i\mu \sin(k\Delta x)\right]. \end{aligned} \tag{5.53}$$

Hence, the amplification factor $g(k, \Delta x, \Delta t)$ is

$$g(k, \Delta x, \Delta t) = \frac{\hat{w}^{n+1}}{\hat{w}^n} = \cos(k\Delta x) - i\mu \sin(k\Delta x). \tag{5.54}$$

Taking into account that the stability condition for the scheme is $|g(\mu, k\Delta x)| \leq 1$, the Lax-Friedrichs scheme must verify that:

$$\begin{aligned}&|g(\mu, k\Delta x)| = [\cos^2(k\Delta x) + \mu^2 \sin^2(k\Delta x)]^{\frac{1}{2}} \\ &= 1 - \sin^2(k\Delta x) + \mu^2 \sin^2(k\Delta x)]^{\frac{1}{2}} = [1 + (\mu^2 - 1)\sin^2(k\Delta x)]^{\frac{1}{2}} \leq 1. \end{aligned} \tag{5.55}$$

Therefore, the stability condition is $\mu^2 - 1 \leq 0 \Rightarrow \mu^2 \leq 1$, this is, $-1 \leq \mu \leq 1$.

Another way to obtain the amplification factor for the Lax-Friedrichs scheme is identifying the coefficients c_{-1}, c_0 and c_1 present in Theorem 5.1. Taking into account the linear expression of the scheme:

$$w_j^{n+1} = \left(\frac{\lambda \Delta t}{2\Delta x} + \frac{1}{2}\right) w_{j-1}^n + \left(\frac{\lambda \Delta t}{2\Delta x} - \frac{1}{2}\right) w_{j+1}^n. \tag{5.56}$$

We have that the coefficients are: $c_{-1} = \frac{\lambda \Delta t}{2\Delta x} + \frac{1}{2} = \frac{1+\mu}{2}$, $c_0 = 0$ and $c_1 = \frac{\lambda \Delta t}{2\Delta x} + \frac{1}{2} = \frac{1-\mu}{2}$. Therefore, applying Theorem 5.1, the expression of the amplification factor is:

$$g(k, \Delta x, \Delta t) = \frac{1}{2}\left[e^{-ik\Delta x} + e^{ik\Delta x} - \mu(e^{ik\Delta x} - e^{-ik\Delta x})\right]. \tag{5.57}$$

Exercise 5.4 Programme the Lax-Friedrichs method for the transport equation and analyse the stability condition numerically. Analyse the convergence to the exact solution by using different grids.

Solution 5.4 Code included in Sect. 8.2.

Figure 5.10 shows the comparison between the exact solution and the numerical solution obtained with the Lax-Friedrichs scheme for the problem presented in Example 5.1. In this case, the correct propagation of the solution is shown. The considered parameters are the same as those employed for the centered scheme defined by (5.36), $\Delta x = 0.1$, $\Delta t = 0.1$ and $\mu = 1$.

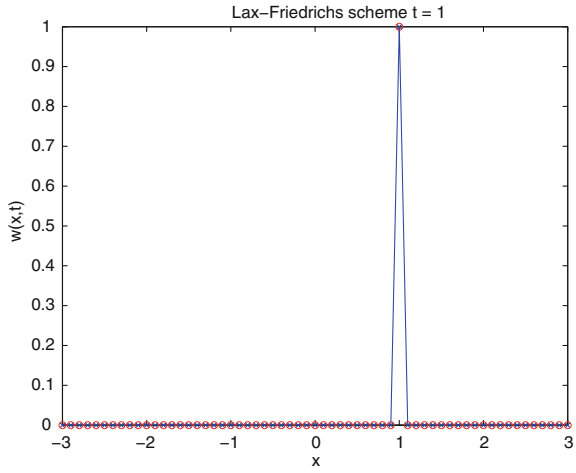


Fig. 5.10 Lax-Friedrichs scheme: exact solution (*red* ∘) and approximate (*blue* ×). $\mu = 1, t = 1$

5.3.1 Local Truncation Error of the Lax-Friedrichs Scheme

The local truncation error of a numerical scheme is obtained by replacing in the equation that defines such scheme the approximate solution w_j^n by the exact solution, which is assumed to be sufficiently smooth, $w(x_j, t^n)$, $\forall j$, n and to which the corresponding Taylor series expansion is applied.

Next, we proceed to the calculation of the truncation error τ_j^n of the Lax-Friedrichs scheme. First, we replace the exact solution in the discretization of the transport equation using the mentioned scheme:

$$\tau_j^n = \frac{w(x_j, t^{n+1}) - \frac{1}{2}(w(x_{j-1}, t^n) + w(x_{j+1}, t^n))}{\Delta t} + \lambda \frac{w(x_{j+1}, t^n) - w(x_{j-1}, t^n)}{2\Delta x}.$$

If we take into account the Taylor series expansion in time and space of $w(x_j, t^{n+1})$, $w(x_{j-1}, t^n))$, $w(x_{j+1}, t^n))$ in (x_j, t^n), we obtain the expression

$$\begin{aligned}\tau_j^n &= \frac{w(x_j, t^{n+1}) - w(x_j, t^n) - \frac{\Delta x^2}{2} w_{xx}(x_j, t^n) + \mathcal{O}(\Delta x^4)}{\Delta t} \\ &\quad + \lambda \left(w_x(x_j, t^n) + \frac{\Delta x^2}{6} w_{xxx}(x_j, t^n) + \mathcal{O}(\Delta x^4) \right) \\ &= w_t(x_j, t^n) + \frac{\Delta t}{2} w_{tt}(x_j, t^n) + \mathcal{O}(\Delta t^2) \\ &\quad - \frac{\Delta x^2}{2\Delta t} w_{xx}(x_j, t^n) + \lambda w_x(x_j, t^n) + \mathcal{O}(\Delta x^2). \end{aligned} \tag{5.58}$$

Further, we consider that $w(x_j, t^n)$ is a solution of the transport equation. Hence, $w_t(x_j, t^n) + \lambda w_x(x_j, t^n) = 0$. Therefore, it is immediate to check that the solution of the transport equation satisfies the relation $w_{tt}(x_j, t^n) = \lambda^2 w_{xx}(x_j, t^n)$. By considering these two equalities in (5.58), the expression of the truncation error is:

$$\begin{aligned}\tau_j^n &= \frac{\Delta t}{2} w_{tt}(x_j, t^n) - \frac{\Delta x^2}{2\Delta t} w_{xx}(x_j, t^n) + \mathcal{O}(\Delta x^2) + \mathcal{O}(\Delta t^2) \\ &= \frac{\Delta x^2}{2\Delta t} \left(\mu^2 - 1 \right) w_{xx}(x_j, t^n) + \mathcal{O}(\Delta x^2) + \mathcal{O}(\Delta t^2), \quad \forall j \in \mathbf{Z},\ \forall n \geq 0. \end{aligned} \tag{5.59}$$

Hence, it is a first-order scheme in space and time.

It is important to highlight that if the Courant number is $\mu = 1$, the first significant term in (5.59) vanishes. As a result, accuracy increases and the approximate solution becomes exact. An equivalent conclusion is obtained if we consider $\mu = 1$ in the expression of the amplification factor (5.57), which becomes 1, as in the case of the exact solution.

5.4 Upwind Schemes

We have been dealing with centered methods in the previous sections. This is, methods that are symmetric with respect to the point where the solution is updated. However, in the case of the hyperbolic problems, we want the information to propagate as waves that move along the characteristics. For hyperbolic systems, different waves will propagate at different velocities and directions. This idea leads us to the *upwind schemes*, that are characterized by the fact that the *information for each characteristic variable is obtained by searching in the direction where the information could be coming.*

The upwind numerical methods have been proven to perfectly fit Computational Fluid Dynamics (CFD) problems, initially in the field of Aerodynamics, where they have been extensively applied.

Indeed, these methods are amongst the numerical techniques that have raised a greater research interest regarding flow simulation modeling.

The upwind schemes are based on the physical concept of the *direction of propagation of the information*, applied to the discretization of the convective spatial derivative. In order to illustrate its basics and design elements, we continue with the canonic case of the hyperbolic equation of transport.

The numerical solution of the transport equation in a uniform grid considering a first-order upwind scheme is based on the approximation of the upwind spatial derivative depending on the sign of λ. Let us analyze the two possible signs:

$$w_t \simeq \frac{w_j^{n+1} - w_j^n}{\Delta t}, \qquad \begin{array}{ll} \text{(i)} & w_x \simeq \dfrac{w_j^n - w_{j-1}^n}{\Delta x}, \quad \text{if} \quad \lambda > 0. \\ \text{(ii)} & w_x \simeq \dfrac{w_{j+1}^n - w_j^n}{\Delta x}, \quad \text{if} \quad \lambda < 0. \end{array}$$

This means that we need to apply upwind discretizations in space and time. If $\lambda > 0$, the expression of the scheme is the following:

$$w_j^{n+1} = w_j^n - \frac{\lambda \Delta t}{\Delta x}\left(w_j^n - w_{j-1}^n\right). \tag{5.60}$$

Or, in terms of the Courant number (see (5.27)):

$$w_j^{n+1} = w_j^n - \mu\left(w_j^n - w_{j-1}^n\right). \tag{5.61}$$

In the same way, if $\lambda < 0$, we have

$$w_j^{n+1} = w_j^n - \mu\left(w_{j+1}^n - w_j^n\right). \tag{5.62}$$

More concisely, it is possible to obtain an expression of the scheme that accounts for both signs of λ:

$$w_j^{n+1} = w_j^n - \frac{\lambda^+ \Delta t}{\Delta x}\left(w_j^n - w_{j-1}^n\right) - \frac{\lambda^- \Delta t}{\Delta x}\left(w_{j+1}^n - w_j^n\right), \tag{5.63}$$

where

$$\lambda^{\pm} = \frac{1}{2}(\lambda \pm |\lambda|).$$

Scheme (5.63) is a first-order upwind scheme and was introduced by Courant et al. [4]. For this reason, it is also known as CIR scheme.

Moreover, if we take into account that the flux of this equation is $f(w) = \lambda w$, another way of presenting the scheme (see Fig. 5.11) is obtained by defining the flux increments at the mid points $x_{i\pm 1/2}$. Namely,

$$w_j^{n+1} = w_j^n - \frac{\Delta t}{\Delta x}(\delta f^{n+}_{j-\frac{1}{2}} + \delta f^{n-}_{j+\frac{1}{2}}), \tag{5.64}$$

with

$$\delta f^{n\pm}_{j+\frac{1}{2}} = \lambda^{\pm} \delta w^n_{j+\frac{1}{2}}, \quad \delta w^n_{j+\frac{1}{2}} = w^n_{j+1} - w^n_j, \tag{5.65}$$

$$\delta f^{n\pm}_{j-\frac{1}{2}} = \lambda^{\pm} \delta w^n_{j-\frac{1}{2}}, \quad \delta w^n_{j-\frac{1}{2}} = w^n_j - w^n_{j-1}. \tag{5.66}$$

In this way, contributions of different sign emerge from the grid boundaries depending on the sign of λ. In addition, they are a function of the closest nodes to $x_{j-\frac{1}{2}}$ and $x_{j+\frac{1}{2}}$. The solution evolves at each time step Δt according with those of the waves on the right and on the left, which is known as *wave formulation*.

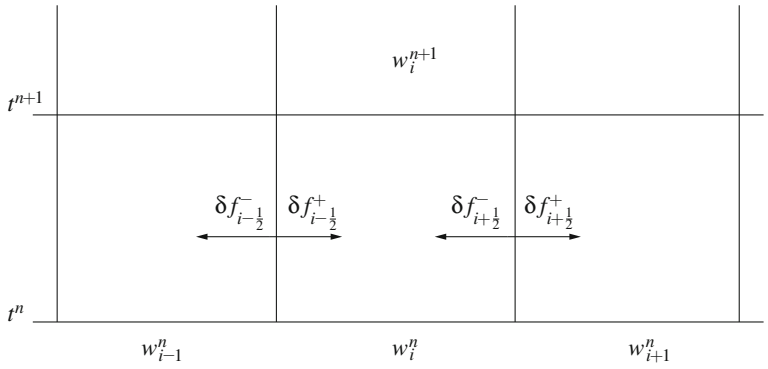


Fig. 5.11 Upwind scheme: wave formulation

As mentioned in the section devoted to the basic concepts, if we use the definitions above to construct the different *finite volume* methods, this scheme admits the following *conservative formulation*:

$$w_j^{n+1} = w_j^n - \frac{\Delta t}{\Delta x}(\phi(w_j^n, w_{j+1}^n) - \phi(w_{j-1}^n, w_j^n)), \tag{5.67}$$

since a *numerical flux* function $\phi(w_j^n, w_{j+1}^n)$ is associated with the numerical scheme, which, in the case of the upwind scheme is given by

$$\begin{aligned}\phi(w_j^n, w_{j+1}^n) &= \frac{1}{2}(\lambda w_j^n + \lambda w_{j+1}^n) - \frac{1}{2}|\lambda|(w_{j+1}^n - w_j^n) \\ &= \frac{1}{2}(f(w_j^n) + f(w_{j+1}^n)) - \frac{1}{2}|\lambda|(w_{j+1}^n - w_j^n),\end{aligned} \tag{5.68}$$

where $\frac{1}{2}(f(w_j^n) + f(w_{j+1}^n))$ represents the centered part of the numerical flux, and the scheme's numerical viscosity is introduced in the second term of the right hand side of (5.68) by means of $-\frac{1}{2}|\lambda|(w_{j+1}^n - w_j^n)$. This is the term that makes the upwind scheme different from the centered scheme defined in (5.37). Hence, this term is responsible for the conditional stability of the method, as we shall see in the corresponding von Neumann analysis.

In contrast with the *wave formulation*, which is centered at the cell boundaries (see Fig. 5.11), the *conservative formulation* focuses on the nodes. Following (5.67), the value of the variable w_j^{n+1} changes depending on the balance of numerical flux that affects its own cell.

Equation (5.64) constitutes an alternative way of considering the same situation. It consists of focusing at the cell boundaries and observe what direction the signals take.

The *stencil* of the scheme if $\lambda > 0$ is provided in Fig. 5.12, where x_p is the *real domain of dependence* and the interval $[x_{j-1}, x_j]$ is the *numerical domain of dependence*. These two domains do not coincide.

The exact value of w in (x_j, t^{n+1}) depends on the exact value at x_p, for being the intersection point of the characteristic at $t = t^n$.

Unfortunately, the only available values at time t^n at the grid points are w_j^n and w_{j-1}^n, that only coincide with x_p if

$$\mu = 1 \Longrightarrow \frac{\Delta x}{\Delta t} = \lambda.$$

However, there is one way of using the available information about the solution at the grid points, consisting of linearly interpolating the values of the solution in the numerical domain of dependence. Let us call d the distance between x_p and x_j. If we take into account that the slope of the characteristic curve $\frac{dx}{dt} = \lambda$ (see Fig. 5.12) is $m = \frac{1}{\lambda} = \frac{\Delta t}{d}$:

$$d = \lambda \Delta t = \frac{\lambda \Delta t}{\Delta x}\Delta x = \mu \Delta x.$$

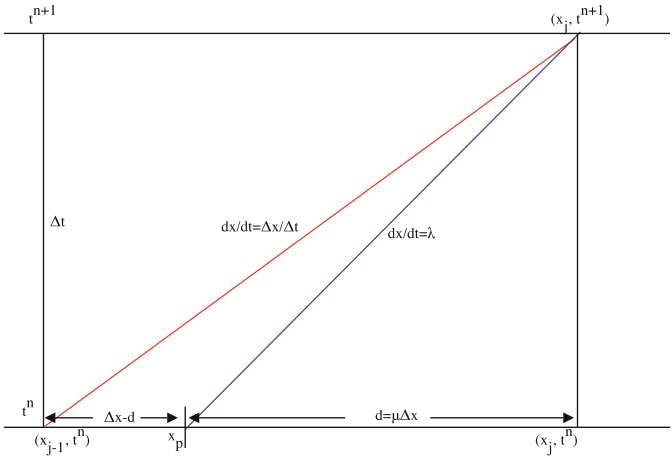


Fig. 5.12 Numerical and exact domains of dependence of the upwind scheme

Hence:

$$x_p = (j-1)\Delta x + (1-\mu)\Delta x.$$

The linear interpolation is given by:

$$\widetilde{w}(x) = w_{j-1}^n + \frac{w_j^n - w_{j-1}^n}{\Delta x}\left(x - x_{j-1}\right),$$

if we take $x = x_p$,

$$\widetilde{w}(x_p) = w_{j-1}^n + \frac{w_j^n - w_{j-1}^n}{\Delta x}(1-\mu)\,\Delta x,$$

and after some algebra we obtain:

$$\widetilde{w}(x_p) = w_j^n - \mu\left(w_j^n - w_{j-1}^n\right),$$

which is, precisely, the first-order upwind scheme (5.68)

$$w_j^{n+1} = \widetilde{w}(x_p),$$

and can be interpreted as a linear interpolation scheme.

5.4.1 von Neumann Analysis of the First-Order Upwind Scheme Defined by (5.63)

Let us consider the initial condition

$$w_j^0 = \widehat{w}^0 \mathrm{e}^{ikx_j}, \quad \forall j \in \mathbf{Z},$$

where $\widehat{w}^0$ represents a complex amplitude and k is a real wave number.

We have the following lemma:

Lemma 5.2 *A solution* $\left\{w_j^n\right\}$ *obtained with the explicit first-order upwind scheme (5.63) with the initial condition*

$$w_j^0 = \widehat{w}^0 \mathrm{e}^{ikx_j}, \ \forall j \in \mathbf{Z},$$

is given by

$$w_j^n = \widehat{w}^n e^{ikx_j}, \quad \forall j \in \mathbf{Z}, \quad \forall n \in \mathbf{N}$$

where the amplitude $\widehat{w}$ *is given by the recurrence relation*

$$\widehat{w}^{n+1} = g(k, \Delta x, \Delta t)\widehat{w}^n,$$

and corresponds to the discrete Fourier transform of w^n*, where*

$$g(k, \Delta x, \Delta t) = 1 - |\mu| + |\mu| \mathrm{e}^{-sig(\lambda)ik\Delta x}, \tag{5.69}$$

is the amplification coefficient of the scheme.

Remark 5.1 In order to obtain the amplification factor it is simpler to split each of the expressions of the upwind scheme as a function of the sign of λ given by (5.61) and (5.62):

$$g(k, \Delta x, \Delta t) = 1 - \mu + \mu \mathrm{e}^{-ik\Delta x} \quad \text{if } \lambda > 0,$$
$$g(k, \Delta x, \Delta t) = 1 + \mu - \mu \mathrm{e}^{ik\Delta x} \quad \text{if } \lambda < 0,$$

that can be expressed uniquely by (5.69), giving as a result relation (iii) of the following Corollary:

Corollary 5.2 *The explicit first-order upwind scheme (5.63) also verifies:*

(i) $\widehat{w}^n = g(k, \Delta x, \Delta t)^n \, \widehat{w}^0$.
(ii) $|\widehat{w}^n| = |g(k, \Delta x, \Delta t)|^n \, |\widehat{w}^0|$.
(iii) $|g(k, \Delta x, \Delta t)| = (1 - 2|\mu|(1 - |\mu|)(1 - cos(k\Delta x)))^{\frac{1}{2}} \leq 1 \Longrightarrow |\mu| \leq 1.$

Therefore, it is a *conditionally stable* scheme.

Exercise 5.5 Obtain the coefficients c_{-1}, c_0 and c_1 of the representation of the upwind scheme as a linear scheme and verify that the same amplification factor is obtained either by applying Theorem 5.1 or considering the one provided by Corollary 5.2.

Solution 5.5 Left to the reader.

Exercise 5.6 Programme the first-order upwind scheme for the transport equation and study the stability condition numerically. Analyze the convergence of the exact solution employing different grids.

Solution 5.6 The code included in Sect. 8.3 solves te problem presented in Example 5.1.

Figure 5.13 shows the comparison between the exact and the numerical solution obtained by means of an upwind scheme for the problem of Example 5.1. We can see that in this case, as in the Lax-Friedrichs scheme, the solution propagates in a correct manner. The considered parameters are the same as the ones used for the centered scheme:

$$\Delta x = 0.1, \quad \Delta t = 0.1, \quad \mu = 1.$$

The accuracy of the upwind scheme in the linear case for $\mu = 1$ is also illustrated with this example.

Exercise 5.7 Prove that for $\mu = 1$, the Lax-Friedrichs and the upwind schemes coincide and are exact, as shown in Figs. 5.10 and 5.13. This is known as *unit CFL condition* (see LeVeque [18]).

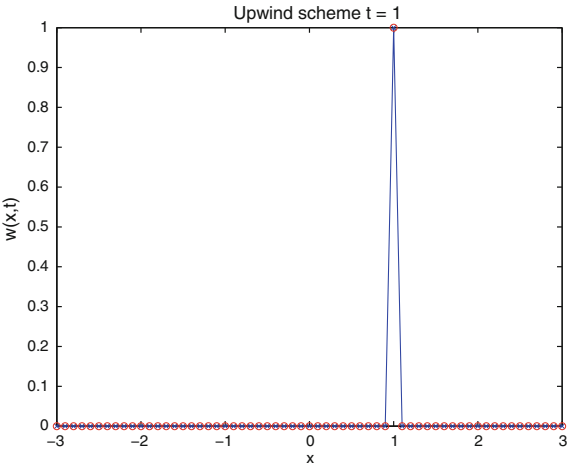


Fig. 5.13 Upwind scheme (5.63): exact solution (*red* ∘) and approximate solution (*blue* ×). $\mu = 1, t = 1$

Solution 5.7 Left to the reader.

Exercise 5.8 Use the codes of the Lax-Friedrichs and upwind schemes to analyze their different behavior if $\mu \neq 1$.

Solution 5.8 Left to the reader.

5.4.2 The First-Order Upwind Scheme Expressed as a Finite Volume Method

In this section we have considered different interpretations of the upwind scheme. The interpretation that best fits the point of view of the finite volume method is depicted in Fig. 5.14. We present the constant solution in cells C_i, $i = 1, \ldots, M$, at time $t = t^n$, which is considered as the initial condition, and the corresponding exact solution that is obtained when solving the transport equation after a time step Δt.

Geometrically, it can be seen that the integral of the exact solution at cell C_i is easily obtained if we subtract to the area of the rectangle of height w_i^n and base Δx the area that decreases (in this figure $\lambda > 0$) in this cell due to the transport of the solution at velocity λ during time Δt. More precisely, the area of the rectangle of base $\lambda \Delta t$ and height $w_i^n - w_i^{n-1}$. The average of the exact solution in every cell C_i gives the value w_i^{n+1}, that will be used to construct the solution of the finite volume method:

$$w_i^{n+1} = \frac{1}{\Delta x} \int_{C_i} w(x, t^{n+1}) dx = \frac{1}{\Delta x} \left(\Delta x w_i^n - \lambda \Delta t \left(w_i^n - w_{i-1}^n \right) \right)$$

$$= w_i^n - \mu \left(w_i^n - w_{i-1}^n \right). \tag{5.70}$$

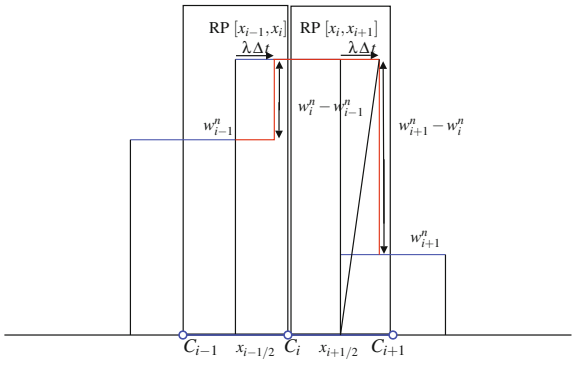


Fig. 5.14 Interpretation of the upwind scheme (5.63) as a finite volume method for $\lambda > 0$

Example 5.2 Representation of the solution obtained by means of the upwind scheme (5.63) expressed as finite volume method, when solving the problem of the propagation of a pulse of Example 5.1. The parameters considered in Figs. 5.15 and 5.16 are $\Delta x = 1$, $\lambda = 1$, $t = 4$ and two choices of the Courant number to illustrate the diffusivity if $\mu = 0.95$ and the exact accuracy if $\mu = 1$.

The code to obtain these plots is provided in Sect. 8.4.

Exercise 5.9 Obtain the expression of the upwind scheme for $\lambda < 0$.

Solution 5.9 Left to the reader.

Exercise 5.10 Carry out a numerical analysis to prove that the first-order upwind scheme (5.63) is conservative. Check that the integral of the initial condition coincides

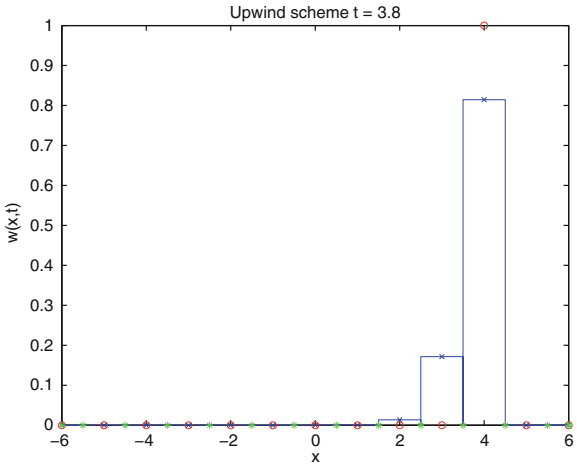


Fig. 5.15 Upwind scheme: exact solution (*red* ∘) and piecewise constant approximate solution (*blue* ×). $\mu = 0.95$. The boundaries of the finite volumes are identified with *green* ∗

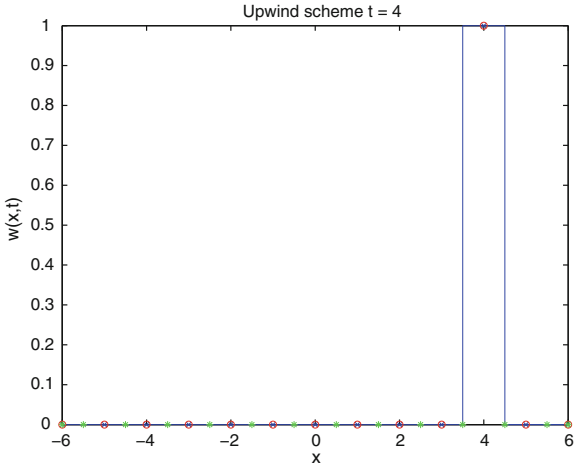


Fig. 5.16 Upwind scheme (5.63): exact solution (*red* ∘) and piecewise constant approximate solution (*blue* ×). $\mu = 1$. The boundaries of the finite volumes are identified with *green* ∗

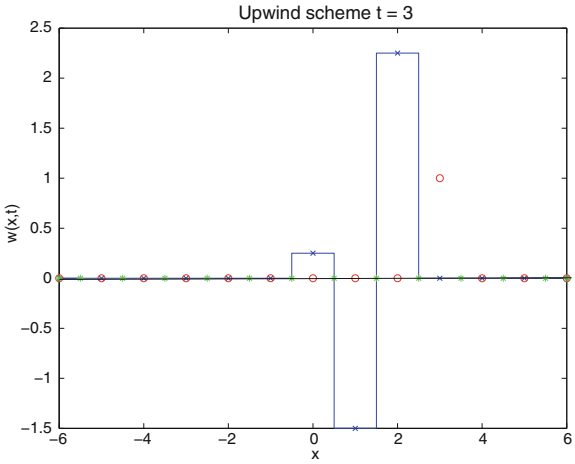


Fig. 5.17 Upwind schme (5.63): exact solution (*red* ∘) and approximated piecewise constant solution (*blue* ×). $\mu = 1.5$. The boundaries of the finite volumes are identified with *green* ∗

with the integral of the solution at every time step although the numerical solution displays the effects of diffusion when the Courant number is reduced (and, hence, the module of the amplification factor of the scheme).

Solution 5.10 The code provided in Sect. 8.4 calculates the integrals that are required in the exercise and allows us to verify that the scheme is conservative, since both of them are 1 for any choice of the Courant number. The results for $\mu = 1.5$ are shown in Fig. 5.17, corresponding to a choice of parameters for which the scheme is unstable.

5.4.3 Local Truncation Error of the First-Order Upwind Scheme

Following the same procedure as in Sect. 5.3.1 for the Lax-Friedrichs scheme, it can be proven that the truncation error τ_j^n of the CIR scheme at each node for $\lambda > 0$ is:

$$\tau_j^n = \frac{w(x_j, t^{n+1}) - w(x_j, t^n))}{\Delta t} + \lambda \frac{w(x_j, t^n) - w(x_{j-1}, t^n)}{\Delta x}$$
$$= \frac{1}{2}\lambda\,(\Delta x - \lambda \Delta t)\, w_{xx}(x_j, t^n) + \mathcal{O}(\Delta x^2) + \mathcal{O}(\Delta t^2), \quad \forall j \in \mathbf{Z}, \quad \forall n \geq 0.$$

Therefore, it is a first-order scheme in space and time. In the same way as for the Lax-Friedrichs scheme, if the Courant number is 1, then $\Delta x = \lambda \Delta t$. Therefore, accouracy increases if we focus on the most representative terms, reaching to the exact accuracy. The same argument can be used if we take into account the expression of the scheme given in (5.69), since $\widetilde{w}(x_p) = w(x_j, t^{n+1})$ if $\Delta x = \lambda \Delta t$.

Exercise 5.11 Study the local truncation error of the upwind scheme for $\lambda < 0$.

Solution 5.11 Left to the reader.

5.5 Analysis of the Order of Accuracy of the First-Order Upwind Scheme and the Lax-Friedrichs Scheme

In this section we will carry out a computational analysis of the order of accuracy of the Lax-Friedrichs and upwind schemes to confirm that they are both first-order in space.

An alternative to determine the order of accuracy of a numerical method consists of comparing the numerical solution with the exact solution for any test for which the exact solution is known. The order of accuracy obtained in this way is not necessarily the same as the formal one, especially when the solution is out of the asympthotic convergence range of the grid.

In order to calculate the order of accuracy of the Lax-Friedrichs and upwind schemes, the scalar equation of transport is solved numerically. We consider a smooth initial condition, in this case $w_0(x) = \sin(2\pi x)$ in the interval $[0, 1]$, with $\lambda = 1$ and periodic boundary conditions, $w(0, t) = w(1, t)$. The exact solution for this test case is $w(x, t) = \sin(2\pi(x - \lambda t))$. We will compare this solution with the ones obtained by means of the two numerical methods mentioned above.

The order of accuracy of each of the two schemes will be determined by observing the behavior of the error $\varepsilon_{\Delta x}$ in the norms $\| \cdot \|_1$ and $\| \cdot \|_\infty$. The spatial error is calculated at time $t^f = N\Delta t$ in a uniformly spaced grid, with $\Delta x = \frac{b-a}{M}$, being M the total number of nodes. The spatial error in each norm is defined as

$$\varepsilon^1_{\Delta x}(t^f) = \Delta x \sum_{j=1}^{M} \| w(x_j, t^f) - w_j^N \|, \tag{5.71}$$

and

$$\varepsilon^\infty_{\Delta x} = \max_j \| w(x_j, t^f) - w_j^N \| . \tag{5.72}$$

Formally, we say that a method is of order p, if for a positive constant c, we have

$$\varepsilon_{\Delta x_k}(t^f) \leq c(\Delta x_k)^p, \tag{5.73}$$

such that if we refine the grid, it holds that

$$\varepsilon_{\Delta x_{k+1}}(t^f) \leq c(\Delta x_{k+1})^p, \tag{5.74}$$

for all grid k.

Table 5.1 Order of accuracy in space of the Lax-Friedrichs scheme depending on the degree of grid refinement

M	Error in $\|\cdot\|_1$	p_k	Error in $\|\cdot\|_\infty$	p_k
60	0.3180	0.77368	0.4993	0.77313
120	0.1860	0.88126	0.2922	0.88103
240	0.1010	0.93918	0.1586	0.93911
480	0.0527	0.96922	0.0827	0.96920
960	0.0269	–	0.0423	–

For these calculations we have considered $\mu = 0.4$

Table 5.2 Order of accuracy in space of the upwind scheme depending on the degree of grid refinement

M	Error in $\|\cdot\|_1$	p_k	Error in $\|\cdot\|_\infty$	p_k
60	0.1140	0.93049	0.1792	0.93095
120	0.0598	0.96484	0.0940	0.96495
240	0.0307	0.98231	0.0482	0.98234
480	0.0155	0.99113	0.0244	0.99113
960	0.0078	–	0.0123	–

For these calculations we have considered $\mu = 0.4$

The selected grids verify that $\frac{\Delta x_k}{\Delta x_{k+1}} = 2$. Therefore, the scheme of order p in space satisfies the relation

$$\mathscr{K}_k = \frac{\varepsilon_{\Delta x_k}(t^f)}{\varepsilon_{\Delta x_{k+1}}(t^f)} \leq \frac{(\Delta x_k)^p}{(\Delta x_{k+1})^p} = 2^p \Rightarrow \log(\mathscr{K}_k) \leq p\log(2). \tag{5.75}$$

Then, in terms of $\mathscr{K}_k$, for two grids $\mathscr{C}_{\Delta x_k}, \mathscr{C}_{\Delta x_{k+1}}$, the order of accuracy in space can be estimated numerically as

$$p_k := \frac{\log(\mathscr{K}_k)}{\log(2)}. \quad k = 1, 2, \ldots \tag{5.76}$$

Tables 5.1 and 5.2 show the errors of each method as well as the order of accuracy of each grid. We have considered 5 grids for these calculations, with an initial grid of 60 nodes.

Figure 5.18 shows the behavior of the Lax-Friedrichs and upwind schemes when approximating the solution of the test problem stated at the beginning of this section. In Sect. 8.5 we provide the code to solve this problem and to obtain Fig. 5.18.

Exercise 5.12 Programme the Lax-Friedrichs scheme and the upwind scheme and check numerically the observed order of accuracy.

Solution 5.12 Code included in Sect. 8.5.

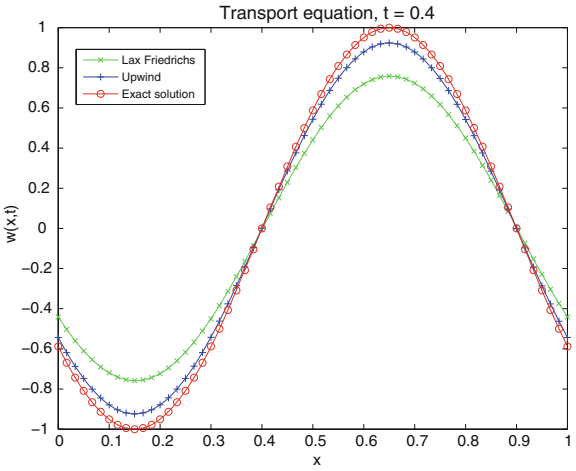


Fig. 5.18 Numerical solution obtained with the Lax-Friedrichs and upwind schemes. Parameters are $M = 120$, $\lambda = 1$, $\mu = 0.4$, $t^f = 1$ and $\Delta t = 0.0033$

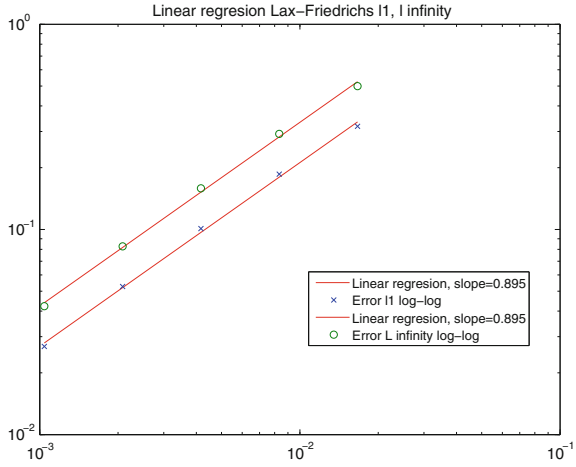


Fig. 5.19 Observed order of accuracy using the regression line for the Lax-Friedrichs scheme with norms $\|\cdot\|_1$ and $\|\cdot\|_\infty$

It is also possible to estimate numerically the order of accuracy of the numerical methods by calculating the slope of the regression line using logarithmic scales for the errors and the discretization steps. Figures 5.19 and 5.20 show the obtained errors for the Lax-Friedrichs and upwind schemes, respectively.

5.5.1 Godunov Method for a Linear System

The first-order Godunov method for a linear system of equations as the one studied in Sect. 2.4:

$$\mathbf{w}_t + \mathbf{A}\mathbf{w}_x = 0, \tag{5.77}$$

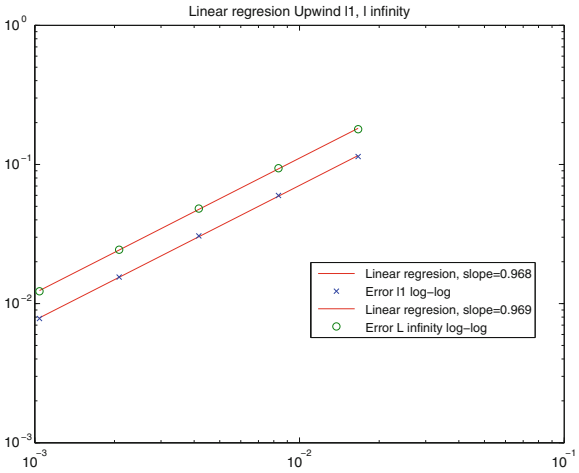


Fig. 5.20 Observed order of accuracy using the regression line for the upwind scheme with norms $\|\cdot\|_1$ and $\|\cdot\|_\infty$

is based on the discrete conservative form

$$\mathbf{w}_i^{n+1} = \mathbf{w}_i^n - \frac{\Delta t}{\Delta x}(\mathbf{f}_{i+\frac{1}{2}}^n - \mathbf{f}_{i-\frac{1}{2}}^n), \tag{5.78}$$

which is defined over a discrete constant distribution by cells, and in the exact solution of the Riemann problem defined for each couple of values $\mathbf{w}_i^n$,$\mathbf{w}_{i+1}^n$ at the boundary $x_{i+1/2}$ for the calculation of the numerical flux. More precisely, the scheme defines the numerical flux as the physical flux evaluated in the solution of the Riemann problem at the interface, this is, at $x = 0$ expressed in a local coordinate system centered at the initial discontinuity, that will be denoted by $\mathbf{w}_{i+\frac{1}{2}}(0)$:

$$\mathbf{f}_{i+\frac{1}{2}}^n = \phi\left(\mathbf{w}_i^n, \mathbf{w}_{i+1}^n\right) = \mathbf{f}(\mathbf{w}_{i+\frac{1}{2}}(0)). \tag{5.79}$$

Locally, the values of the solution at x_i and x_{i+1} play the role of the states $\mathbf{w}_L = \mathbf{w}_i^n = \sum_{k=1}^2 \alpha_k^L \mathbf{e}_k$ and $\mathbf{w}_R = \mathbf{w}_{i+1}^n = \sum_{k=1}^2 \alpha_k^R \mathbf{e}_k$ explained in Sect. 2.4.2, devoted to the solution of the Riemann problem associated with a linear hyperbolic system (Fig. 5.21). These values will be used to define the Godunov scheme. We recall the expressions of $\mathbf{w}_{i+\frac{1}{2}}(0)$ obtained from the general solution $\mathbf{w}(x,t)$ defined in (2.23)

$$\mathbf{w}(x,t) = \sum_{k=1}^{I} \alpha_k^R \mathbf{e}_k + \sum_{k=I+1}^{p} \alpha_k^L \mathbf{e}_k, \tag{5.80}$$

where I is the index that splits the positive and negative eigenvalues. This is, $\lambda_1 < \cdots < \lambda_I < 0 < \lambda_{I+1} < \cdots < \lambda_p$. In the present case, we will calculate the solution in the local reference system $x/t = 0$. Therefore, the solution admits the expression

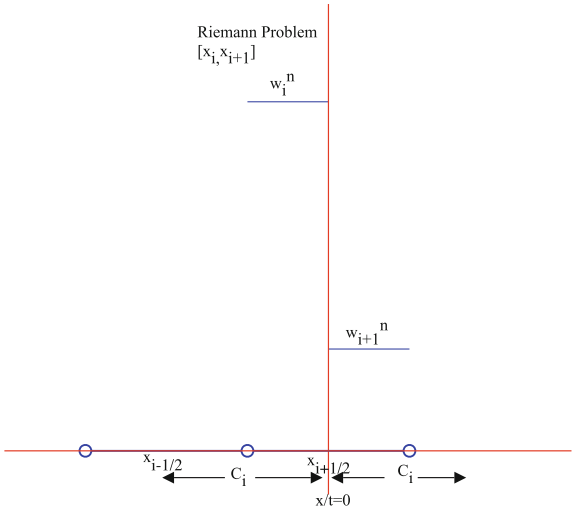


Fig. 5.21 Local coordinate system centered at the initial discontinuity

$$\begin{aligned}\mathbf{w}_{i+\frac{1}{2}}(0) &= \sum_{k=1}^{p} \alpha_k^R \mathbf{e}_k - \sum_{k=I+1}^{p} \alpha_k^R \mathbf{e}_k + \sum_{k=I+1}^{p} \alpha_k^L \mathbf{e}_k \\ &= \mathbf{w}_{i+1}^n - \sum_{k=I+1}^{p} sig(\lambda_k)(\alpha_k^R - \alpha_k^L)\mathbf{e}_k, \end{aligned} \tag{5.81}$$

and, analogously

$$\begin{aligned}\mathbf{w}_{i+\frac{1}{2}}(0) &= \sum_{k=1}^{I} \alpha_k^R \mathbf{e}_k + \sum_{k=1}^{p} \alpha_k^L \mathbf{e}_k - \sum_{k=1}^{I} \alpha_k^L \mathbf{e}_k \\ &= \mathbf{w}_i^n - \sum_{k=1}^{I} sig(\lambda_k)(\alpha_k^R - \alpha_k^L)\mathbf{e}_k. \end{aligned} \tag{5.82}$$

Averaging both versions (5.81) and (5.82)

$$\mathbf{w}_{i+\frac{1}{2}}(0) = \frac{1}{2}(\mathbf{w}_i^n + \mathbf{w}_{i+1}^n) - \frac{1}{2}\sum_{k=1}^{p} sig(\lambda_k)(\alpha_k^R - \alpha_k^L)\mathbf{e}_k. \tag{5.83}$$

Hence, the numerical flux of the Godunov method is

$$\begin{aligned}\mathbf{f}^n_{i+\frac{1}{2}} &= \mathbf{f}(\mathbf{w}_{i+\frac{1}{2}}(0)) = \mathbf{A}\mathbf{w}_{i+\frac{1}{2}}(0) \\ &= \frac{1}{2}(\mathbf{A}\mathbf{w}^n_i + \mathbf{A}\mathbf{w}^n_{i+1}) - \frac{1}{2}\sum_{k=1}^{p} sig(\lambda_k)(\alpha^R_k - \alpha^L_k)\mathbf{A}\mathbf{e}_k \\ &= \frac{1}{2}(\mathbf{f}(\mathbf{w}^n_i) + \mathbf{f}(\mathbf{w}^n_{i+1})) - \frac{1}{2}\sum_{k=1}^{p} |\lambda_k|\,(\alpha^R_k - \alpha^L_k)\mathbf{e}_k. \end{aligned} \tag{5.84}$$

If we define the absolute value matrix of $\mathbf{A}$, $|\mathbf{A}|$, as the matrix whose eigenvalues are the absolute value of the eigenvalues of $\mathbf{A}$

$$|\mathbf{A}| = \mathbf{P}\,|\Lambda|\,\mathbf{P}^{-1}, \tag{5.85}$$

being $|\Lambda|$ the diagonal matrix defined by $|\Lambda|_{ii} = |\lambda_{ii}|$, $i = 1, \ldots, p$. Taking into account that matrix $\mathbf{A}$ is constant in (5.84), we obtain the expression

$$\mathbf{f}^n_{i+\frac{1}{2}} = \frac{1}{2}\left(\mathbf{f}(\mathbf{w}^n_i) + \mathbf{f}(\mathbf{w}^n_{i+1})\right) - \frac{1}{2}|\mathbf{A}|\sum_{k=1}^{p}(\alpha^R_k - \alpha^L_k)\mathbf{e}_k. \tag{5.86}$$

Finally, using Eq. (2.30):

$$\mathbf{f}^n_{i+\frac{1}{2}} = \frac{1}{2}\left(\mathbf{f}(\mathbf{w}^n_i) + \mathbf{f}(\mathbf{w}^n_{i+1})\right) - \frac{1}{2}|\mathbf{A}|(\mathbf{w}^n_{i+1} - \mathbf{w}^n_i). \tag{5.87}$$

This expression of the numerical flux is in accordance with the one corresponding to the family of the *Q-schemes*. For these schemes the numerical flux comprises a *centered* part, represented by the average of the flux at the two states, and a *non-centered* part, characterized by the absolute value of a matrix $\mathbf{Q}$, which is the one that contains the information relative to the sign of the eigenvalues, as detailed in the previous calculations:

$$\mathbf{f}^n_{i+\frac{1}{2}} = \frac{1}{2}(\mathbf{f}(\mathbf{w}^n_i) + \mathbf{f}(\mathbf{w}^n_{i+1})) - \frac{1}{2}\,|\mathbf{Q}|\,(\mathbf{w}^n_{i+1} - \mathbf{w}^n_i). \tag{5.88}$$

In (5.87) matrix $\mathbf{A}$ coincides with $\mathbf{Q}$ and, in the non-linear case, we will have to precisely specify the point where it is evaluated. It is important to highlight that upwinding techniques are applied to the difference between states. This is why it is known in the literature as *flux-difference*.

In order to complete the formulation, the Godunov flux can also be expressed using (2.23) as follows:

$$\mathbf{f}^n_{i+\frac{1}{2}} = \mathbf{A}\mathbf{w}_{i+\frac{1}{2}}(0) = \sum_{k=1}^{I}\alpha^R_k\mathbf{A}\mathbf{e}_k + \sum_{k=I+1}^{p}\alpha^L_k\mathbf{A}\mathbf{e}_k. \tag{5.89}$$

If we replace the products of the Jacobian $\mathbf{A}$ and the eigenvalues, $\mathbf{A}\mathbf{e}_k$ by the products of the eigenvectors and eigenvalues, $\lambda_k \mathbf{e}_k$, we have:

$$\mathbf{f}^n_{i+\frac{1}{2}} = \mathbf{A}\mathbf{w}_{i+\frac{1}{2}}(0) = \sum_{k=1}^{I} \alpha_k^R \lambda_k \mathbf{e}_k + \sum_{k=I+1}^{p} \alpha_k^L \lambda_k \mathbf{e}_k. \tag{5.90}$$

Moreover, if we take advantage of the definition of I as the index that splits the characteristics of opposite sign, then we can complete the first summatory from $k = I + 1$ to p replacing λ_k with λ_k^-, and the second summatory from $k = 1$ to $k = I$ replacing λ_k with λ_k^+. Thus, the numerical flux can be written as:

$$\mathbf{f}^n_{i+\frac{1}{2}} = \sum_{k=1}^{p} \alpha_k^R \lambda_k^- \mathbf{e}_k + \sum_{k=1}^{p} \alpha_k^L \lambda_k^+ \mathbf{e}_k. \tag{5.91}$$

If we define the *positive* part and *negative* part matrices of matrix $\mathbf{A}$, $\mathbf{A}^+$ and $\mathbf{A}^-$ on the basis of its eigenvalues $\lambda_k^{\pm}$, $k = 1, \ldots, p$:

$$\mathbf{A}^{\pm} = \mathbf{P}\Lambda^{\pm}\mathbf{P}^{-1} = \mathbf{P}\left[\frac{1}{2}\left(\Lambda \pm |\Lambda|\right)\right]\mathbf{P}^{-1}, \tag{5.92}$$

then we reconstruct the conservative variables and obtain the expression of the numerical flux according with the *flux splitting techniques*:

$$\mathbf{f}^n_{i+\frac{1}{2}} = \mathbf{A}^- \sum_{k=1}^{p} \alpha_k^R \mathbf{e}_k + \mathbf{A}^+ \sum_{k=1}^{p} \alpha_k^L \mathbf{e}_k = \mathbf{A}^- \mathbf{w}^n_{i+1} + \mathbf{A}^+ \mathbf{w}^n_i. \tag{5.93}$$

It is interesting to highlight that in this case the flux is *split* into a positive part $\mathbf{A}^+\mathbf{w}^n_i$, which is multiplied by the left state (*the information that comes from the left is associated with positive eigenvalues*), and a negative part $\mathbf{A}^-\mathbf{w}^n_{i+1}$, which is multiplied by the right state (*the information that comes from the right is associated with negative eigenvalues*).

If we replace this expression of the numerical flux in the corresponding conservative scheme:

$$\begin{aligned}\mathbf{w}_i^{n+1} &= \mathbf{w}_i^n - \frac{\Delta t}{\Delta x}\left[\mathbf{A}^-\mathbf{w}^n_{i+1} + \mathbf{A}^+\mathbf{w}^n_i - \mathbf{A}^-\mathbf{w}^n_i - \mathbf{A}^+\mathbf{w}^n_{i-1}\right] \\ &= \mathbf{w}_i^n - \frac{\Delta t}{\Delta x}\left[\mathbf{A}^+\left(\mathbf{w}^n_i - \mathbf{w}^n_{i-1}\right) + \mathbf{A}^-\left(\mathbf{w}^n_{i+1} - \mathbf{w}^n_i\right)\right].\end{aligned}$$

This scheme is written as an extension of (5.64) in the *wave formulation*:

$$\mathbf{w}_i^{n+1} = \mathbf{w}_i^n - \frac{\Delta t}{\Delta x}(\delta\mathbf{f}^+_{i-\frac{1}{2}} + \delta\mathbf{f}^-_{i+\frac{1}{2}}), \tag{5.94}$$

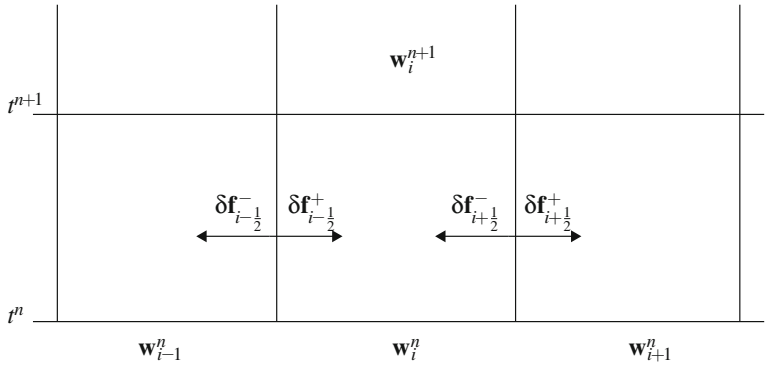


Fig. 5.22 Godunov method applied to a linear system: wave formulation

with

$$\delta\mathbf{f}^{\pm}_{i-\frac{1}{2}} = \mathbf{A}^{\pm}\delta\mathbf{w}^n_{i-\frac{1}{2}},\ \delta\mathbf{f}^{\pm}_{i+\frac{1}{2}} = \mathbf{A}^{\pm}\delta\mathbf{w}^n_{i+\frac{1}{2}},$$
$$\delta\mathbf{w}^n_{i-\frac{1}{2}} = \mathbf{w}^n_i - \mathbf{w}^n_{i-1},\ \delta\mathbf{w}^n_{i+\frac{1}{2}} = \mathbf{w}^n_{i+1} - \mathbf{w}^n_i. \tag{5.95}$$

This expression of the scheme implies centering at the cell boundaries (see Fig. 5.22) and send *information packages* to the cell on the left or on the right depending on the sign of the eigenvalues at the boundaries:

- The information that enters the cell from the left is associated with positive eigenvalues, $\delta\mathbf{f}^+_{i-\frac{1}{2}}$.
- The information that enters the cell from the right is associated with negative eigenvalues, $\delta\mathbf{f}^-_{i+\frac{1}{2}}$.

Exercise 5.13 Given the system of partial differential equations that represent a linearization of the shallow water equations studied in Exercise 2.5:

$$h_t + \tilde{u}h_x + \tilde{h}u_x = 0,$$
$$u_t + gh_x + \tilde{u}u_x = 0,$$

where $\tilde{h},\ \tilde{u},\ g$ are constants and $h = h(x,t),\ u = u(x,t)$.

1. Develop and implement in a code the discretization of the problem using the finite volume method and the Godunov scheme in any of the versions presented in this section. Consider that $\tilde{h}$, $\tilde{u}$ and $g = 9.8\,\mathrm{ms}^{-2}$ are given as well as the initial conditions of the Riemann problem h_l, u_l and h_r, u_r.

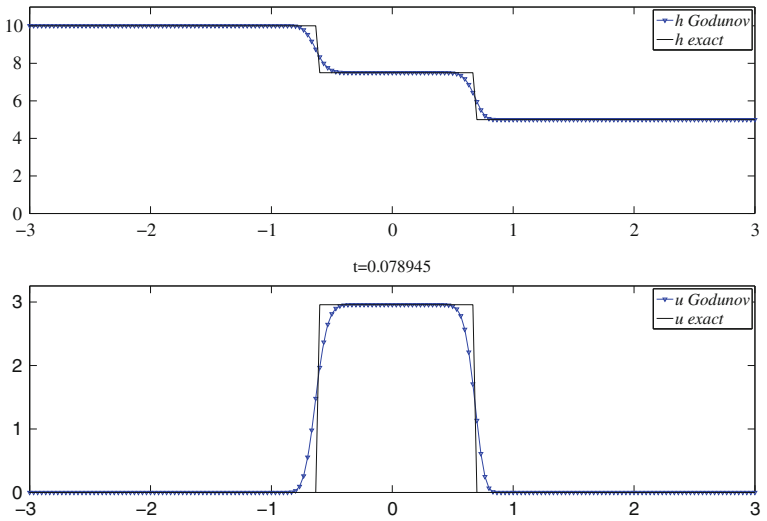


Fig. 5.23 Solution of the Riemann problem for the linear shallow water equations obtained by means of the Godunov method for $t = 0.08\,\mathrm{s}$. $M = 180$, $CFL = 0.8$

2. Taking into account that the stability condition $|\mu| \leq 1$, that was estudied for the transport equation and the upwind scheme, leads us to the stability condition for systems:

$$\max_{k=1,2} |\lambda_k| \frac{\Delta t}{\Delta x} \leq 1,$$

carry out different executions inside and outside the stability region.

Solution 5.13 Figure 5.23 shows the obtained results using the first-order Godunov method for the resolution of the Riemann problem presented in Exercise 2.5. We recall that the values that define the problem are:

$$\tilde{h} = 7\,\mathrm{m},\ \tilde{u} = 0.5\,\mathrm{ms}^{-1},\ \mathbf{w}_L = (h_L, u_L)^T = (10, 0)^T,\ \mathbf{w}_R = (h_R, u_R)^T = (5, 0)^T. \tag{5.96}$$

Exercise 5.14 Given the system of partial differential equations studied in Sect. 1.4 that represents an acoustic wave propagation model:

$$\frac{\partial u}{\partial t} + \frac{1}{\rho_0} \frac{\partial p}{\partial x} = 0, \tag{5.97}$$

$$\frac{\partial p}{\partial t} + \rho_0 c_0^2 \frac{\partial u}{\partial x} = 0, \tag{5.98}$$

where ρ_0 represents the density of the medium at rest and c_0 the compressibility of the medium at rest, and the conservative variables $(u,\ p)'$ refer to the velocity and the pressure of the perturbed medium.

1. Develop and implement in a code the discretization of the problem using the finite volume method and the Godunov scheme in any of the versions presented in this section. Consider that ρ_0 and c_0 are given as well as the initial conditions of the Riemann problem $u_l,\ p_l$ and $u_r,\ p_r$.
2. If we take into account that the stability condition $|\mu| \leq 1$, studied for the transport equation and the upwind scheme, leads us to the stability condition for systems:

$$\max_{k=1,2} |\lambda_k| \frac{\Delta t}{\Delta x} \leq 1,$$

carry out different executions inside and outside the stability region.

Solution 5.14 Left to the reader.

Chapter 6
Numerical Resolution of One-Dimensional Non-linear Hyperbolic Systems

Abstract This chapter is devoted to the development of the *finite volume method* to solve numerically one-dimensional non-linear conservation laws. We will construct the method on the basis of the progresses done in the previous chapter for linear systems. Non-linearity introduces new problems that are not present in the discretization of the linear problems, being mostly related to the difficulty of the non-linear problems that were studied in the first part of the book. We will analyse the relevance of using *conservative schemes* in order to obtain approximations of the weak solutions of the conservations laws. Moreover, we need to guarantee that the methods converge to the *correct* weak solution, since we have already studied that it is not unique. This difficulty implies that we require the *numerical method* to be *consistent* with an appropriate *entropy condition*. The mathematical models we will work with are the Burgers equation, for the scalar case and the shallow water equations for systems. The exact solutions for the corresponding Riemann problems will contribute both to the construction of the *Godunov method* and the corresponding approximations that constitute the *approximate Riemann solvers* and the *flux-splitting techniques*, as well as to the validation of the methods for this kind of problems.

6.1 Conservative Schemes

When we discretize a scalar non-linear conservation law of the form

$$w_t + f(w)_x = 0, \tag{6.1}$$

a necessary condition for the numerical method to correctly approximate the weak solutions is that it is written in conservative form.

We recall that the conservative expression is obtained from the integral form of the Eq. (6.1). There are two possibilities:

$$(i) \int\int_R \left(\frac{\partial w}{\partial t} + \frac{\partial}{\partial x} (f(w)) \right) dxdt = \oint_{\partial R} (wdx - f(w)dt) = 0. \tag{6.2}$$

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UNITEXT - La Matematica per il 3+2 90, DOI 10.1007/978-3-319-14784-0_6

$$(\ddot{u}) \int_{x_1}^{x_2} w(x,t_2)dx = \int_{x_1}^{x_2} w(x,t_1)dx + \int_{t_1}^{t_2} f(w(x_1,t))dt - \int_{t_1}^{t_2} f(w(x_2,t))dt, \tag{6.3}$$

for any control volume $R = [x_1, x_2] \times [t_1, t_2]$. The previous identities are derived from the Green's theorem and Fubini's theorem.

The numerical methods that approximate the conservation equation appropriately are called *conservative methods*. They can be defined in several ways, but we will follow the one proposed by Godlewski and Raviart [6].

Definition 6.1 Definition of conservative scheme.

A conservative scheme for the scalar conservation law (6.1) is

$$w_j^{n+1} = w_j^n + \frac{\Delta t}{\Delta x}\left(f_{j-\frac{1}{2}}^n - f_{j+\frac{1}{2}}^n\right), \tag{6.4}$$

where

$$f_{j+\frac{1}{2}}^n = f_{j+\frac{1}{2}}^n\left(w_{j-l_1}^n, \ldots, w_{j+l_2}^n\right), \tag{6.5}$$

with l_1 and l_2 two non-negative integers. The term $f_{j+\frac{1}{2}}^n$ is called numerical flux and, according to the literature, it is also denoted by ϕ. It is an approximation of the physical flux, $f(w)$, in (6.1). As we mentioned before, for each choice of the numerical flux function we have a different conservative scheme.

A fundamental requirement for a conservative scheme is the *consistency condition*, that was already introduced for linear problems in Sect. 5.1.4:

$$f(\overline{w}) = f_{j+\frac{1}{2}}^n(\overline{w}, \ldots, \overline{w}). \tag{6.6}$$

If $l_1 = 0$ and $l_2 = 1$, we have a *three*-point scheme:

$$w_j^{n+1} = w_j^n - \frac{\Delta t}{\Delta x}\left(\phi(w_j^n, w_{j+1}^n) - \phi(w_{j-1}^n, w_j^n)\right). \tag{6.7}$$

As we mentioned in the previous chapter, the derivation of the scheme from the integral form of the conservation law in $R = C_j \times [t_1, t_2]$, where C_j denotes the *finite volume*, leads us to reconsider the finite difference grid. If we assume that our solution is piecewise constant:

$$w_j^n \simeq \frac{1}{\Delta x}\int_{x_{j-1/2}}^{x_{j+1/2}} w(x,t^n)\,dx, \tag{6.8}$$

then we have a *Riemann problem* at each of the finite volume boundaries. Moreover, the expression of the numerical scheme can be interpreted as:

$$w_j^{n+1} \simeq \frac{1}{\Delta x} \int_{x_{j-1/2}}^{x_{j+1/2}} w(x, t^{n+1})\, dx, \tag{6.9}$$

$$f_{j+\frac{1}{2}}^n = \phi(w_j^n, w_{j+1}^n) \simeq \frac{1}{\Delta t} \int_{t^n}^{t^{n+1}} f(w(x_{j+\frac{1}{2}}, t))\, dt, \tag{6.10}$$

$$f_{j-\frac{1}{2}}^n = \phi(w_{j-1}^n, w_j^n) \simeq \frac{1}{\Delta t} \int_{t^n}^{t^{n+1}} f(w(x_{j-\frac{1}{2}}, t))\, dt. \tag{6.11}$$

We can find different *discrete flux* or *numerical flux* functions in the bibliography proposed by several authors (see, for instance, Harten et al. [15], Toro [30] and LeVeque [17]). We will specially focus on the Godunov method, the Q-schemes devised by Roe and van Leer and also the schemes obtained by means of the flux-splitting techniques of Steger-Warming and Vijayasundaram.

6.1.1 Conservative Version of the Godunov Method

The Godunov method was already presented in the chapter devoted to linear systems. According to this method the numerical flux associated with each of the boundaries of the finite volume is defined as the physical flux evaluated in the exact solution of the Riemann problem associated with the corresponding boundary $w_{j\pm\frac{1}{2}}(0)$:

$$w_j^{n+1} = w_j^n - \frac{\Delta t}{\Delta x} \left(f(w_{j+\frac{1}{2}}(0)) - f(w_{j-\frac{1}{2}}(0)) \right). \tag{6.12}$$

The Godunov method provides a numerical solution that corresponds to an exact evolution at each time step Δt. Using this method, we can also obtain the expression of the approximate solution by following the procedure in Sect. 5.5.1, this is

1. The two Riemann problems associated with the boundaries of the finite volume C_j are solved:

   $RP(w_{j-1}^n, w_j^n)$, whose solution is represented by $w_{j-1/2}\left(\frac{x}{t}\right)$, and
   $RP(w_j^n, w_{j+1}^n)$, whose solution is represented by $w_{j+1/2}\left(\frac{x}{t}\right)$.

   If we consider the piecewise constant function $\widetilde{w}(x, 0)$, then the initial conditions at $t = t^n$ of these problems are defined by the function:

   $$\widetilde{w}(x, 0) = w_j^n, \text{ if } x \in C_j,$$

   where we denote by $\widetilde{w}(x, t)$ the exact solution constructed from the solutions of the two mentioned Riemann problems:

$$\widetilde{w}(x,t) = \begin{cases} w_{j-1/2}\left(\frac{x-x_{j-1/2}}{t}\right) \text{ if } x \in [x_{j-1/2}, x_j), \\ \\ w_{j+1/2}\left(\frac{x-x_{j+1/2}}{t}\right) \text{ if } x \in (x_j, x_{j+1/2}]. \end{cases} \tag{6.13}$$

2. The value of the solution at cell C_j is the average value of the two solutions of the Riemann problems:

$$w_j^{n+1} = \frac{1}{\Delta x}\int_{x_{j-1}}^{x_{j+1}} \widetilde{w}(x, \Delta t)\, dx, \tag{6.14}$$

where $\widetilde{w}$ is the solution of the Riemann problems. Hence, it verifies the integral form of the conservation law:

$$\frac{1}{\Delta x}\int_{x_{j-1/2}}^{x_{j+1/2}} \widetilde{w}(x, \Delta t)\, dx = \frac{1}{\Delta x}\int_{x_{j-1/2}}^{x_{j+1/2}} \widetilde{w}(x, 0)\, dx$$
$$+ \frac{1}{\Delta x}\left[\int_0^{\Delta t} f(\widetilde{w}(x_{j-\frac{1}{2}}, t))\, dt - \int_0^{\Delta t} f(\widetilde{w}(x_{j+\frac{1}{2}}, t))\, dt\right].$$

This last relation allows us to recover the expression of the numerical flux of the Godunov method, see Fig. 6.1, and prove that the obtained numerical solution corresponds to an exact evolution of the solution in a time step Δt:

$$\frac{1}{\Delta t}\int_0^{\Delta t} f(\widetilde{w}(x_{j\pm\frac{1}{2}}, t)\, dt = \frac{1}{\Delta t}\int_0^{\Delta t} f(w_{j\pm\frac{1}{2}}(0))\, dt = f(w_{j\pm\frac{1}{2}}(0)), \tag{6.15}$$

where the equalities in (6.15) are obtained by taking into account that the value of the solution $(\widetilde{w})$ at points $(x_{j\pm\frac{1}{2}}, t)$ is equivalent to evaluating the corresponding solution

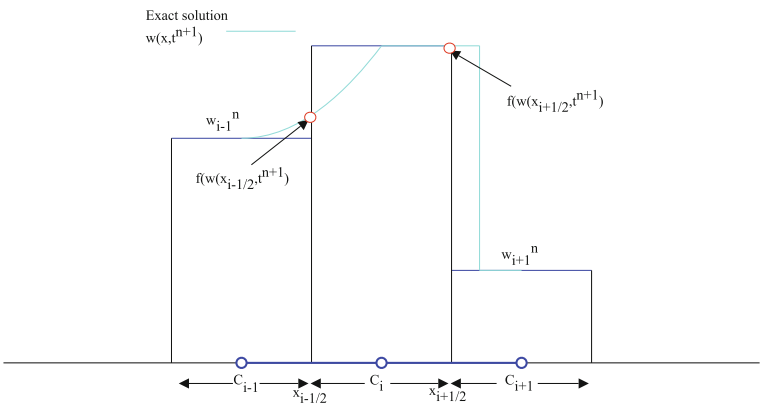


Fig. 6.1 Numerical flux of the Godunov method

$(w_{j\pm\frac{1}{2}})$ at the characteristic $\frac{x}{t} = 0$ of the local reference system that we locate at each boundary $x_{j\pm\frac{1}{2}}$ of the finite volume. This is:

$$\widetilde{w}(x_{j\pm\frac{1}{2}}, t) = w_{j\pm\frac{1}{2}}\left(\frac{0}{t}\right) = w_{j\pm\frac{1}{2}}(0). \tag{6.16}$$

Exercise 6.1 Verify that the numerical flux of the Godunov scheme is consistent.

Solution 6.1 Left to the reader.

Exercise 6.2 Give the expression of the Godunov flux for the scalar linear case $f(w) = \lambda w$.

Solution 6.2 Left to the reader.

Exercise 6.3 If we take into account that the *similar* entropy solution of the Burgers equation with initial conditions

$$w_0(x) = \begin{cases} w_l & \text{if } x < 0 \\ w_r & \text{if } x > 0, \end{cases}$$

is an expansion wave if $w_l \leq w_r$ and a shock wave if $w_l > w_r$, calculate the Godunov numerical flux in the following cases, represented in Fig. 6.2:

1. Supersonic to the right ($w_l > 0,\ w_r > 0$).
2. Supersonic to the left ($w_l < 0,\ w_r < 0$).
3. Transonic expansion ($w_l \leq 0 \leq w_r$).
4. Transonic shock ($w_l > s > 0 \geq w_r$) or ($w_l \geq 0 > s > w_r$),

with s the jump propagation velocity.

Solution 6.3 Left to the reader.

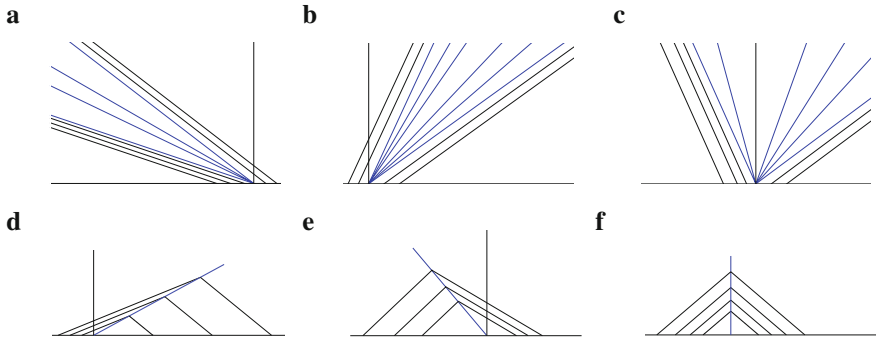


Fig. 6.2 **a** Supersonic to the left ($w_l < 0,\ w_r < 0$); **b** supersonic to the right ($w_l > 0,\ w_r > 0$); **c** transonic expansion ($w_l \leq 0 \leq w_r$); **d** transonic shock ($w_l > s > 0 \geq w_r$); **e** transonic shock ($w_l \geq 0 > s > w_r$); **f** stationary shock ($w_l > 0 = s > w_r = -w_l$)

6.1.2 Lax-Friedrichs Scheme

The Lax-Friedrichs scheme in the non-linear case is defined by:

$$w_j^{n+1} = \frac{1}{2}\left(w_{j-1}^n + w_{j+1}^n\right) - \frac{\Delta t}{2\Delta x}\left(f(w_{j+1}^n) - f(w_{j-1}^n)\right). \tag{6.17}$$

It is a *centered* method. We already mentioned for the linear case that the corresponding numerical flux function that allows us to prove that it is a conservative scheme is:

$$\phi(w_j^n, w_{j+1}^n) = \frac{f(w_j^n) + f(w_{j+1}^n)}{2} - \frac{\Delta x}{2\Delta t}(w_{j+1}^n - w_j^n). \tag{6.18}$$

6.1.3 A Non-conservative Scheme for the Burgers Equation

For the following numerical scheme associated with the Burgers equation

$$w_j^{n+1} = w_j^n - \frac{\Delta t}{\Delta x} w_j^n \left(w_j^n - w_{j-1}^n\right), \quad \text{if} \quad w_j^n \geq 0, \tag{6.19}$$

$$w_j^{n+1} = w_j^n - \frac{\Delta t}{\Delta x} w_j^n \left(w_{j+1}^n - w_j^n\right), \quad \text{if} \quad w_j^n \leq 0, \tag{6.20}$$

it is not possible to obtain the numerical flux function. Therefore, it is a numerical scheme that does not allow for a conservative formulation to approximate the Burgers equation.

One way to illustrate the unsuitable behavior of the method in an interval $[a, b]$ is to consider as initial condition:

$$w_0(x) = \begin{cases} 1 \text{ if } x < (a+b)/2, \\ 0 \text{ if } x \geq (a+b)/2, \end{cases} \tag{6.21}$$

and see that, for any t^n, the approximate solution is the initial condition:

$$w_j^n = w_0(x_j), \quad \forall j, \quad \forall n. \tag{6.22}$$

This solution is not correct, since the exact solution of this Riemann problem is known to be a shock wave that propagates at velocity $s = \frac{1}{2}$ (see Fig. 6.3).

Exercise 6.4 Solve the Riemann problem associated with the Burgers equation with the non-conservative scheme (6.19) and the initial condition (6.21).

Solution 6.4 In Chap. 9 we provide the Matlab code to implement the scheme. The results obtained with this code are shown in Fig. 6.3 for a choice of the interval $[a, b] = [0, 5]$.

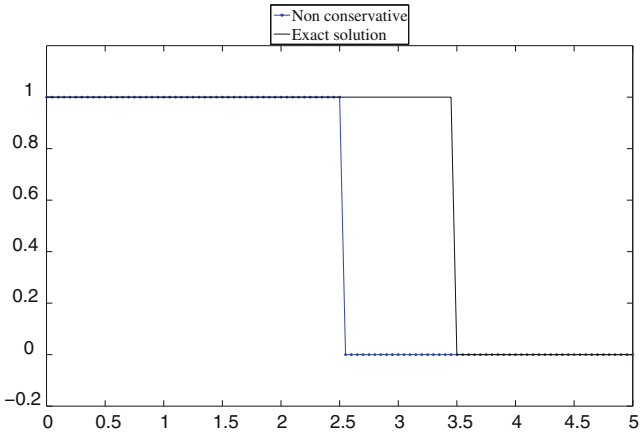


Fig. 6.3 Exact solution and and numerical results obtained with the non-conservative scheme defined in (6.19) for the Burgers equation with initial conditions $w_L = 1$, $w_R = 0$. The Courant number employed is $\mu = 0.9$ and $\Delta x = 0.05$

Exercise 6.5 Compare the solution of the Riemann problem associated with the Burgers equation with the code of the non-conservative scheme (6.19) and the initial condition

$$w_0(x) = \begin{cases} 0.5 \text{ if } x < (a+b)/2, \\ 0.1 \text{ if } x \geq (a+b)/2. \end{cases} \tag{6.23}$$

Solution 6.5 In Chap. 9 we provide the Matlab code to implement the mentioned schemes. Figure 6.4 shows the comparison of the numerical solution with the exact

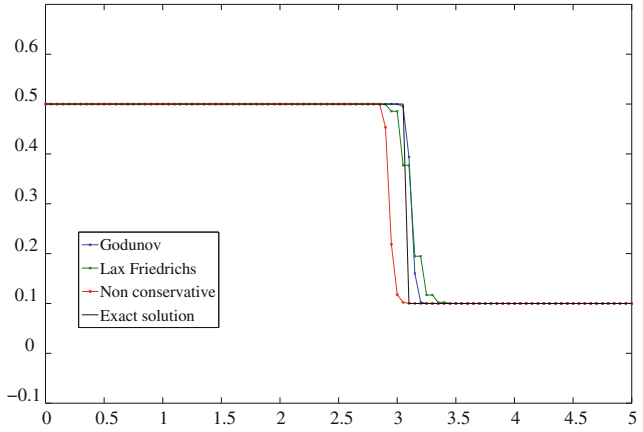


Fig. 6.4 Comparison of the conservative schemes (Lax-Friedrichs and Godunov method) and the non-conservative scheme defined in (6.19) for the Burgers equation with initial conditions $w_L = 0.5$, $w_R = 0.1$ and considering the Courant number $\mu = 0.9$ and $\Delta x = 0.05$ in $t = 2$

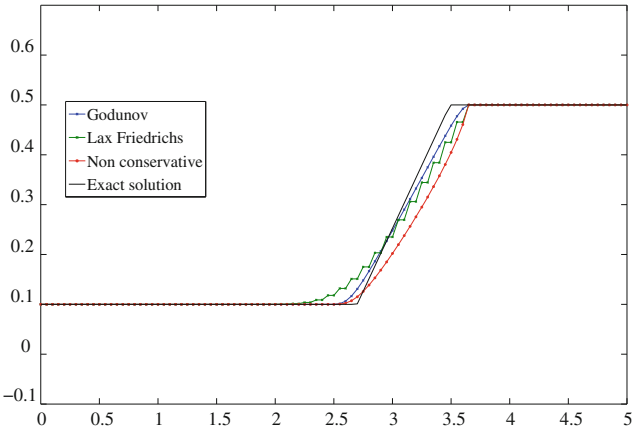


Fig. 6.5 Comparison of the conservative schemes (Lax-Friedrichs and Godunov method) and the non-conservative scheme defined in (6.19) for the Burgers equation with initial conditions $w_L = 0.1,\ w_R = 0.5$, using the Courant number $\mu = 0.9$ and $\Delta x = 0.05$ in $t = 2$

solution for the conservative schemes (Lax-Friedrichs, **Q**-scheme defined in (6.5) and the Godunov method) and the non-conservative scheme defined by (6.19) for the Burgers equation with initial conditions $w_L = 0.5$, $w_R = 0.1$ and the initial discontinuity located at $x = 2.5$.

It can be clearly seen how the *non-conservative scheme* displaces the location of the discontinuity. In other words, it does not provide the correct weak solution. However, if we change the initial conditions in such a way that we obtain an expansion wave, $w_L = 0.1,\ w_R = 0.5$, this problem is overcome, as we see in Fig. 6.5.

6.2 Lax-Wendroff Theorem

Since conservative finite volume methods are based on the integral form of the conservation laws, it seems reasonable that they would properly approximate the weak solutions of the conservation law. This was proven by Lax and Wendorff [16], at least in the sense that if the approximation converges to a certain function $w(x, t)$ when the grid is refined, by means of the sequences $\Delta t_k,\ \Delta x_k \longrightarrow 0$, then this function will be, indeed, a weak solution of the conservation law.

The theorem does not guarantee convergence. Therefore, we need to introduce some kind of stability, since if more than a weak solution exists, it may happen that a sequence of approximations converges to a weak solution, whereas another could converge to a different weak solution.

Theorem 6.1 Lax-Wendroff Theorem

Let us consider a sequence of grids denoted by $k = 1, 2, \ldots,$ *with discretization parameters* $\Delta t_k,\ \Delta x_k \longrightarrow 0$ *as* $k \longrightarrow \infty$. *Let* w_k *be the numerical approximation*

calculated with a consistent and conservative method on the k-th grid. We assume that w_k converges to a function w as $k \longrightarrow \infty$. Then, $w(x,t)$ is a weak solution of the conservation law.

Proof It can be found in Godlewski and Raviart [8].

6.3 Approximate Riemann Solvers

Godunov method only takes into account an approximation of the exact solution that is obtained when solving the local Riemann problems. Therefore, it seems convenient to consider approximations of the local Riemann problem if they present an appropriate behavior and simplify the resolution. The Godunov method can be generalized in two ways:

1. In the conservative form of the Godunov method the numerical flux is defined as:

$$\phi(w_l, w_r) = f(\widetilde{w}(w_l, w_r)), \tag{6.24}$$

   where $\widetilde{w}(w_l, w_r)$ is the value of the solution of the Riemann problem with initial conditions w_l, w_r at $x/t = 0$. This generalization of the method consists of replacing this solution $\widetilde{w}$ by an approximation $\widehat{w}$. Therefore, the resulting method is

$$w_j^{n+1} = w_j^n - \frac{\Delta t}{\Delta x}\left(f(\widehat{w}(w_j^n, w_{j+1}^n)) - f(\widehat{w}(w_{j-1}^n, w_j^n))\right). \tag{6.25}$$

   It is important to highlight that this method is consistent for any choice of $\widehat{w}$, since $\widehat{w}(\overline{w}, \overline{w}) = \overline{w}$.
2. The second option is derived starting from the integral:

$$w_j^{n+1} = \frac{1}{\Delta x}\int_{x_{j-1}}^{x_{j+1}} \widetilde{w}^n(x, t^{n+1})\, dx, \tag{6.26}$$

   where $\widetilde{w}^n$ is the solution of the Riemann problem starting from piecewise constant solutions w_j^n at time t^n, then it is evaluated at time t^{n+1} and averaged over each finite volume.
   This generalization of the method consists of defining the states

$$w_j^{n+1} = \frac{1}{\Delta x}\int_{x_{j-1}}^{x_{j+1}} \widehat{w}^n(x, t^{n+1})\, dx. \tag{6.27}$$

   We should remark that we do not obtain the same method as in the first case. This second approximation is more common but it requires the definition of an

approximate solution of the Riemann problem, since arbitrary solutions do not guarantee that the scheme is conservative. This generalization leads to the so-called *Godunov-type methods* or *approximate Riemann solvers*, since the Godunov method is known to be the *exact Riemann solver*.

In contrast with the previous case in which the definition of the approximate Riemann solver was explicitly associated with a numerical flux and, therefore, the resulting scheme was conservative, in the second option, the approximate Riemann solver will be conservative if the approximate solution $\widehat{w}(x,t) = \widehat{u}(\frac{x}{t})$ satisfies, for a sufficiently large M, the following condition:

$$\int_{-M}^{M} \widehat{u}(\xi)\, d\xi = M(w_l + w_r) + f(w_l) - f(w_r), \tag{6.28}$$

that is interpreted in terms of the conservation: the integral of the approximation coincides with the *initial mass* given by $M(w_l + w_r)$ plus the *mass that enters through the left boundary* $f(w_l)$, minus the *mass that goes out through the right boundary* $f(w_r)$.

Remark 6.1 Interpretation of the condition (6.28).

The exact solution verifies the previous expression if we consider the integral form of the conservation law in the rectangle $[-M, M] \times [0, 1]$:

$$\int_{-M}^{M} w(x,1)\, dx = \int_{-M}^{M} w(x,0)\, dx - \int_0^1 [f(w(M,t)) - f(w(-M,t))]\, dt. \tag{6.29}$$

If M is sufficiently large in the following sense:

$$-\frac{M}{t} < \lambda_{\min}, \quad \frac{M}{t} > \lambda_{\max},$$

with $\lambda_{\min}$ and $\lambda_{\max}$ the smallest and largest eigenvalues, respectively, of the Jacobian matrix of the flux f, then

$$\int_{-M}^{M} w(x,0)\, dx = M(w_l + w_r), \tag{6.30}$$

$$\int_0^1 f(w(M,t))\, dt = \int_0^1 f(w_r)\, dt = f(w_r), \tag{6.31}$$

$$\int_0^1 f(w(-M,t))\, dt = \int_0^1 f(w_l)\, dt = f(w_l), \tag{6.32}$$

therefore, (6.28) holds for the exact solution.

If the approximate solution $\widehat{w}(x,t) = \widehat{u}(\frac{x}{t})$ verifies (6.28), then we can define the resulting method in conservative form by introducing the numerical flux function ϕ. For this purpose, we integrate the approximation of the solution in the interval

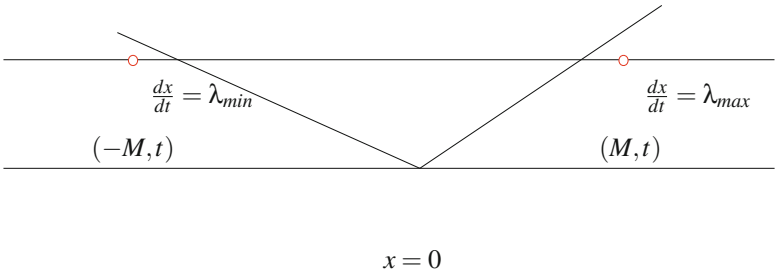


Fig. 6.6 Interpretation of the choice of M in the consistency condition given by Eq. (6.28)

$[0, M]$, for M sufficiently large, in the sense we previously described, see Fig. 6.6. Hence, we have

$$\begin{aligned}\int_0^M \widehat{u}(\xi)\, d\xi &= \int_0^M w(x,0)\, dx - \int_0^1 [f(w(M,t)) - f(w(0,t))]\, dt \\ &= Mw_r - f(w_r) + \phi(w_l, w_r). \end{aligned} \tag{6.33}$$

Therefore, the *numerical flux* is defined as:

$$\phi(w_l, w_r) = f(w_r) - Mw_r + \int_0^M \widehat{u}(\xi)\, d\xi, \tag{6.34}$$

or, analogously, if we integrate in $[-M, 0]$,

$$\phi(w_l, w_r) = f(w_l) + Mw_l - \int_{-M}^0 \widehat{u}(\xi)\, d\xi. \tag{6.35}$$

These two equations are based on the equality required in (6.28) for the approximation of the solution.

Exercise 6.6 Verify that the numerical flux defined for the approximate Riemann solvers employing relations (6.34) or (6.35) is consistent.

Solution 6.6 Left to the reader.

6.4 Upwind Schemes

We have already studied upwind schemes in the case of linear systems, $\mathbf{f}(\mathbf{w}) = \mathbf{A}\mathbf{w}$. The numerical flux of the upwind scheme is given, in one of its equivalent expressions, by

$$\phi(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{A}^+\mathbf{w}_L + \mathbf{A}^-\mathbf{w}_R. \tag{6.36}$$

We will formalize next what we understand by upwinding in the non-linear case following the definition proposed by Harten et al. [15].

Definition 6.2 Upstream[1] schemes

A conservative scheme is said to be an upstream scheme if the numerical flux function verifies the following properties:

(i) For two nearby states $\mathbf{w}_L$ and $\mathbf{w}_R$, (6.36) is a linear approximation to the numerical flux $\phi(\mathbf{w}_L, \mathbf{w}_R)$. This is, for a state $\mathbf{w}^*$ close to $\mathbf{w}_L$ and $\mathbf{w}_R$,

$$\phi(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{f}(\mathbf{w}^*) + \mathbf{A}^+(\mathbf{w}^*)(\mathbf{w}_L - \mathbf{w}^*) + \mathbf{A}^-(\mathbf{w}^*)(\mathbf{w}_R - \mathbf{w}^*) + o\left(\left|\mathbf{w}_L - \mathbf{w}^*\right| + \left|\mathbf{w}_R - \mathbf{w}^*\right|\right),$$

with $\mathbf{A}^+(\mathbf{w})$ and $\mathbf{A}^-(\mathbf{w})$ the positive and negative parts of the Jacobian matrix of the flux $\mathbf{f}(\mathbf{w})$.

(ii) When the characteristic velocities, this is, the eigenvalues of the Jacobian matrix of the flux, are all positive, then

$$\phi(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{f}(\mathbf{w}_L).$$

(iii) If the characteristic velocities are all negative, then

$$\phi(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{f}(\mathbf{w}_R).$$

Exercise 6.7 Prove that the Lax-Friedrichs scheme is not an upwind scheme.

Solution 6.7 Hint: apply it to the Burgers equation.

6.5 Q-schemes

Q-schemes are a family of upwind schemes, introduced by van Leer [15], whose discrete flux is of the form:

$$\phi(\mathbf{w}_L, \mathbf{w}_R) = \frac{\mathbf{f}(\mathbf{w}_L) + \mathbf{f}(\mathbf{w}_R)}{2} - \frac{1}{2}\left|\mathbf{Q}\left(\mathbf{w}_L, \mathbf{w}_R\right)\right|\left(\mathbf{w}_R - \mathbf{w}_L\right), \tag{6.37}$$

where $\mathbf{Q}$ is a characteristic matrix of each **Q**-scheme that depends continuously on $\mathbf{w}_L$ and $\mathbf{w}_R$.

In (6.37) it can be seen that the numerical flux is formed by a centered part

$$\frac{\mathbf{f}(\mathbf{w}_L) + \mathbf{f}(\mathbf{w}_R)}{2}, \tag{6.38}$$

[1] Upstream is the original term used by Harten et al. [15].

and by a viscosity term or *numerical viscosity*, that corresponds to the non-centered part of the flux, since by means of matrix $|\mathbf{Q}|$ we take into account the signs of the eigenvalues as we analyzed in the linear case:

$$d(\mathbf{w}_L, \mathbf{w}_R) = -\frac{1}{2} |\mathbf{Q}(\mathbf{w}_L, \mathbf{w}_R)| (\mathbf{w}_R - \mathbf{w}_L). \tag{6.39}$$

The $\mathbf{Q}$ matrix for the two $\mathbf{Q}$-schemes we will work with coincides with the Jacobian matrix evaluated in a state denoted by $\mathbf{w}_Q$, whose details are given below:

$$\mathbf{w}_Q(\mathbf{w}_L, \mathbf{w}_R) = \begin{cases} \dfrac{\mathbf{w}_L + \mathbf{w}_R}{2} & \text{van Leer } \mathbf{Q}\text{-scheme,} \\ \widetilde{\mathbf{w}}(\mathbf{w}_L, \mathbf{w}_R) & \text{Roe scheme,} \end{cases} \tag{6.40}$$

where the state $\widetilde{\mathbf{w}}(\mathbf{w}_L, \mathbf{w}_R)$ is the *Roe's average* [22], that is defined by

$$\mathbf{f}(\mathbf{w}_R) - \mathbf{f}(\mathbf{w}_L) = \mathbf{A}(\widetilde{\mathbf{w}}(\mathbf{w}_L, \mathbf{w}_R))(\mathbf{w}_R - \mathbf{w}_L), \tag{6.41}$$

with $\mathbf{A}(\mathbf{w})$ the Jacobian matrix of $\mathbf{f}(\mathbf{w})$.

Exercise 6.8 Prove that the numerical flux of the $\mathbf{Q}$-schemes is consistent independently of the expression of matrix $\mathbf{Q}$.

Solution 6.8 Left to the reader.

Example 6.1 The van Leer $\mathbf{Q}$-scheme is an upwind scheme.

It suffices to take as $\mathbf{w}^* = \dfrac{\mathbf{w}_L + \mathbf{w}_R}{2}$ and take into account the following relations:

$$\mathbf{f}\left(\frac{\mathbf{w}_L + \mathbf{w}_R}{2}\right) = \frac{\mathbf{f}(\mathbf{w}_L) + \mathbf{f}(\mathbf{w}_R)}{2} + o(|\mathbf{w}_R - \mathbf{w}_L|),$$
$$\mathbf{A}^+(\mathbf{w}) - \mathbf{A}^-(\mathbf{w}) = |\mathbf{A}(\mathbf{w})|.$$

Exercise 6.9 Prove that the numerical flux of the Roe and van Leer $\mathbf{Q}$-schemes are the same for the Burgers equation.

Solution 6.9 Left to the reader.

6.6 Schemes Consistent with the Entropy Condition

Lax-Wendroff theorem assures that a convergent, consistent and conservative scheme will take us to the weak solutions of our problem. Now, among all the weak solutions we must select the ones with a physical meaning. Therefore, an analogous condition to the one considered in the continuous problem must be introduced.

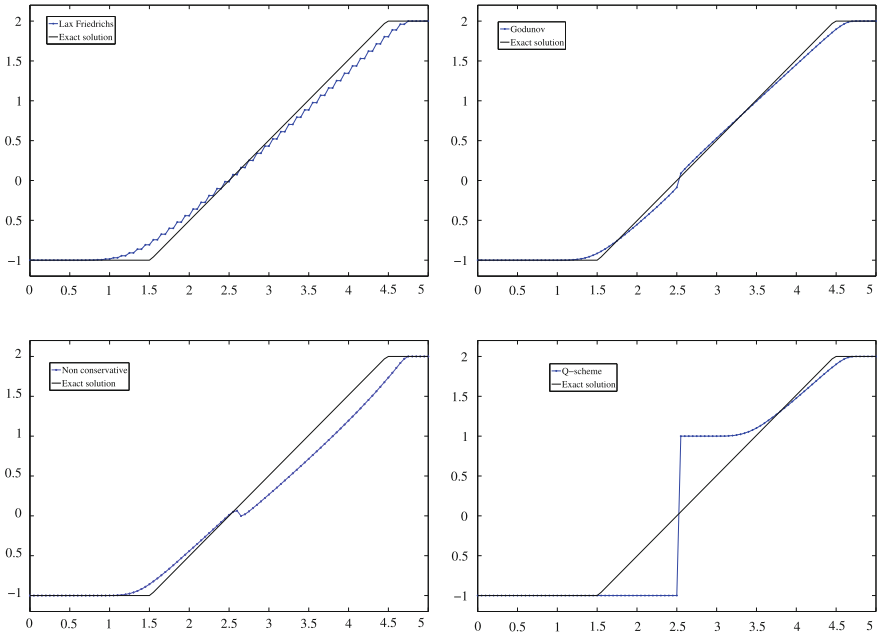


Fig. 6.7 Comparison among Lax-Friedrichs scheme, Godunov method, the non-conservative scheme studied and van Leer **Q**-scheme solving the Burgers equation with the discontinuous initial condition at $x = 2.5$: $w_L = -1,\ w_R = 2,\ \mu = 0.9$ at $t = 1$ and the exact solution of the problem

Example 6.2 The need for this numerical entropy condition is motivated in Fig. 6.7.

Let us recall that the Riemann problem associated with the Burgers equation when $w_L < w_R$ has an *entropy shock* as weak solution, whereas the solution that verifies the entropy condition is an *expansion wave*.

We ran the conservative methods presented in this text with this particular problem and compared the approximate solutions with the exact solution calculated in the first part of this book.

More precisely, we considered $w_L = -1,\ w_R = 2$, a Courant number that guarantees stability of the linear problem $\mu = 0.9$, and compared the approximate solutions of Lax-Friedrichs scheme, Godunov method, the non-conservative scheme studied and van Leer Q-scheme.

It is clearly noticed how the Q-scheme provides a weak solution that is not the entropy solution corresponding to the rarefaction wave (also depicted in Fig. 6.7).

The *numerical viscosity* of the different schemes produces an effect similar to the viscosity term on the right hand side of the *viscous Burgers equation*:

$$\frac{\partial w}{\partial t} + w\frac{\partial w}{\partial x} = \nu\frac{\partial^2 w}{\partial x^2}, \tag{6.42}$$

and drives the scheme to reproduce the expansion wave. Nevertheless, unlike the viscosity that depends on the physical quantity ν, present in (6.42), the amount of numerical viscosity $d(w_{i-1}^n, w_i^n)$ depends on the numerical method. In particular, if $d(w_{i-1}, w_i) = 0$, then there is no *smearing* of the discontinuity and no numerical viscosity. This situation arises in the presence of a *sonic point*[2] of the Burgers equation $f'(w) = A(w) = w = 0$, which occurs at $x = 2.5$ in the example solved in Fig. 6.7.

For the Q-schemes and the Burgers equation, the numerical viscosity is:

$$d(w_{j-1}^n, w_j^n) = -\frac{1}{2}\left|\frac{w_{j-1}^n + w_j^n}{2}\right|(w_j^n - w_{j-1}^n). \tag{6.43}$$

Therefore d is null if $w_{j-1}^n = -w_j^n$, in which case the shock-expansion weak solution is stationary, which is the situation shown in Fig. 6.7.

On the contrary, in the Lax-Friedrichs scheme

$$d(w_{j-1}^n, w_j^n) = -\frac{\Delta x}{2\Delta t}(w_j^n - w_{j-1}^n); \tag{6.44}$$

i.e., the part multiplied by $(w_j^n - w_{j-1}^n)$ does not depend on the eigenvalues ($\lambda = w$ in this case), but only on a grid-dependant fixed constant $\dfrac{\Delta x}{\Delta t}$. This is the reason why the numerical viscosity does not vanish near the sonic point.

Exercise 6.10 Analyse the behaviour of the Lax-Friedrichs scheme, Godunov method, the non-conservative sheme and **Q**-scheme of van Leer studied if we take $w_L = 2,\ w_R = -1$ as initial conditions to solve the Burgers equation, and a Courant number that assures the stability of the linear problem $\mu = 0.9$. Is the numerical viscosity relevant in this case for van Leer Q-scheme to obtain the entropy solution?

Solution 6.10 Left to the reader.

Definition 6.3 Entropy scheme.

We will say that a conservative numerical scheme is consistent with the entropy condition if it verifies an inequality of the form

$$\mathscr{U}(\mathbf{w}_j^{n+1}) \leq \mathscr{U}(\mathbf{w}_j^n) - \frac{\Delta t}{\Delta x}\left\{\mathbf{\Psi}(\mathbf{w}_j^n, \mathbf{w}_{j+1}^n) - \mathbf{\Psi}(\mathbf{w}_{j-1}^n, \mathbf{w}_j^n)\right\}, \tag{6.45}$$

where $\mathscr{U}$ is the entopy function and $\mathbf{\Psi}$ is the numerical entropy flux, that must be consistent with the entropy flux $\mathscr{F}$:

$$\mathbf{\Psi}(\mathbf{w}, \mathbf{w}) = \mathscr{F}(\mathbf{w}).$$

The previous inequality is known as *discrete entropy condition* and a numerical scheme satisfying (6.45) is called *entropic*.

[2] A sonic point is characterized by the fact that an eigenvalue is null at such point.

If the numerical method is convergent and verifies the entropy condition, the numerical solution that is obtained with that method converges to an *entropy solution*.

We shall see that there are simpler ways to verify if a scheme is entropic.

Example 6.3 Godunov method is an entropy scheme.

It is easy to prove that the numerical approximation obtained by this method verifies the entropy condition, taking into account that the solution of the Riemann problem employed to define the numerical flux at the boundary of each cell verifies the entropy condition (see LeVeque [18]). Moreover, Fig. 6.7 shows a correct approximation of the entropy solution obtained for the Burgers equation.

6.6.1 Analysis of the Numerical Viscosity. Rusanov Scheme

We have analysed in the previous example that the numerical viscosity term corresponding to the uncoupled equation vanishes if any of the eigenvalues vanishes, and, as a consequence, the convergence to the entropy solution is lost. This can introduce oscillations in the computed approximations that lead to incorrect solutions.

In order to avoid this effect, a strategy known as *entropy fix* can be used, consisting in assuring the presence of non-null numerical viscosity. This is done redefining the absolute value of the eigenvalues in the corresponding part of the numerical flux.

6.6.1.1 Harten Regularization

We will analyse first the *parabolic regularization* of the absolute value of the eigenvalues, proposed by Harten [14], that will be used in the calculation of the matrix absolute value of **Q**:

$$|\lambda|_\varepsilon = |\lambda| + \frac{1}{2}\left(1 + sig\left(\varepsilon - |\lambda|\right)\right)\left(\frac{\lambda^2 + \varepsilon^2}{2\varepsilon} - |\lambda|\right), \tag{6.46}$$

where ε is a small parameter. It is immediate to verify that with this redefinition the minimum value of $|\lambda|_\varepsilon$ is $\varepsilon/2$, and $|\lambda|$ is only replaced by $\dfrac{\lambda^2 + \varepsilon^2}{2\varepsilon}$ when $|\lambda| < \varepsilon$. The drawback of this approximation is that, in general, the ε parameter must be adjusted for each problem (Fig. 6.8).

6.6.1.2 Lax-Friedrichs Local Regularization (LLF). Rusanov Scheme

The numerical integration can also be carried out by applying the *local Lax-Friedrichs method* (LLF) to the **Q**-schemes, also called Rusanov method [24]. According to this method, the correction of the absolute value of the eigenvalues at each boundary

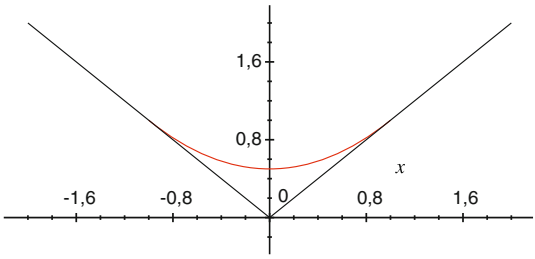


Fig. 6.8 Harten regularization (*red*)

$x_{j-1/2}$ is performed by taking the maximum of both of them at each side of the boundary, this is:

$$|\lambda|_{j-1/2} = \max\left\{|\lambda|_{j-1}, |\lambda|_j\right\}. \tag{6.47}$$

Like in the previous case, this is a redefinition of the absolute value of the eigenvalues, that will be introduced in the numerical viscosity of the **Q**-scheme by means of the matrix absolute value of **Q**.

A drawback of this approximation is that it generally adds more viscosity in all the fields, regardless of whether it is a transonic expansion or not. Nevertheless, wherever the solution is smooth it follows that $\lambda_{i-1/2} \sim \lambda_{i-1} \sim \lambda_i$, and therefore the **Q**-scheme in these cases behaves as if no regularization had been applied.

Applying the *local Lax-Friedrichs* (LLF) regularization method to the **Q**-schemes is equivalent in the scalar case to employing the Rusanov method that uses

$$\phi(\mathbf{w}_L, \mathbf{w}_R) = \frac{\mathbf{f}(\mathbf{w}_L) + \mathbf{f}(\mathbf{w}_R)}{2} - \frac{1}{2}\alpha_{Rus}(\mathbf{w}_L, \mathbf{w}_R)(\mathbf{w}_R - \mathbf{w}_L), \tag{6.48}$$

as numerical flux, where the constant α_R of this method is defined as

$$\alpha_{Rus}(\mathbf{w}_L, \mathbf{w}_R) = \max\left\{\max_{k=1,p}\left\{|\lambda_k(\mathbf{w}_L)|\right\}, \max_{k=1,p}\left\{|\lambda_k(\mathbf{w}_R)|\right\}\right\}. \tag{6.49}$$

Example 6.4 Comparison of the results using Harten regularization and local Lax-Friedrichs (LLF) regularization or the Rusanov method.

We will start analysing the Harten regularization. Figure 6.9 shows the results obtained for two different choices of ε in the Riemann problem in Example 6.2. The convergence to the entropy solution as a result of the introduction of numerical viscosity using the **Q**-scheme, and the dependence on the value of ε are clearly seen.

Remark 6.2 The corresponding codes to the implementation of the aforementioned regularizations for the Burgers equation are provided in Sect. 9.3.

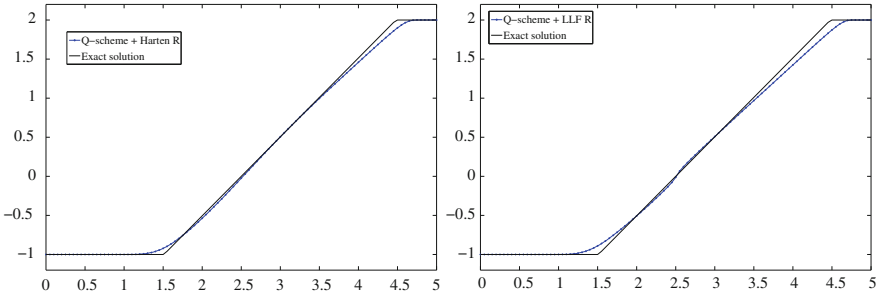


Fig. 6.9 Results obtained by the **Q**-scheme with the Harten regularization using $\varepsilon = 0.9$ and by the **Q**-scheme with the local Lax-Friedrichs regularization, or Rusanov scheme, solving the Burgers equation with the discontinuous initial condition at $x = 2.5$: $w_L = -1,\ w_R = 2,\ \mu = 0.9$ at $t = 1$

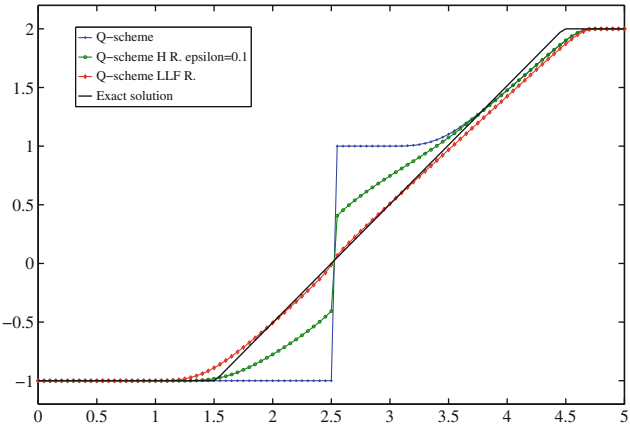


Fig. 6.10 Comparison between the **Q**-scheme with the Harten regularization using $\varepsilon = 0.1$ and the **Q**-scheme with the local Lax-Friedrichs regularization, or Rusanov scheme, solving the Burgers equation with the discontinuous initial condition at $x = 2.5$: $w_L = -1,\ w_R = 2,\ \mu = 0.9$ at $t = 1$

Finally, Fig. 6.10 shows the comparison between the methods using the **Q**-scheme and the *local Lax-Friedrichs (LLF)* method. Notice the convergence to the entropy solution without the need to adjust the ε parameter for the numerical viscosity.

Example 6.5 A dam-break problem on a dry bed.

As described with the Burgers equation, van Leer **Q**-scheme needs the application of any of the regularizations mentioned before in the presence of sonic points. We show in Fig. 6.12 the results of the application of the studied schemes to the shallow water equations, with both regularizations. The schemes were applied to a dam-break problem over a dry bed. The details of the Riemann problem solved in Fig. 6.11 can be examined in Sect. 3.3 of [12].

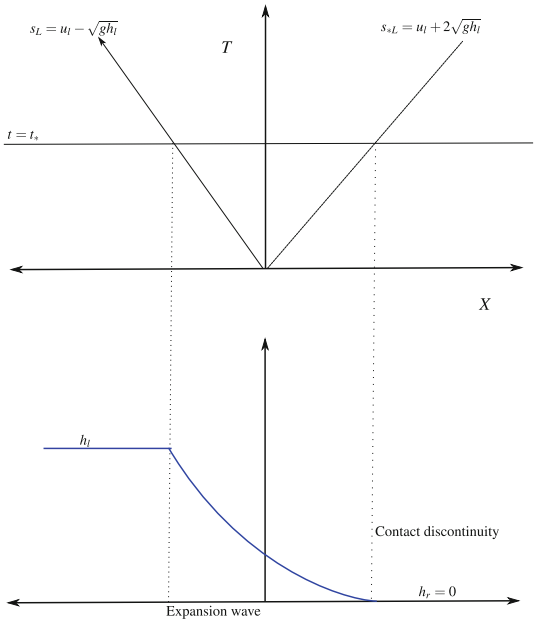


Fig. 6.11 Scheme of the solution of the Riemann problem for the shallow water equations

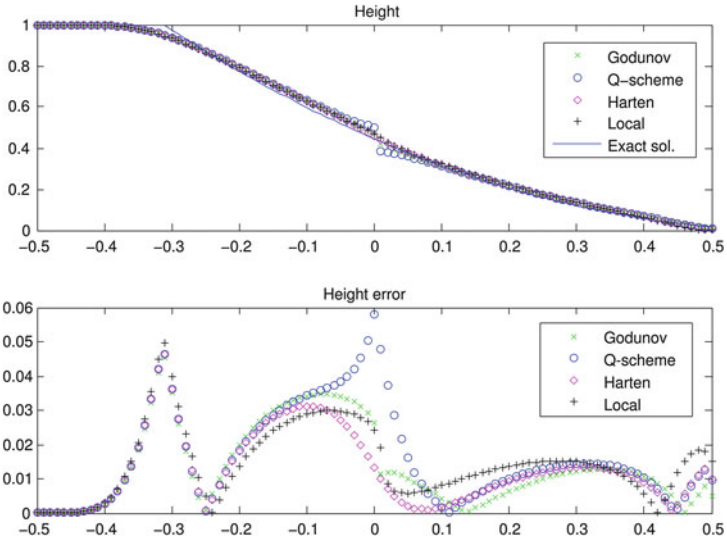


Fig. 6.12 Results for water height in the Riemann problem for the shallow water equations using Godunov flux schemes, the flux of the **Q**-schemes without regularization, with Harten regularization and with local Lax-Friedrichs regularization

6.7 The Roe Scheme

One of the most renowned approximate Riemann solvers in the literature was devised by Roe in 1981. The idea behind this solver is to determine $\widehat{\mathbf{w}}$, introduced in (6.27), solving a linear system of constant coefficient conservation laws at the boundaries of the cells, instead of the original non-linear system. More precisely, the coefficient matrix employed at the boundary $x_{j-1/2}$ must depend on $\mathbf{w}_L = \mathbf{w}_{j-1}^n$ and $\mathbf{w}_R = \mathbf{w}_j^n$, and the linear system will be given by

$$\frac{\partial \widehat{\mathbf{w}}}{\partial t} + \widehat{\mathbf{A}}(\mathbf{w}_L, \mathbf{w}_R)\frac{\partial \widehat{\mathbf{w}}}{\partial x} = 0. \tag{6.50}$$

Roe-type linearisation is based on the construction of an approximate Jacobian matrix $\widehat{\mathbf{A}}(\mathbf{w}_L, \mathbf{w}_R)$. Once $\widehat{\mathbf{A}}$ is constructed we can consider its eigenvalues as the velocities of the particles of the Riemann problem. The projection of $\mathbf{w}_R - \mathbf{w}_L$ onto its eigenvectors are the jumps that take place at the intermediate states:

$$\mathbf{w}_R - \mathbf{w}_L = \mathbf{P}(\mathbf{u}_R - \mathbf{u}_L) = \sum_k \widehat{\alpha}_k \widehat{\mathbf{e}}_k, \tag{6.51}$$

where $\widehat{\lambda}_k$ and $\widehat{\mathbf{e}}_k$ are the eigenvalues and eigenvectors corresponding to the matrix $\widehat{\mathbf{A}}(\mathbf{w}_L, \mathbf{w}_R)$, and the coefficients $\widehat{\alpha}_k$, as we show in Eq. 6.51, are the coordinates of $\mathbf{w}_R - \mathbf{w}_L$ in the basis of eigenvectors of that matrix. For simplicity, in the rest of the present section, the dependency of the neighbouring nodes of each boundary $x_{j-1/2}$ will not be explicitly stated, this is, $\widehat{\alpha}_k = (\widehat{\alpha}_k)_{j-1/2}$, $\widehat{\mathbf{e}}_k = (\widehat{\mathbf{e}}_k)_{j-1/2}$, $\widehat{\lambda}_k = (\widehat{\lambda}_k)_{j-1/2}$.

Let us recall the expression obtained for the solution of the Riemann problem in the linear case (see Sect. 2.4.2):

$$\widehat{\mathbf{w}}(x,t) = \widehat{\mathbf{u}}\left(\frac{x}{t}\right) = \widehat{\mathbf{u}}(\xi) = \mathbf{w}_r - \sum_{\widehat{\lambda}_k > \xi} \widehat{\alpha}_k \widehat{\mathbf{e}}_k, \tag{6.52}$$

where the notation introduced in the previous summation means that we include the addends corresponding to the waves $\widehat{\alpha}_k \widehat{\mathbf{e}}_k$ associated to the eigenvalues with characteristics that intersect the OX-axis in the negative side, this is:

$$x - \widehat{\lambda}_k t < 0 \Longrightarrow \frac{x}{t} - \widehat{\lambda}_k < 0 \Longrightarrow \xi - \widehat{\lambda}_k < 0 \Longrightarrow \widehat{\lambda}_k > \xi. \tag{6.53}$$

In order to determine the matrix $\widehat{\mathbf{A}}$, Roe suggests that it must verify the following properties:

(P1) (*Hiperbolicity*) $\widehat{\mathbf{A}}(\mathbf{w}_L, \mathbf{w}_R)$ is diagonalizable and has real eigenvalues and a complete set of eigenvectors.

(P2) (*Consistency*) $\widehat{\mathbf{A}}(\mathbf{w}_L, \mathbf{w}_R) \longrightarrow \mathbf{f}'(\mathbf{w})$ when $\mathbf{w}_L,\ \mathbf{w}_R \longrightarrow \mathbf{w}$ smoothly.

(P3) (*Conservation*) $\widehat{\mathbf{A}}(\mathbf{w}_L, \mathbf{w}_R)(\mathbf{w}_R - \mathbf{w}_L) = \mathbf{f}(\mathbf{w}_R) - \mathbf{f}(\mathbf{w}_L)$.

The first property (P1) is necessary for the problem (6.50) to be hyperbolic and solvable. Property (P2) guarantees that the method is consistent with the original conservation law. Property (P3), which characterizes the Roe-type linearisation, assures that the condition required for a conservative scheme is verified, as we will detail in Sect. 6.7.1 devoted to the numerical flux of the Roe scheme.

Moreover, this property leads us to the exact solution of the problem of departure in the special case that $\mathbf{w}_L$ and $\mathbf{w}_R$ are connected by a shock wave or a contact discontinuity. The latter follows from the fact that the Rankine-Hugoniot condition is verified for $\mathbf{w}_L$ and $\mathbf{w}_R$ in this case:

$$\mathbf{f}(\mathbf{w}_R) - \mathbf{f}(\mathbf{w}_L) = s(\mathbf{w}_R - \mathbf{w}_L),$$

with s the shock propagation velocity or the contact discontinuity propagation velocity. If we apply property (P3) at this point we can prove that, in this situation, $\mathbf{w}_R - \mathbf{w}_L$ must be an eigenvector of $\widehat{\mathbf{A}}$, and s the corresponding eigenvalue:

$$\mathbf{f}(\mathbf{w}_R) - \mathbf{f}(\mathbf{w}_L) = \widehat{\mathbf{A}}(\mathbf{w}_L, \mathbf{w}_R)(\mathbf{w}_R - \mathbf{w}_L) = s(\mathbf{w}_R - \mathbf{w}_L),$$

and therefore the approximate solution $\widehat{\mathbf{w}}(x, t)$ also consists in a simple jump that propagates at velocity s.

One way to make sure that properties (P1) and (P2) are verified is to take

$$\widehat{\mathbf{A}}(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{A}(\mathbf{w}_m),$$

for some intermediate value $\mathbf{w}_m$. We could use the arithmetic mean of $\mathbf{w}_R$ and $\mathbf{w}_L$, but this choice is not always compatible with property (P3). Nevertheless, Harten et al. proved in [15] that for all hyperbolic system with a strictly convex entropy function, there exists at least one Roe-type linearisation.

Example 6.6 Roe-type linearisation for the one-dimensional shallow water equations.

Glaister [7] gives the following expressions for $\widetilde{\mathbf{w}}$ and $\mathbf{Q}$ for the Saint-Venant equations:

$$\mathbf{Q}(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{A}\left(\widetilde{\mathbf{w}}(\mathbf{w}_L, \mathbf{w}_R)\right) = \begin{pmatrix} 0 & 1 \\ -\widetilde{u}^2 + \widetilde{c}^2 & 2\widetilde{u} \end{pmatrix}, \tag{6.54}$$

where

$$\mathbf{w}_L = \begin{pmatrix} h_\mathrm{L} \\ h_\mathrm{L}\, u_\mathrm{L} \end{pmatrix}, \quad \mathbf{w}_R = \begin{pmatrix} h_\mathrm{R} \\ h_\mathrm{R}\, u_\mathrm{R} \end{pmatrix}, \quad \widetilde{\mathbf{w}}(\mathbf{w}_L, \mathbf{w}_R) = \begin{pmatrix} \widetilde{h} \\ \widetilde{h}\widetilde{u} \end{pmatrix}, \tag{6.55}$$

$$\widetilde{c} = \sqrt{g\left(\frac{h_\mathrm{L} + h_\mathrm{R}}{2}\right)}, \quad \widetilde{h} = \sqrt{h_\mathrm{L} h_\mathrm{R}}, \quad \widetilde{u} = \frac{\sqrt{h_\mathrm{L}}u_\mathrm{L} + \sqrt{h_\mathrm{R}}u_\mathrm{R}}{\sqrt{h_\mathrm{L}} + \sqrt{h_\mathrm{R}}}. \tag{6.56}$$

6.7.1 Numerical Flux of the Roe Scheme

In this section we will introduce the approximate solver or Godunov-type solver as a conservative method by constructing the associated numerical flux.

For the Roe solver, the function $\widehat{\mathbf{u}}(\frac{x}{t})$ can also be interpreted as the exact solution of the Riemann problem for the hyperbolic conservation law, whose numerical flux is

$$\widehat{\mathbf{f}}(\mathbf{w}) = \widehat{\mathbf{A}}\mathbf{w}. \tag{6.57}$$

As we shall see, if the condition (6.28) is verified when we introduce the Godunov approximate solvers, we could define the numerical flux by means of the expression

$$\phi(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{f}(\mathbf{w}_R) - M\mathbf{w}_R + \int_0^M \widehat{\mathbf{w}}(\xi)\, d\xi, \tag{6.58}$$

for M sufficiently large. In order to obtain the expression for ϕ we take into account that $\widehat{\mathbf{u}}(\frac{x}{t})$ is an exact solution of the corresponding conservation law, and therefore it verifies

$$\int_0^M \widehat{\mathbf{u}}(\xi)\, d\xi = M\mathbf{w}_R + \widehat{\mathbf{f}}(\widehat{\mathbf{w}}(0)) - \widehat{\mathbf{f}}(\mathbf{w}_R).$$

Hence, if we replace this expression of $\int_0^M \widehat{\mathbf{u}}(\xi)\, d\xi$ in (6.58), we get

$$\begin{aligned}\phi(\mathbf{w}_L, \mathbf{w}_R) &= \mathbf{f}(\mathbf{w}_R) - M\mathbf{w}_R + \left[M\mathbf{w}_R + \widehat{\mathbf{f}}(\widehat{\mathbf{w}}(0)) - \widehat{\mathbf{f}}(\mathbf{w}_R)\right] \\ &= \mathbf{f}(\mathbf{w}_R) + \widehat{\mathbf{f}}(\widehat{\mathbf{u}}(0)) - \widehat{\mathbf{f}}(w_R).\end{aligned}$$

After this, taking into account the definition of $\widehat{f}$ given by (6.57),

$$\phi(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{f}(\mathbf{w}_R) + \widehat{\mathbf{A}}\left(\widehat{\mathbf{u}}(0) - \mathbf{w}_R\right) = \mathbf{f}(\mathbf{w}_R) - \widehat{\mathbf{A}}\left(\mathbf{w}_R - \widehat{\mathbf{u}}(0)\right).$$

If we employ the solution of the Riemann problem at $\frac{x}{t} = \xi = 0$, $\widehat{\mathbf{u}}(0)$, given by (6.52), the expression of $\mathbf{w}_R - \widehat{\mathbf{u}}(0)$ is

$$\mathbf{w}_R - \widehat{\mathbf{u}}(0) = \sum_{\widehat{\lambda}_k > 0} \widehat{\alpha}_k\, \widehat{\mathbf{e}}_k,$$

what leads us to a definition of the numerical flux of the form

$$\phi(\mathbf{w}_l, \mathbf{w}_r) = \mathbf{f}(\mathbf{w}_r) - \widehat{A} \sum_{\widehat{\lambda}_k > 0} \widehat{\alpha}_k\, \widehat{\mathbf{e}}_k.$$

Introducing the matrix A in the summation and applying the characterization of the corresponding eigenvalues and eigenvectors:

$$\phi(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{f}(\mathbf{w}_R) - \sum_{\widehat{\lambda}_k > 0} \widehat{\alpha}_k \, \widehat{\lambda}_k \widehat{\mathbf{e}}_k.$$

Operating analogously to the linear case, we can complete the previous summation taking into account the expressions of the positive and negative parts of the eigenvalues:

$$\phi(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{f}(\mathbf{w}_R) - \sum_{k}^{p} \widehat{\alpha}_k \, \widehat{\lambda}_k^{+} \widehat{\mathbf{e}}_k, \tag{6.59}$$

and, similarly,

$$\phi(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{f}(\mathbf{w}_L) + \sum_{k}^{p} \widehat{\alpha}_k \, \widehat{\lambda}_k^{-} \widehat{\mathbf{e}}_k. \tag{6.60}$$

The sum of these two expressions, together with the definition of the matrix $\left|\widehat{\mathbf{A}}(\mathbf{w}_L, \mathbf{w}_R)\right|$ and its eigenvalues,[3] allows us to define the numerical flux of the Roe scheme, as a **Q**-scheme, taking the **Q** matrix as the corresponding Roe-type linearization:

$$\phi(\mathbf{w}_L, \mathbf{w}_R) = \frac{1}{2}\left(\mathbf{f}(\mathbf{w}_L) + \mathbf{f}(\mathbf{w}_R)\right) - \frac{1}{2}\left|\widehat{\mathbf{A}}(\mathbf{w}_L, \mathbf{w}_R)\right|(\mathbf{w}_R - \mathbf{w}_L). \tag{6.61}$$

6.8 Flux-Splitting Techniques

These techniques also allow to generalize the differences scheme (5.93) to the case of linear conservation laws (see [15]).

In the beginning they were devised to solve the Euler equations. Since for these equations the flux is an homogeneous function of degree one, the following equality holds

$$\mathbf{f}(\mathbf{w}) = \mathbf{A}(\mathbf{w})\mathbf{w},$$

with **A** the Jacobian matrix of the flux. Hence, **f** has one direct splitting

$$\mathbf{f}(\mathbf{w}) = \mathbf{f}^{+}(\mathbf{w}) + \mathbf{f}^{-}(\mathbf{w}), \tag{6.62}$$

in terms of the *outgoing flux* $\mathbf{f}^{+}$ and the *incoming flux* $\mathbf{f}^{-}$, with

$$\mathbf{f}^{\pm}(\mathbf{w}) = \mathbf{A}^{\pm}(\mathbf{w})\,\mathbf{w}. \tag{6.63}$$

[3] For further detais, see how the Godunov flux is obtained for the linear case in (5.86).

Generally speaking, the flux does not have to be a homogeneous function, as is the case of the Saint-Venant equations.

In order to define the discrete flux in the homogeneous case, by analogy with the Euler equations, we take

$$\phi(\mathbf{w}_L, \mathbf{w}_R) = \phi^+(\mathbf{w}_L, \mathbf{w}_R) + \phi^-(\mathbf{w}_L, \mathbf{w}_R), \tag{6.64}$$

with

$$\phi^+(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{A}^+(\mathbf{w}_1)\mathbf{w}_L, \quad \phi^-(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{A}^-(\mathbf{w}_2)\mathbf{w}_R \tag{6.65}$$

where the states $\mathbf{w}_1$ and $\mathbf{w}_2$ depend on $\mathbf{w}_L$ and $\mathbf{w}_R$ in such way that if $\mathbf{w}_L = \mathbf{w}_R = \mathbf{w}$, then $\mathbf{w}_i(\mathbf{w}, \mathbf{w}) = \mathbf{w}$, $i = 1, 2$.

Exercise 6.11 Prove that the physical flux of the one-dimensional Euler equations is a homogeneous function of degree one.

Solution 6.11 Left to the reader.

Exercise 6.12 Prove that the physical flux of the one-dimensional shallow water equations is not a homogeneous function of degree one.

Solution 6.12 Left to the reader.

6.8.1 Steger-Warming Scheme

Steger and Warming introduced in 1981 the notion of flux-splitting for gas dynamics equations. As we mentioned in the beginning of the present section, the advantage of these equations is that their flux function is a homogeneous function of degree one. The expression of the numerical flux for the Steger-Warming method [25] is obtained by replacing the following states $\mathbf{w}_1$ and $\mathbf{w}_2$ in the expressions given in Sect. 6.8:

$$\mathbf{w}_1(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{w}_L, \quad \mathbf{w}_2(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{w}_R. \tag{6.66}$$

Therefore, the numerical flux for this scheme is

$$\phi^+(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{A}^+(\mathbf{w}_L)\mathbf{w}_L + \mathbf{A}^-(\mathbf{w}_R)\mathbf{w}_R. \tag{6.67}$$

6.8.2 Vijayasundaram Scheme

Vijayasundaram proposes in [37], in 1985, a similar splitting to Steger-Warming, but considering centered evaluations of the matrix, this is, the states $\mathbf{w}_1$ and $\mathbf{w}_2$ coincide

with the average of $\mathbf{w}_L$ and $\mathbf{w}_R$,

$$\mathbf{w}_1(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{w}_2(\mathbf{w}_L, \mathbf{w}_R) = \frac{\mathbf{w}_L + \mathbf{w}_R}{2}. \tag{6.68}$$

Therefore, the numerical flux of this scheme is given by

$$\phi^+(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{A}^+ \left(\frac{\mathbf{w}_L + \mathbf{w}_R}{2} \right) \mathbf{w}_L + \mathbf{A}^- \left(\frac{\mathbf{w}_L + \mathbf{w}_R}{2} \right) \mathbf{w}_R. \tag{6.69}$$

6.8.3 Numerical Viscosity Analysis

In the section devoted to the **Q**-schemes we analysed the need to perform a regularization of the absolute value of the eigenvalues of the matrix **Q**. This regularization can also be applied to the flux-splitting techniques, since the matrices $\mathbf{A}^+$ and $\mathbf{A}^-$ are given by

$$\mathbf{A}^+ = \frac{1}{2}(\mathbf{A} + |\mathbf{A}|), \quad \mathbf{A}^- = \frac{1}{2}(\mathbf{A} - |\mathbf{A}|).$$

Hence, by applying *Harten regularization* to the $|\mathbf{A}|$ matrix we obtain the corresponding *regularizations*

$$\mathbf{A}^+_\varepsilon = \frac{1}{2}(\mathbf{A} + |\mathbf{A}|_\varepsilon), \quad \mathbf{A}^-_\varepsilon = \frac{1}{2}(\mathbf{A} - |\mathbf{A}|_\varepsilon).$$

Exercise 6.13 Prove that the numerical flux of the Steger-Warming and Vijayasundaram flux-splitting techniques are consistent with the the physical flux.

Solution 6.13 Left to the reader.

6.8.4 Flux Factorization for the Case of a Non-homogeneous Flux Function

In the case that the flux function is a non-homogeneous function of degree one, in [2] Bermúdez and Vázquez-Cendón extend the techniques of flux-splitting to the case in which there exists a matrix $\mathbf{A}^*$, that can be different to the Jacobian matrix, that verifies the relation

$$\mathbf{f}(\mathbf{w}) = \mathbf{A}^*(\mathbf{w})\mathbf{w}. \tag{6.70}$$

In order to define the discrete flux in the non-homogeneous case, by analogy with the Euler equations, we take

$$\phi^+(\mathbf{w}_L, \mathbf{w}_R) = \phi^+(\mathbf{w}_L, \mathbf{w}_R) + \phi^-(\mathbf{w}_L, \mathbf{w}_R), \tag{6.71}$$

with

$$\phi^+(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{A}^{*+}(\mathbf{w}_1)\mathbf{w}_L, \quad \phi^-(\mathbf{w}_L, \mathbf{w}_R) = \mathbf{A}^{*-}(\mathbf{w}_2)\mathbf{w}_R, \tag{6.72}$$

where the states $\mathbf{w}_1$ and $\mathbf{w}_2$ are those presented in the corresponding sections for the two flux-splitting techniques analysed.

Example 6.7 The flux of the Burgers equation is not a homogeneous function of degree one:

$$f(w) = \frac{w^2}{2} \neq f'(w)w = w^2. \tag{6.73}$$

Nevertheless, there exists a matrix $\mathbf{A}^*$ that verifies the factorization that allows for the flux-splitting:

$$f(w) = \frac{w^2}{2} = \mathbf{A}^*(w)w = \frac{1}{2}w^2, \text{ where } \mathbf{A}^*(w) = \frac{1}{2}w. \tag{6.74}$$

Example 6.8 Comparison of the results obtained with Steger-Warming and Vijayasundaram flux decomposition techniques when solving the Burgers equation.

Figure 6.13 shows the results obtained with Steger-Warming and Vijayasundaram flux decomposition techniques when solving Example 6.4.

Remark 6.3 The codes that correspond to the implementation of the flux decomposition techniques for the Burgers equation are provided in Sect. 9.4.

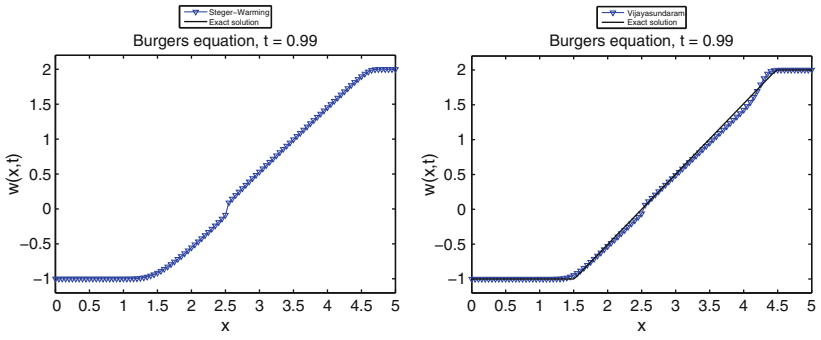


Fig. 6.13 Results obtained with the Seteger-Warming and Vijayasundaram flux decomposition techniques, when solving the Burgers equation with the discontinuous initial condition at $x = 2.5$: $w_L = -1$, $w_R = 2$, $\mu = 0.9$ at $t = 1$

Example 6.9 The flux of the Saint-Venant equations is not a homogeneous function of degree one.

To overcome this problem we have constructed a matrix $\mathbf{A}^*$, that verifies

$$\mathbf{f}(\mathbf{w}) = \mathbf{A}^*(\mathbf{w})\mathbf{w}. \tag{6.75}$$

Following, we detail the expression of the matrix and its eigenvalues and eigenvectors:

$$\mathbf{A}^*(\mathbf{w}) = \begin{pmatrix} 0 & 1 \\ -\dfrac{q^2}{h^2} + \dfrac{1}{2}gh & \dfrac{2q}{h} \end{pmatrix}, \tag{6.76}$$

$$\text{Eigenvalues} \quad \begin{cases} \lambda_1^* = \dfrac{q}{h} + c^*, \\ \lambda_2^* = \dfrac{q}{h} - c^*, \end{cases}$$

$$\text{Eigenvectors} \quad \left\{ \mathbf{e}_1^* = \begin{pmatrix} 1 \\ \dfrac{q}{h} + c^* \end{pmatrix}, \ \mathbf{e}_2^* = \begin{pmatrix} 1 \\ \dfrac{q}{h} - c^* \end{pmatrix}, \right.$$

where $c^* = \sqrt{\dfrac{gh}{2}}$. As can be noticed, the matrix $\mathbf{A}^*$ is similar to the Jacobian matrix, but the term gh had to be multiplied by a factor $\frac{1}{2}$ for the matrix to verify (6.75).

Chapter 7
Biographical Summary of Professor Sergei Konstantinovich Godunov

Abstract Professor Sergei Konstantinovich Godunov is one of the most remarkable references in finite volume methodology. His contributions to Mathematics involve a diversity of fields.

Professor Sergei Konstantinovich Godunov (Fig. 7.1) was born the 17th of July of 1929, to the family of Konstantin Dimitrievich Godunov, a soviet *stratonaut* and airship designer, famous for his flight height record, onboard of the URSS-1 stratosphere, a balloon designed by himself in 1934. Possibly under the influence of his father's occupation, Godunov attended the first Moscow school of the Air Forces,



Fig. 7.1 Professor Sergei Konstantinovich Godunov

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M.E. Vázquez-Cendón, *Solving Hyperbolic Equations with Finite Volume Methods*,
UNITEXT - La Matematica per il 3+2 90, DOI 10.1007/978-3-319-14784-0_7

from which he graduated in 1946. Godunov's family was also known for another mathematician, Elena Konstantínovna Godunov, after who an famous inequality was named.

Since school, Godunov showed great interest in mathematics, and participated in the MSU olympics, displaying an enormous aptitude towards this field. He graduated in 1951 in the *MSU Faculty of Mechanics and Mathematics* under the speciality of Mathematics, and started his postgraduate studies at the *Steklov Institute of Mathematics (MIAN)*.

His most renowned idea is to employ the solution of local Riemann problems in the construction of numerical schemes, in particular the Godunov method that we have studied in this part of the book. In addition, this idea also leads to the approximate Riemann solvers, also known as *Godunov Methods*.

From the present biographical summary, I want to invite the readers to review [10]. Godunov himself describes in Section 1 of this reference the process of development of this scheme. I hope that the review of this work, narrated in first person, 55 years after being published, is as motivational for the reader as it has been for me.

By the autumn 1953 M.V. Keldysh and I.M. Gelfand were demanding a new method. Only one year after, in response to this challenge, Godunov defended his PhD dissertation under the supervision of I.G. Petrovskii and I.M. Gelfand. The level of excellence shown by Godunov must be taken as a source of admiration for young researchers, rather than as a source of stress. Godunov also explains that he only knew Lax's work after finishing his thesis, and stated that, had he had access to it before, Godunov method would have never been constructed.

The brilliant idea that leads to Godunov Methods is not present only in finite volume methodology. As Toro precises in the preface of [28], these techniques are also used in approximations such as *front tracking* and in Galerkin discontinuous finite element methods. Godunov's theorem (see [30]), not presented in this book since only first order schemes are considered, is one of the most important contributions to this methodology.

I had the opportunity to meet Prof. Godunov in the International Congress organized to celebrate his 70th birthday, held in Oxford in 1999. If the book [10] is in all sense three-dimensional (it has more than one thousand pages) is because, as Toro mentions in the preface of this book, the feeling of the 140 participants from 20 different countries is summarised in the sentence "*everyone is here*". The contributions from all the participants gave place to an unvaluable and extense volume. I could personally experiment in this event the power of a great idea to generate knowledge, and the deep humble feeling I perceived in the words of Prof. Godunov in the course of a short conversation, and many other participants I only knew previously through their works.

The commemoration of his 80th birthday was held in the framework of an International Conference held in Novosibirsk in 2009. We invite the interested readers to consult the review done by Brushlinskii et al., from the *Russian Academy of Sciences*, in which he summarises his personal life and professional career [3]. The relation of his contributions is very numerous, more that 300 scientific works, mentions and honours. Among those awarded in the USSR, it is worth mention the outstanding

Lenin Prize (1959). Other details about his personal history and his work can be read in the following links:

- Sergei K. Godunov in MathSciNet: http://www.ams.org/mathscinet/search/author.html?mrauthid=201628&Submit=Search
- Sergei K. Godunov in Mathematics Genealogy Project: http://www.genealogy.ams.org/id.php?id=55067
- Sergei K. Godunov in Wikipedia: http://en.wikipedia.org/wiki/Sergei_K._Godunov
- Sergei K. Godunov in his personal web site: http://www.math.nsc.ru/LBRT/d1/english/godunov.htm

Part III
MATLAB Codes for the Studied Methods

Chapter 8
Codes for the Linear Transport Equation

Abstract This chapter contains the codes to solve the Exercises related with the linear transport equation. The different numerical schemes studied in Chap. 5 are implemented. The pulse defined in Example 5.1 is considered as initial condition and also the continuous sine function considered in Sect. 5.5 to analyze the accuracy of the Lax Friedrichs and the upwind schemes. Transmissive and periodic boundary conditions are also implemented. The conservation of the schemes is also checked.

8.1 Function to Solve the Transport Equation with the Centered Scheme (5.37): TEcentered.m

This code corresponds to Solution 5.1 and solves the problem presented in Example 5.1 with the centered scheme.

```
function TEcentered
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
%   Centered scheme for the transport equation
%
%       w +lambda w =0
%        t         x
%
%   Domain: [a,b]
%   Initial condition: Pulse
%
%   w0(x)=wp if x=xp=(b+a)/2; else w0(x)=0
%
%   Transmissive boundary conditions
%
%   Exact solution:
%   w(x,t)=w0(x-lambda*t)
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
 clear all
 clc
 disp('Centered scheme')
```

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```
 disp('-----------------------------------------')
  disp('w +lambda w =0')
 disp(' t         x')
 disp('Domain: [a,b]')
 disp('Transmissive boundary conditions')
 disp('Initial condition: Pulse')
 disp('w0(x)=wp if x=xp=(b+a)/2; else w0(x)=0')
 disp('Exact solution')
 disp('w(x,t)=w0(x-lambda*t)')
 disp('-----------------------------------------')
 a=-3;
 disp(['Lower end of the interval a = ', num2str(a)])
 b=3;
 disp(['Upper end of the interval b = ', num2str(b)])
 lambda=1;
 disp(['Velocity of propagation =', num2str(lambda)])
 m=60;
 disp(['Number of nodes =', num2str(m)])
 deltax=(b-a)/(m);
 disp(['deltax =', num2str(deltax)])
 x=[a:deltax:b];
 deltat=0.1;
 disp(['Time step =', num2str(deltat)])
 tmax=1;
 disp(['Time end = ', num2str(tmax)])
 mt=tmax/deltat;
% Courant number
 cfl=lambda*deltat/deltax;
 disp(['Courant number = ', num2str(cfl)])
 wp=1;
 disp(['Value of the pulse wp=w0(xp) =',num2str(wp)])
disp('-----------------------------------------')
%
% Plot the initial condition at [a,b]
%
 w0=zeros(1,m+1);
 w0(m/2+1)=wp;
 figure(1)
 plot(x,w0,'or')
 xlabel('x'); ylabel('w(x,0)');
title('Initial condition');
%
% Initialization
%
wa=w0;
we=w0;
%
%
for n=1:mt
%
wam(1:m)=wa(2:m+1);
wan(2:m+1)=wa(1:m);
```

```
%
wn(2:m)=wa(2:m)-cfl*0.5*(wam(2:m)-wan(2:m));
%
% Exact solution
%
% Exact position of the pulse
%       at time t=n*deltat
%
xp=lambda*n*deltat;
%
    for i=1:m+1
%
        if(abs(x(i)-xp)<=deltax/2)
        we(i)=wp;
        else
        we(i)=0;
        end
    end
%
% Transmissive boundary conditions
%
    wn(1)=wn(2);
    wn(m+1)=wn(m);
%
figure(2)
plot(x,wn,'xb',x,wn,'b',x,we,'or')
xlabel('x'); ylabel('w(x,t)');
title(['Centered scheme t = ',num2str(n*deltat)]);
pause(1)
% Update
wa=wn;
end
```

8.2 Function to Solve the Transport Equation with the Lax-Friedrichs Scheme (5.50): TELaxFriedrichs.m

This code corresponds to Solution 5.4 and solves the problem presented in Example 5.1 with the Lax-Friedrichs scheme.

```
function TELaxFriedrichs
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
%   Lax-Friedrichs scheme for the transport equation
%
%       w +lambda w =0
%        t         x
%
%   Domain: [a,b]
```

```
%   Initial condition: Pulse
%
%   w0(x)=wp if x=xp=(b+a)/2; else w0(x)=0
%
%   Transmissive boundary conditions
%
%   Exact solution:
%   w(x,t)=w0(x-lambda*t)
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
 clear all
 clc
 disp('Lax-Friedrichs scheme')
 disp('-------------------------------------------')
 disp('w +lambda w =0')
 disp(' t         x')
 disp('Domain: [a,b]')
 disp('Transmissive boundary conditions')
 disp('Initial condition: Pulse')
 disp('w0(x)=wp if x=xp=(b+a)/2; else w0(x)=0')
 disp('Exact solution')
 disp('w(x,t)=w0(x-lambda*t)')
 disp('-------------------------------------------')
 a=-3;
 disp(['Lower end of the interval a = ', num2str(a)])
 b=3;
 disp(['Upper end of the interval b = ', num2str(b)])
 lambda=1;
 disp(['Velocity of propagation =', num2str(lambda)])
 m=60;
 disp(['Number of nodes =', num2str(m)])
 deltax=(b-a)/(m);
 disp(['deltax =', num2str(deltax)])
 x=[a:deltax:b];
 deltat=0.1;
 disp(['Time step =', num2str(deltat)])
 tmax=1;
 disp(['Time end = ', num2str(tmax)])
 mt=tmax/deltat;
% Courant number
 cfl=lambda*deltat/deltax;
 disp(['Courant number = ', num2str(cfl)])
 wp=1;
 disp(['Value of the pulse wp=w0(xp) =',num2str(wp)])
disp('-------------------------------------------')
%
% Plot the initial condition at [a,b]
%
 w0=zeros(1,m+1);
 w0(m/2+1)=wp;
 figure(1)
 plot(x,w0,'or')
```

```
 xlabel('x'); ylabel('w(x,0)');
title('Initial condition');
%
% Initialization
%
wa=w0;
we=w0;
%
%
for n=1:mt
%
wam(1:m)=wa(2:m+1);
wan(2:m+1)=wa(1:m);
%
wn(2:m)=0.5*(wam(2:m)+wan(2:m))-...
              cfl*0.5*(wam(2:m)-wan(2:m));
%
% Exact solution
%
% Exact position of the pulse
%       at time t=n*deltat
%
xp=lambda*n*deltat;
%
    for i=1:m+1
%
        if(abs(x(i)-xp)<=deltax/2)
        we(i)=wp;
        else
        we(i)=0;
        end
    end
%
% Transmissive boundary conditions
%
    wn(1)=wn(2);
    wn(m+1)=wn(m);
%
figure(2)
plot(x,wn,'xb',x,wn,'b',x,we,'or')
xlabel('x'); ylabel('w(x,t)');
title(['Lax-Friedrichs scheme t = ',num2str(n*deltat)]);
pause(1)
% Update
wa=wn;
end
```

8.3 Function to Solve the Transport Equation with the Upwind Scheme (5.63): TEupwind.m

This code corresponds to Solution 5.12 and solves the problem presented in Example 5.1 with the upwind scheme.

```
function TEupwind
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
%   Upwind scheme for the transport equation
%
%       w +lambda w =0
%        t         x
%
%   Domain: [a,b]
%   Initial condition: Pulse
%
%   w0(x)=wp if x=xp=(b+a)/2; else w0(x)=0
%
%   Transmissive boundary conditions
%
%   Exact solution:
%   w(x,t)=w0(x-lambda*t)
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
 clear all
 clc
 disp('Upwind scheme')
 disp('-------------------------------------------')
 disp('w +lambda w =0')
 disp(' t         x')
 disp('Domain: [a,b]')
 disp('Transmissive boundary conditions')
 disp('Initial condition: Pulse')
 disp('w0(x)=wp if x=xp=(b+a)/2; else w0(x)=0')
 disp('Exact solution')
 disp('w(x,t)=w0(x-lambda*t)')
 disp('-------------------------------------------')
 a=-3;
 disp(['Lower end of the interval a = ', num2str(a)])
 b=3;
 disp(['Upper end of the interval b = ', num2str(b)])
 lambda=1;
 disp(['Velocity of propagation lambda =', num2str(lambda)])
 m=60;
 disp(['Number of nodes =', num2str(m)])
 deltax=(b-a)/(m);
 disp(['deltax =', num2str(deltax)])
 x=[a:deltax:b];
 deltat=0.1;
 disp(['Time step =', num2str(deltat)])
```

```
 tmax=1;
 disp(['Time end = ', num2str(tmax)])
 mt=tmax/deltat;
% Courant number
 cfl=lambda*deltat/deltax;
 disp(['Courant number = ', num2str(cfl)])
 wp=1;
 disp(['Value of the pulse wp=w0(xp) =',num2str(wp)])
disp('-----------------------------------------')
%
% Plot the initial condition at [a,b]
%
 w0=zeros(1,m+1);
 w0(m/2+1)=wp;
 figure(1)
 plot(x,w0,'or')
 xlabel('x'); ylabel('w(x,0)');
title('Initial condition');
%
% Initialization
%
wa=w0;
we=w0;
%
cflp=0.5*(lambda+abs(lambda))*deltat/deltax;
cfln=0.5*(lambda-abs(lambda))*deltat/deltax;
%
for n=1:mt
%
wam(1:m)=wa(2:m+1);
wan(2:m+1)=wa(1:m);
%
wn(2:m)=wa(2:m)-cflp*(wa(2:m)-wan(2:m))-...
                cfln*(wam(2:m)-wa(2:m));
%
% Exact solution
%
% Exact position of the pulse
%       at time t=n*deltat
%
xp=lambda*n*deltat;
%
    for i=1:m+1
%
        if(abs(x(i)-xp)<=deltax/2)
        we(i)=wp;
        else
        we(i)=0;
        end
    end
%
```

```
% Transmissive boundary conditions
%
    wn(1)=wn(2);
    wn(m+1)=wn(m);
%
figure(2)
plot(x,wn,'xb',x,wn,'b',x,we,'or')
xlabel('x'); ylabel('w(x,t)');
title(['Upwind scheme t = ',num2str(n*deltat)]);
pause(1)
% Update
wa=wn;
end
```

8.4 Function to Solve the Transport Equation with the Upwind Scheme (5.63) Expressed as a Finite Volume Method: TEupwindFV.m

This code corresponds to Solution 5.10 and solves the problem presented in Example 5.1 with the upwind scheme. In this code the CFL number is an input and Δt is obtained: $\Delta t = \frac{\mu \Delta x}{\lambda}$. This code also checks that the numerical scheme is conservative.

```
function TEupwindFV
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
%   Upwind scheme for the transport equation
%
%       w +lambda w =0
%        t         x
%
%   Domain: [a,b]
%   Initial condition: Pulse
%
%   w0(x)=wp if x=xp=(b+a)/2; else w0(x)=0
%
%  Transmissive boundary conditions
%
%   Exact solution:
%   w(x,t)=w0(x-lambda*t)
%
%  Approximated solution constant per finite volume
%
%  Deltat is obtained from CFL number
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
 clear all
 clc
 disp('-------------------------------------------')
```

```
 disp('Upwind scheme to solve the transport equation')
 disp('-------------------------------------------')
 disp('w +lambda w =0')
 disp(' t         x')
 disp('Domain: [a,b]')
 disp('Transmissive boundary conditions')
 disp('Initial condition: Pulse')
 disp('w0(x)=wp if x=xp=(b+a)/2; else w0(x)=0')
 disp('Exact solution')
 disp('w(x,t)=w0(x-lambda*t)')
 disp('-------------------------------------------')
 a=-6;
 disp(['Lower end of the interval a = ', num2str(a)])
 b=6;
 disp(['Upper end of the interval b = ', num2str(b)])
 lambda=1;
 disp(['Velocity of propagation lambda =', num2str(lambda)])
 m=12;
 disp(['Number of nodes m=', num2str(m)])
 deltax=(b-a)/(m);
 disp(['deltax =', num2str(deltax)])
 x=[a:deltax:b];
 xm=linspace(a+deltax/2,b-deltax/2,m);
 xm=[a,xm,b];
 cfl=1.5;
 disp(['Courant number = ', num2str(cfl)])
 % Deltat from Courant number
 deltat=cfl*deltax/lambda;
 disp(['Time step from Courant number =', num2str(deltat)])
 tmax=4;
 disp(['Time end = ', num2str(tmax)])
 mt=tmax/deltat;
%
 wp=1;
disp(['Value of the pulse wp=w0(xp) =',num2str(wp)])
disp('-------------------------------------------')
%
% Plot the initial condition at [a,b]
%
 w0=zeros(1,m+1);
 w0(m/2+1)=wp;
 figure(1)
 %
 for i=1:1:m
    plot([xm(i),xm(i),xm(i+1),xm(i+1)],[0,w0(i),w0(i),0],'b')
 hold on
    end
    plot(xm,zeros(1,m+2),'*g')
 xlabel('x'); ylabel('w(x,0)');
title('Initial condition');
hold off
%
```

```
% Initialization
%
wa=w0;
we=w0;
%
cflp=0.5*(lambda+abs(lambda))*deltat/deltax;
cfln=0.5*(lambda-abs(lambda))*deltat/deltax;
%
for n=1:mt
%
wam(1:m)=wa(2:m+1);
wan(2:m+1)=wa(1:m);
%
wn(2:m)=wa(2:m)-cflp*(wa(2:m)-wan(2:m))-...
                cfln*(wam(2:m)-wa(2:m));
%
% Exact solution
%
% Exact position of the pulse
%       at time t=n*deltat
%
xp=lambda*n*deltat;
%
    for i=1:m
%
        if(abs(x(i)-xp)<=deltax/2)
        we(i)=wp;
        else
        we(i)=0;
        end
        % Approximated solution constant per finite volume
        figure(2)
plot([xm(i),xm(i),xm(i+1),xm(i+1)],[0,wn(i),wn(i),0],'b')
hold on
    end
%
% Transmissive boundary conditions
%
    wn(1)=wn(2);
    wn(m+1)=wn(m);
%
    we(m+1)=wa(m+1);
%
plot([xm(1),xm(1),xm(2),xm(2)],[0,wn(1),wn(1),0],'b')
hold on
    wn(m+1)=wa(m+1);
plot([xm(m),xm(m),x(m+1),xm(m+1)],[0,wn(m+1),wn(m+1),0],'b')
hold on
%
plot(x,wn,'xb',x,we,'or',xm,zeros(1,m+2),'*g')
xlabel('x'); ylabel('w(x,t)');
t=n*deltat;
```

```
title(['Upwind scheme t = ',num2str(t)]);
pause(1)
% Update
wa=wn;
hold off
end
%
%    Mass conservation
%
intw0=deltax*sum(w0);
intwn=deltax*sum(wn);
errorconservation=abs(intw0-intwn);
disp('Test Conservation property =')
disp(['deltax*sum(w0) =', num2str(intw0)])
disp(['deltax*sum(wn) =', num2str(intwn)])
disp(['error=abs(deltax*sum(w0)-deltax*sum(wn)) =',...
                  num2str(errorconservation)])
disp('-------------------------------------------')
```

8.5 Functions to Solve the Transport Equation with the Lax Friedrichs and the Upwind Schemes: lxf_tpbc.m, upwind_tpbc.m and TEupwindLxFCIsinMCPl1lm

These codes are the Solution 5.12 and they solve the problem presented in Sect. 5.5 to analyze the accuracy of the Lax Friedrichs and the upwind schemes. In these codes periodic boundary conditions are considered.

```
function TEupwindLxFCIsinMCPl1lm
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
%   Lax Friedrichs and Upwind schemes
%
%       w +lambda w =0
%        t         x
%
%   Domain: [a,b]
%   Initial condition: sin
%
%   w0(x)=sin(2*pi*)
%
%   Periodic boundary conditions
%   w(x_1,tn)=w(x_(m+1),tn)
%
%   Exact solution:
%   w(x,t)=w0(x-lambda*t)
%
% Observed Order of accuracy in space and time L1 Lm
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
 clear all
```

```
 clc
 disp('Lax Friedrichs and Upwind schemes')
 disp('-----------------------------------------')
 disp('w +lambda w =0')
 disp(' t         x')
 disp('Domain: [a,b]')
 disp('Periodic boundary conditions')
 disp('Initial condition: sin')
 disp('w0(x)=sin(2*pi*x)')
 disp('Exact solution')
 disp('w(x,t)=w0(x-lambda*t)')
 disp('-----------------------------------------')
 a=0;
 disp(['Lower end of the interval = ', num2str(a)])
 b=1;
 disp(['Upper end of the interval = ', num2str(b)])
 lambda=1;
 disp(['Velocity of propagation =', num2str(lambda)])
 mi=60
 disp(['Number of nodes mesh M(1)=',num2str(mi)])
 cfl=0.9;
 disp(['Courant number = ', num2str(cfl)])
 tmax=1;
 disp(['Time end = ', num2str(tmax)])
 disp('-----------------------------------------')
%
clf
% Mesh definition
m1vals = [mi mi*2 mi*4 mi*8 mi*16]'
ntest=size(m1vals,1) % Number of meshes

for jtest=1:ntest
     m=m1vals(jtest);
     disp(['Number of nodes mesh M(',...
         num2str(jtest),')=',num2str(m)])
     deltaxvals(jtest)=(b-a)/(m);
     deltax=deltaxvals(jtest);
     disp(['deltax =', num2str(deltax)])
     x=[a:deltax:b];
    % Plot the initial condition at [a,b]
    %
     w0=sin(2*pi*x);
     figure(1)
     plot(x,w0,'or')
     hold on
     xlabel('x'); ylabel('w(x,0)');
    title('Initial condition');
    %
    % Initialization
    %
    walxf=w0;% Lax Friedrichs
    wau=w0;% Upwind
```

```
we=w0;
%
% Deltat from Courant number
 deltatvals(jtest)=cfl*deltax/lambda;
 deltat=deltatvals(jtest);
 disp(['Time step from Courant number ='...
      , num2str(deltat)])
 mt=tmax/deltat;
      %
    cflp=0.5*(lambda+abs(lambda))*deltat/deltax;
    cfln=0.5*(lambda-abs(lambda))*deltat/deltax;
%
for n=1:mt
%
% Lax Friedrichs scheme
wnlxf=lxf_tpbc(walxf,cfl,m);
% Upwind scheme
wnu=upwind_tpbc(wau,cflp,cfln,m);
%
%
% Exact solution
%
we=sin(2*pi*(x-lambda*n*deltat));
%
%
figure(2)
plot(x,wnlxf,'xb',x,wnlxf,'b',x,wnu,'+b',...
    x,wnu,'b',x,we,'or')
xlabel('x'); ylabel('w(x,t)');
title(['Lax Friedrichs and Upwind schemes t ='...
    ,num2str(n*deltat)]);
% Update
walxf=wnlxf;
wau=wnu;
end
%
%    Mass conservation
%
intw0=deltax*sum(w0);
intwnlxf=deltax*sum(wnlxf);
intwnu=deltax*sum(wnu);
disp('Test Conservation property =')
disp(['deltax*sum(w0) =', num2str(intw0)])
disp(['Lax Friedrichs deltax*sum(wn)=',...
    num2str(intwnlxf)])
errorconservation=abs(intw0-intwnlxf);
disp(['error=abs(deltax*sum(w0)-deltax*sum(wn))=',...
                    num2str(errorconservation)])
disp(['Upwind deltax*sum(wn)=', num2str(intwnu)])
errorconservation=abs(intw0-intwnu);
disp(['error=abs(deltax*sum(w0)-deltax*sum(wn))=',...
                    num2str(errorconservation)])
```

```
    disp('-------------------------------------------')
    %
    %  Error L1
    %
    errorLxFl1(jtest)=deltax*sum(abs(wnlxf-we));
    errorul1(jtest)=deltax*sum(abs(wnu-we));
    %
    %  Error L infinity
    %
    errorLxFlm(jtest)=max(abs(wnlxf-we));
    errorulm(jtest)=max(abs(wnu-we));
    %
    coefoxLxFl1(jtest)=errorLxFl1(jtest)...
        /(deltaxvals(jtest));
    coefotLxFl1(jtest)=errorLxFl1(jtest)...
        /(deltatvals(jtest));
    %
    coefoxLxFlm(jtest)=errorLxFlm(jtest)...
        /(deltaxvals(jtest));
    coefotLxFlm(jtest)=errorLxFlm(jtest)...
        /(deltatvals(jtest));
    %
    coefoxul1(jtest)=errorul1(jtest)...
        /(deltaxvals(jtest));
    coefotul1(jtest)=errorul1(jtest)...
        /(deltatvals(jtest));
    %
    coefoxulm(jtest)=errorulm(jtest)...
        /(deltaxvals(jtest));
    coefotulm(jtest)=errorulm(jtest)...
        /(deltatvals(jtest));
end
%
disp(['   Deltax   errorLxFl1   coefoxLxFl1 '])
LxFs=[deltaxvals  ; errorLxFl1  ; coefoxLxFl1  ]'
disp(['   Deltat   errorLxFl1   coefotLxFl1   '])
LxFt=[deltatvals  ; errorLxFl1  ; coefotLxFl1  ]'
%
disp(['   Deltax   errorLxFlm   coefoxLxFlm '])
LxFs=[deltaxvals  ; errorLxFlm  ; coefoxLxFlm  ]'
disp(['   Deltat   errorLxFlm   coefotLxFlm   '])
LxFt=[deltatvals  ; errorLxFlm  ; coefotLxFlm  ]'
 %
disp(['   Deltax   errorul1     coefoxul1  '])
Us= [deltaxvals  ;  errorul1  ; coefoxul1  ]'
disp(['   Deltat   errorul1     coefotul1  '])
Ut= [deltatvals  ;  errorul1  ; coefotul1  ]'
 %
disp(['   Deltax   errorulm     coefoxulm  '])
Us= [deltaxvals  ;  errorulm  ; coefoxulm  ]'
disp(['   Deltat   errorulm     coefotulm  '])
Ut= [deltatvals  ;  errorulm  ; coefotulm  ]'
```

```
%
% Lax-Friedrish
%
 cerrorLxFl1=errorLxFl1(1:(ntest-1))./...
     errorLxFl1(2:ntest);
 ordenLxFl1=log(cerrorLxFl1)./log(2);
disp('Observed Order of accuracy in space')
disp(['Lax-Friedrichs l1=',...
      num2str(ordenLxFl1)])
% Linear regresion to obtain the Observed Order
% of accuracy in space L1
pxLxFl1=polyfit(log(deltaxvals),log(errorLxFl1),1);
disp('Observed Order of accuracy in space')
disp('Lax-Friedrichs l1')
disp('via Linear regresion pxLxFl1(1)=');
disp(pxLxFl1(1))
%
cerrorLxFlm=errorLxFlm(1:(ntest-1))./...
             errorLxFlm(2:ntest);
ordenLxFlm=log(cerrorLxFlm)./log(2);
disp(['Observed Order Lax-Friedrichs linfinity=',...
       num2str(ordenLxFlm)])
% Linear regresion to obtain the Observed Order
% of accuracy in space Lm
pxLxFlm=polyfit(log(deltaxvals),log(errorLxFlm),1);
disp('Observed Order of accuracy in space')
disp('Lax-Friedrichs linfinity')
disp('via Linear regresion pxLxFlm(1)=');
disp(pxLxFlm(1))
% Representation Linear regresion
% Order of accuracy in space L1, Lm LxF
figure(3)
rgxLxFl1=@(x) exp(pxLxFl1(2))*x.^pxLxFl1(1);
rgxLxFlm=@(x) exp(pxLxFlm(2))*x.^pxLxFlm(1);
loglog(deltaxvals,rgxLxFl1(deltaxvals),'r',...
       deltaxvals,errorLxFl1,'x'...
      ,deltaxvals,rgxLxFlm(deltaxvals),'r'...
      ,deltaxvals,errorLxFlm,'o')
title('Linear regresion Lax-Friedrichs l1,linfinity')
legend(sprintf('Linear regresion, slope=%0.3g',...
    pxLxFl1(1)),'Error l1 log-log',...
       sprintf('Linear regresion, slope=%0.3g',...
    pxLxFlm(1)),'Error L infinity log-log')
%
% Upwind scheme
%
 cerrorul1=errorul1(1:(ntest-1))./errorul1(2:ntest);
 ordenul1=log(cerrorul1)./log(2);
disp('Observed Order of accuracy in space')
disp(['Upwind scheme l1=',...
      num2str(ordenul1)])
% Linear regresion to obtain the Observed Order
```

```
% of accuracy in space L1
pxul1=polyfit(log(deltaxvals),log(errorul1),1);
disp('Observed Order of accuracy in space')
disp('Upwind scheme l1')
disp('via Linear regresion pxul1(1)=');
disp(pxul1(1))
%
 cerrorulm=errorulm(1:(ntest-1))./...
     errorulm(2:ntest);
 ordenulm=log(cerrorulm)./log(2);
 disp(['Observed Order Upwind scheme linfinity=',...
        num2str(ordenulm)])
 % Linear regresion to obtain the Observed Order
 % of accuracy in space L1
pxulm=polyfit(log(deltaxvals),log(errorulm),1);
disp('Observed Order of accuracy in space')
disp('Upwind scheme linfinity')
disp('via Linear regresion pxulm(1)=');
disp(pxulm(1))
% Representation Linear regresion Order
% of accuracy in space L1, linfinity Upwind
figure(4)
rgxul1=@(x) exp(pxul1(2))*x.^pxul1(1);
rgxulm=@(x) exp(pxulm(2))*x.^pxulm(1);
loglog(deltaxvals,rgxul1(deltaxvals),'r',...
       deltaxvals,errorul1,'x',...
       deltaxvals,rgxulm(deltaxvals),'r',...
       deltaxvals,errorulm,'o')
title('Linear regresion Upwind l1, linfinity')
legend(sprintf('Linear regresion, slope=%0.3g',...
       pxul1(1)),'Error l1 log-log',...
       sprintf('Linear regresion, slope=%0.3g',...
       pxulm(1)),'Error L infinity log-log')
```

8.5.1 lxf_tpbc.m

Function with the Lax Friedrichs scheme (5.50) for the transport equation.

```
function [wn] = lxf_tpbc(wa,cfl,m)
%
%  Lax Friedrichs scheme for the transport equation
%
wam=zeros(1,m+1);
wan=zeros(1,m+1);
%
wam(1:m)=wa(2:m+1);
% Periodic boundary conditions
wam(m+1)=wam(1);
%
wan(2:m+1)=wa(1:m);
```

```
%  Periodic boundary conditions
wan(1)=wan(m+1);
%
wn(1:m+1)=1/2*(wam(1:m+1)+wan(1:m+1))-...
       cfl/2*(wam(1:m+1)-wan(1:m+1));
end
```

8.5.2 *upwind_tpbc.m*

Function with the upwind scheme (5.63) for the transport equation.

```
function [wn] = upwind_tpbc(wa,cflp,cfln,m)
%
%  Upwind scheme for the transport equation

%
wam=zeros(1,m+1);
wan=zeros(1,m+1);
%
wam(1:m)=wa(2:m+1);
% Periodic boundary conditions
wam(m+1)=wam(1);
%
wan(2:m+1)=wa(1:m);
%  Periodic boundary conditions
wan(1)=wan(m+1);
%
wn(1:m+1)=wa(1:m+1)-cflp*(wa(1:m+1)-wan(1:m+1))-...
                    cfln*(wam(1:m+1)-wa(1:m+1));
end
```

Chapter 9
Codes for the Burgers Equation

Abstract This chapter contains the codes to solve the Exercises related with the Burgers equation. The different numerical schemes studied in Chap. 6 are implemented. The methods studied are compared with the exact solution of the Riemann problem explained in Sect. 3.4. Moreover, other initial conditions are considered.

9.1 Function Exact Entropy Solution of the Riemann Problem for the Burgers Equation: exact_burgers.m

The methods studied are compared with the exact solution of the Riemann problem for the Burgers equation presented in Sect. 3.4, which is implemented in this function.

```
function [w] = exact_burgers(x,t,xb,m,wl,wr,s)
%
% Entropic solution for Burgers equation
%
if wl >= wr
%
% Entropy shock
%
  for i=1:m+1
      if (x(i)-xb)/t < s
         w(i) = wl ;
      else
         w(i) = wr ;
     end
  end
else
%
% Expansion wave
%
  for i=1:m+1
      if (x(i)-xb)/t <wl
         w(i) = wl ;
      elseif (x(i)-xb)/t >=wr
```

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```
            w(i) = wr;
        else
            w(i) = (x(i)-xb)/t ;
        end
    end
end
```

9.2 Functions to Solve Riemann Problem for the Burgers Equation to Study the Conservation of the Schemes: BEconservation.m, god_btbc.m, lxf_btbc.m, qscheme_btbc.m and ncon_btbc.m

This code solves the Burgers equation with the schemes:

- Godunov method (see Sect. 6.1.1).
- Lax Friedrichs scheme (see Sect. 6.1.2).
- Q-scheme of van Leer/Roe (see Sect. 6.5).
- Non conservative scheme (see Sect. 6.1.3).

With this code the conservative methods studied are compared with the non conservative one. This is the solution of Exercise 6.5.

```
function BEconservation
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
%   Finite volume methods for Burgers equation
%
%       w + w w =0
%        t         x
%
%   Domain: [a,b]
%
%   Riemann Problem
%   w0(x)=wl if x<xb=(b+a)/2; else w0(x)=wr
%
%   Transmissive boundary conditions
%   Deltat is obtained from CFL number
%
%   Numerical methods:
%     * Godunov
%     * Lax Friedrichs
%     * Q-scheme of  van Leer / Roe
%     * Non conservative
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

```
clear all
clc
disp('-------------------------------------------')
disp('Finite volume methods for Burgers equation')
disp('-------------------------------------------')
disp('w + w w =0')
disp(' t     x')
disp('Domain: [a,b]')
disp('Transmissive boundary conditions')
disp('Riemann problem for Burgers equation')
disp('w0(x)=wl if x<xb=(b+a)/2; else w0(x)=wr')
disp('Exact solution')
disp('-------------------------------------------')
a=0;
disp(['Lower end of the interval a = ', num2str(a)])
b=5;
disp(['Upper end of the interval b = ', num2str(b)])
m=100;
disp(['Number of nodes m=', num2str(m)])
deltax=(b-a)/(m);
xb=(b+a)/2;
disp(['Initial location of the discontinuity, xb = '...
     , num2str(xb)])
disp(['deltax =', num2str(deltax)])
x=[a:deltax:b];
cfl=0.8;
disp(['Courant number = ', num2str(cfl)])
%
wl=0.5;
wr=0.1;
%
% Deltat from Courant number
deltat=cfl*deltax/max(abs(wl),abs(wr));
%
disp(['Time step from Courant number = ',...
      num2str(deltat)])
tmax=2;
disp(['Time end = ', num2str(tmax)])
mt=tmax/deltat;
disp('-------------------------------------------')
disp(['Initial condition:'])
disp(['w_l= ',num2str(wl)])
disp(['w_r= ',num2str(wr)])
disp(['w0(x)=',num2str(wl),' if x<=',num2str(xb)])
disp(['w0(x)=',num2str(wr),' if x>',num2str(xb)])
%
s=(wl+wr)/2;
disp(['The speed of the shock'])
disp(['from the Rankine Hugoniot conditions:',...
      num2str(s)])
```

```
if wl>wr
    disp(['The exact solution is an entropy shock'])
    disp(['w(x,t)=',num2str(wl),...
         ' if (x-',num2str(xb),')/t<=',num2str(s)])
    disp(['w(x,t)=',num2str(wr),...
         ' if (x-',num2str(xb),')/t>',num2str(s)])
elseif wl<wr
    disp(['The exact solution is a rarefaction wave'])
    disp(['w(x,t)=',num2str(wl),...
        ' if (x-',num2str(xb),')/t<=',num2str(wl)])
    disp(['w(x,t)= (x-',num2str(xb),')/t',...
         ' if ', num2str(wl),...
         '<= (x-',num2str(xb),')/t<=',num2str(wr)])
    disp(['w(x,t)=',num2str(wr),...
        ' if (x-',num2str(xb),')/t>',num2str(wr)])
elseif wl==wr
    disp('The exact solution is')
    disp('a constant solution w(x,t)=wl')
end
disp('-----------------------------------------------')
disp(['Numerical methods:'])
disp([' * Godunov'])
disp([' * Lax Friedrichs'])
disp([' * Q-scheme van Leer / Roe'])
disp([' * Non conservative'])
disp('-----------------------------------------------')
%
% Plot the initial condition at [a,b]
%
 for i=1:m+1
     if x(i)<=xb
        w0(i)=wl;
     else
        w0(i)=wr;
     end
 end
%
 wmin=min(wl,wr)-0.2; %lower limit for the y axi
 wmax=max(wl,wr)+0.2; %upper limit for the y axi
 figure(1)
%
    plot(x,w0,'ok')
    axis([a b wmin wmax])
 xlabel('x'); ylabel('w(x,0)');
title('Initial condition');
hold off
%
% Initialization
%
t=0;
%
```

```
we=w0;%Exact solution
wagod=w0;% Godunov method
walxf=w0;% Lax Friedrichs
waq=w0;% Q-scheme (Roe=van leer for Burgers equation)
wanc=w0;% Non conservative scheme
%
dtdx=deltat/deltax;
%
for n=1:mt
% Godunov method
wngod=god_btbc(wagod,dtdx,m);
% Lax Friedrichs scheme
wnlxf=lxf_btbc(walxf,dtdx,m);
% Q-scheme
wnq=qscheme_btbc(waq,dtdx,m);
% Non conservative scheme
wnnc=ncon_btbc(wanc,dtdx,m);
%
% Exact solution
%
t=t+deltat;
we=exact_burgers(x,t,xb,m,wl,wr,s);
%
%
figure(2)
%    plot(x,wngod,'-xb',x,wnlxf,'-xg',x,wnq,'-+r',...
%         x,wnnc,'-*m',x,we,'-ok')
     plot(x,wngod,'-x',x,wnlxf,'-x',x,wnq,'-+',...
         x,wnnc,'-*',x,we,'-ok')
    axis([a b wmin wmax])
    xlabel('x'); ylabel('w(x,t)');
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Godunov','Lax Friedrichs','Q-scheme',...
           'Non conservative','Exact solution')


    pause(1)
%
figure(3)
    plot(x,abs(we-wngod),'-x',x,abs(we-wnlxf),'-x',...
         x,abs(we-wnq),'-+',x,abs(we-wnnc),'-*')
    xlabel('x'); ylabel('|we(x,t)-w(x,t)|');
    axis([a b wmin wmax])
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Error Godunov','Error Lax Friedrichs',...
           'Error Q-scheme','Error Non conservative')
    pause(1)
% Update
walxf=wnlxf;
wagod=wngod;
```

```
waq=wnq;
wanc=wnnc;
end
disp('--------------------------------------------')
disp(['Error at time t=',num2str(t)])
disp(['* Godunov (m\'{a}x(abs(we-wngod)))=',...
                      num2str( max(abs(we-wngod)) )])
disp(['* Lax Friedrichs (m\'{a}x(abs(we-wnlxf)))=',...
                      num2str( max(abs(we-wnlxf)) )])
disp(['* Q-scheme (m\'{a}x(abs(we-wnlxf)))=',...
                      num2str( max(abs(we-wnq))   )])
disp(['* Non conservative (m\'{a}x(abs(we-wnnc)))=',...
                      num2str( max(abs(we-wnnc))  )])
disp('--------------------------------------------')
disp('--------------------------------------------')
```

The schemes considered are implemented also as functions, and are presented in the following sections.

9.2.1 god_btbc.m

Godunov method for Burgers equation with transmissive boundary conditions (see Sect. 6.1.1).

```
function [wn]=god_btbc(wa,dtdx,m)
%
%  Godunov scheme for the Burgers equation
%
fw=zeros(1,m+1);
fwn=zeros(1,m+1);
wn=zeros(1,m+1);
fw(1:m+1)=0.5*wa(1:m+1).^2;
%
  for i=1:m
      wl=wa(i);
      wr=wa(i+1);
      s=(wl+wr)/2;
      if wl > wr
        % Entropy shock
        if s < 0
         fw(i) = 0.5*wr^2
        end
      elseif wl < wr
        % Expansion wave
        if wr < 0
         fw(i) = 0.5*wr^2 ;
```

```
        elseif wl > 0.
         fw(i) = 0.5*wl^2 ;
        else
         fw(i) = 0;
        end
      end
  end
%
fwn(2:m+1)=fw(1:m);
% Transmissive boundary conditions
fwn(1)=fw(1);
%
wn(1:m+1)=wa(1:m+1)-dtdx*(fw(1:m+1)-fwn(1:m+1));
%
end
```

9.2.2 lxf_btbc.m

Lax Friedrichs scheme with transmissive boundary conditions (see Sect. 6.1.2).

```
function [wn] = lxf_btbc(wa,dtdx,m)
%
%  Lax Friedrichs scheme for the Burgers equation
%
wam=zeros(1,m+1);
wan=zeros(1,m+1);
%
wam(1:m)=wa(2:m+1);
% Transmissive boundary conditions
wam(m+1)=wa(m);
%
wan(2:m+1)=wa(1:m);
% Transmissive boundary conditions
wan(1)=wa(2);
%
wn(1:m+1)=0.5*(wam(1:m+1)+wan(1:m+1))-...
        dtdx*0.25*(wam(1:m+1).^2-wan(1:m+1).^2);
end
```

9.2.3 qscheme_btbc.m

Q-scheme of van Leer/Roe with transmissive boundary conditions (see Sect. 6.5).

```
function [wn] = qscheme_btbc(wa,dtdx,m)
%
%  Q-scheme scheme for the Burgers equation
%
```

```
wam=zeros(1,m+1);
wan=zeros(1,m+1);
dm=zeros(1,m+1);
dn=zeros(1,m+1);
%
wam(1:m)=wa(2:m+1);
% Transmissive boundary conditions
wam(m+1)=wa(m);
%
wan(2:m+1)=wa(1:m);
% Transmissive boundary conditions
wan(1)=wa(2);
%
% Central flux
%
cf(1:m+1)=0.25*dtdx*(wam(1:m+1).^2-wan(1:m+1).^2);
%
% Numerical viscosity Q-scheme
%
dm(1:m+1)=(0.25*dtdx)*abs(wa(1:m+1)+wam(1:m+1)).*(wam(1:m+1)-wa(1:m+1));
dn(1:m+1)=(0.25*dtdx)*abs(wa(1:m+1)+wan(1:m+1)).*(wa(1:m+1)-wan(1:m+1));
%
wn(1:m+1)=wa(1:m+1)-cf(1:m+1)+dm(1:m+1)-dn(1:m+1);
end
```

9.2.4 ncon_btbc.m

Non conservative scheme with transmissive boundary conditions (see Sect. 6.1.3).

```
function [wn] = ncon_btbc(wa,dtdx,m)
%
%  Non conservative scheme for the Burgers equation
%
wam=zeros(1,m+1);
wan=zeros(1,m+1);
cflw=zeros(1,m+1);
%
wam(1:m)=wa(2:m+1);
% Transmissive boundary conditions
wam(m+1)=wa(m);
%
wan(2:m+1)=wa(1:m);
% Transmissive boundary conditions
wan(1)=wa(2);
%
cflw(1:m+1)=dtdx.*wa(1:m+1);
%
for i=1:m+1
    if(wa(i)>=0)
```

```
        wn(i)=wa(i)-...
              cflw(i)*( wa(i)-wan(i) );
        else
        wn(i)=wa(i)-...
              cflw(i)*( wam(i)-wa(i) );
        end
    end
```

9.3 Functions to Solve Riemann Problem for the Burgers Equation to Study Entropy Schemes: BEentropy.m, qscheme_llfr_btbc.m and qscheme_hr_btbc.m

This code solves the Burgers equation with entropy and non-entropy schemes for a test problem with a sonic point:

- Godunov method, entropy scheme (see Sect. 6.1.1).
- Lax Friedrichs scheme, entropy scheme (see Sect. 6.1.2).
- Q-scheme of van Leer/Roe (see Sect. 6.5):
  - Without regularization (non-entropy scheme).
  - With Harten regularization (entropy scheme) (see (6.46)).
  - With Local Lax Friedrichs regularization (entropy scheme) (see (6.47)).
- Non conservative scheme, entropy scheme (see Sect. 6.1.3).

This property, schemes consistent with the entropy condition, is analyzed in Sect. 6.6.

```
function BEentropy
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
%   Finite volume methods for Burgers equation
%
%       w + w w =0
%        t         x
%
%   Domain: [a,b]
%
%   Riemann Problem
%
%   w0(x)=wl if x<xb=(b+a)/2; else w0(x)=wr
%
%   Transmissive boundary conditions
%   Deltat is obtained from CFL number
%
%   Numerical methods:
%     * Godunov
%     * Lax Friedrichs
```

```
%     * Q-scheme van Leer / Roe
%     * Q-scheme + Harten regularization
%     * Q-scheme + Local Lax Friedrichs regularization
%     * Non conservative
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
 clear all
 clc
 disp('-------------------------------------------')
 disp('Finite volume methods for Burgers equation')
 disp('-------------------------------------------')
 disp('w + w w =0')
 disp(' t     x')
 disp('Domain: [a,b]')
 disp('Transmissive boundary conditions')
 disp('Riemann problem for Burgers equation')
 disp('w0(x)=wl if x<xb=(b+a)/2; else w0(x)=wr')
 disp('Exact solution')
 disp('-------------------------------------------')
 a=0;
 disp(['Lower end of the interval a = ',num2str(a)])
 b=5;
 disp(['Upper end of the interval b = ',num2str(b)])
 m=200;
 disp(['Number of nodes m=', num2str(m)])
 deltax=(b-a)/(m);
 xb=(b+a)/2;
 disp(['Initial location of the discontinuity, xb = '...
      , num2str(b)])
 disp(['deltax =', num2str(deltax)])
 x=[a:deltax:b];
 cfl=0.8;
 disp(['Courant number = ', num2str(cfl)])
%
 wl=-1;
 wr=2;
%
% Deltat from Courant number
 deltat=cfl*deltax/max(abs(wl),abs(wr));
%
 disp(['Time step from Courant number = ',...
       num2str(deltat)])
 tmax=1;
 disp(['Time end = ', num2str(tmax)])
 mt=tmax/deltat;
disp('-------------------------------------------')
disp(['Initial condition:'])
disp(['w_l= ',num2str(wl)])
disp(['w_r= ',num2str(wr)])
disp(['w0(x)=',num2str(wl),' if x<=',num2str(xb)])
```

```
disp(['w0(x)=',num2str(wr),' if x>',num2str(xb)])
%
disp('-------------------------------------------')
if wl>wr
        s=(wl+wr)/2;
    disp(['The speed of the shock from'])
    disp(['the Rankine Hugoniot condition s=',...
           num2str(s)])
    disp(['The exact solution is an entropy shock'])
    disp(['w(x,t)=',num2str(wl),...
         ' if (x-',num2str(xb),')/t<=',num2str(s)])
    disp(['w(x,t)=',num2str(wr),...
         ' if (x-',num2str(xb),')/t>',num2str(s)])
elseif wl<wr
        s=(wl+wr)/2;
    disp(['The exact solution is a rarefaction wave'])
    disp(['w(x,t)=',num2str(wl),...
         ' if (x-',num2str(xb),')/t<=',num2str(wl)])
    disp(['w(x,t)= (x-',num2str(xb),')/t',...
         ' if ',num2str(wl),'<= (x-',num2str(xb),...
                                ')/t<=',num2str(wr)])
    disp(['w(x,t)=',num2str(wr),...
        ' if (x-',num2str(xb),')/t>',num2str(wr)])
elseif wl==wr
    disp('The exact solution is')
    disp('a constant solution w(x,t)=wl')
end
disp('-------------------------------------------')
disp(['Numerical methods:'])
disp([' * Godunov'])
disp([' * Lax Friedrichs'])
disp([' * Q-scheme van Leer / Roe'])
disp([' * Q-scheme with'])
disp(['Harten regularization (HR)'])
epshr=0.9;
disp(['      eps for Harten regularization= ',...
      num2str(epshr)])
disp([' * Q-scheme with'])
disp(['Local Lax Friedrichs regularization (LLFR)'])
disp('-------------------------------------------')
%
% Plot the initial condition at [a,b]
%
 for i=1:m+1
     if x(i)<=xb
        w0(i)=wl;
     else
        w0(i)=wr;
     end
 end
```

```
%
 wmin=min(wl,wr)-0.2; %lower limit for the y axi
 wmax=max(wl,wr)+0.2; %upper limit for the y axi
 figure(1)
 %
    plot(x,w0,'ok')
%    plot(xm,zeros(1,m+2),'-')
    axis([a b wmin wmax])
 xlabel('x'); ylabel('w(x,0)');
title('Initial condition');
hold off
%
% Initialization
%
t=0;
%
we=w0;%Exact solution
wagod=w0;% Godunov method
walxf=w0;% Lax Friedrichs
waq=w0;% Q-scheme (Roe=van leer for Burgers equation)
waqhr=w0;% Q-scheme Harten regularization
waqllfr=w0;% Q-scheme LLF regularization
wanc=w0;% Non conservative scheme
%
dtdx=deltat/deltax;
%
for n=1:mt
% Godunov method
wngod=god_btbc(wagod,dtdx,m);
% Lax Friedrichs scheme
wnlxf=lxf_btbc(walxf,dtdx,m);
% Q-scheme
wnq=qscheme_btbc(waq,dtdx,m);
% Non conservative scheme
wnnc=ncon_btbc(wanc,dtdx,m);
% Q-scheme Harten regularization
wnqhr=qscheme_hr_btbc(waqhr,dtdx,m,epshr);
% Q-scheme LLF regularization
wnqllfr=qscheme_llfr_btbc(waqllfr,dtdx,m);
%
% Exact solution
%
t=t+deltat;
we=exact_burgers(x,t,xb,m,wl,wr,s);
%
%
figure(2)
%
     plot(x,wngod,'-x',x,wnlxf,'-x',x,wnq,'-+',...
          x,wnnc,'-*',x,wnqhr,'-o',x,wnqllfr,'-d',...
          x,we,'-ok')
```

```
    axis([a b wmin wmax])
    xlabel('x'); ylabel('w(x,t)');
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Godunov','Lax Friedrichs','Q-scheme',...
           'Non conservative','Q-scheme Harten R.',...
           'Q-scheme LLF R.','Exact solution',...
           'Location','NorthOutside')
    pause(1)
%
figure(3)
    plot(x,abs(we-wngod),'-x',x,abs(we-wnlxf),'-x',...
         x,abs(we-wnq),'-+',x,abs(we-wnnc),'-*',...
         x,abs(we-wnqhr),'-o',x,abs(we-wnqllfr),'-d')
    xlabel('x'); ylabel('|we(x,t)-w(x,t)|');
    axis([a b wmin wmax])
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Error Godunov','Error Lax Friedrichs',...
           'Error Q-scheme','Error Non conservative',...
           'Error Q-scheme Harten R.',...
           'Error Q-scheme LLF R.',...
           'Location','NorthOutside')
    pause(1)
% Update
walxf=wnlxf;
wagod=wngod;
waq=wnq;
waqhr=wnqhr;
waqllfr=wnqllfr;
wanc=wnnc;
end
disp('-------------------------------------------')
disp(['Error at time t=',num2str(t)])
disp(['* Godunov max(|we-wngod|)=.........',...
                      num2str( max(abs(we-wngod)) )])
disp(['* Lax Friedrichs m\'{a}x|we-wnlxf|=.....',...
                      num2str( max(abs(we-wnlxf)) )])
disp(['* Non conservative max|we-wnnc|=....',...
                      num2str( max(abs(we-wnnc))  )])
disp(['* Q-scheme max|we-wnlxf|=..........',...
                      num2str( max(abs(we-wnq))   )])
disp(['* Q-scheme + Harten R. max|we-wnnc|=',...
                     num2str( max(abs(we-wnqhr))  )])
disp(['* Q-scheme + LLF R. max|we-wnqllfr|=',...
                     num2str( max(abs(we-wnqllfr)))])
disp('-------------------------------------------')
disp('-------------------------------------------')
```

The considered regularizations of the Q-schemes are implemented also as functions, and are presented in the following sections.

9.3.1 qscheme_hr_btbc.m

Function with the Q-scheme for the Burgers equation with the Harten regularization (6.46).

```
function [wn] = qscheme_hr_btbc(wa,dtdx,m,epshr)
%
%  Q-scheme for the Burgers equation
%  Harten regularization
%
wam=zeros(1,m+1);
wan=zeros(1,m+1);
dm=zeros(1,m+1);
dn=zeros(1,m+1);
epsr=zeros(1,m+1);
ablanm=zeros(1,m+1);
regm=zeros(1,m+1);
ablanm=zeros(1,m+1);
regn=zeros(1,m+1);
%
wam(1:m)=wa(2:m+1);
% Transmissive boundary conditions
wam(m+1)=wa(m);
%
wan(2:m+1)=wa(1:m);
% Transmissive boundary conditions
wan(1)=wa(2);
%
% Central flux
%
cf(1:m+1)=0.25*dtdx*(wam(1:m+1).^2-wan(1:m+1).^2);
%
% Numerical viscosity Q-scheme with Harten regularization
%
epsr=epshr*ones(1,m+1);
%
ablanm=0.5*abs(wa(1:m+1)+wam(1:m+1));
regm=ablanm+0.5*(1+sign(epsr-ablanm)).*...
             ((ablanm.^2+epsr.^2)./(2*epsr)-ablanm);
%
dm(1:m+1)=(0.5*dtdx)*regm.*(wam(1:m+1)-wa(1:m+1));
%
ablann=0.5*abs(wa(1:m+1)+wan(1:m+1));
regn=ablann+0.5*(1+sign(epsr-ablann)).*...
             ((ablann.^2+epsr.^2)./(2*epsr)-ablann);
%
dn(1:m+1)=(0.5*dtdx)*regn.*(wa(1:m+1)-wan(1:m+1));
%
wn(1:m+1)=wa(1:m+1)-cf(1:m+1)+dm(1:m+1)-dn(1:m+1);
end
```

9.3.2 qscheme_llfr_btbc.m

Function with the Q-scheme for the Burgers equation and the Local Lax-Friedrichs (LLF) regularization (6.47).

```
function [wn] = qscheme_llfr_btbc(wa,dtdx,m)
%
%  Q-scheme  for the Burgers equation
%  Local Lax-Friedrichs (LLF) regularization
%
wam=zeros(1,m+1);
wan=zeros(1,m+1);
dm=zeros(1,m+1);
dn=zeros(1,m+1);
regm=zeros(1,m+1);
regn=zeros(1,m+1);
%
wam(1:m)=wa(2:m+1);
% Transmissive boundary conditions
wam(m+1)=wa(m);
%
wan(2:m+1)=wa(1:m);
% Transmissive boundary conditions
wan(1)=wa(2);
%
% Central flux
%
cf(1:m+1)=0.25*dtdx*...
          (wam(1:m+1).^2-wan(1:m+1).^2);
%
% Numerical viscosity Q-scheme with Harten regularization
%
regm(1:m+1)=max(abs(wa(1:m+1)),abs(wam(1:m+1)));
regn(1:m+1)=max(abs(wa(1:m+1)),abs(wan(1:m+1)));
%
dm(1:m+1)=0.5*dtdx*regm.*(wam(1:m+1)-wa(1:m+1));
%
dn(1:m+1)=0.5*dtdx*regn.*(wa(1:m+1)-wan(1:m+1));
%
wn(1:m+1)=wa(1:m+1)-cf(1:m+1)+dm(1:m+1)-dn(1:m+1);
end
```

9.3.3 Output of This Code

Figure 9.1 shows the results at final time.

```
This code provides the following information at the command prompt:
-------------------------------------------
Finite volume methods for Burgers equation
-------------------------------------------
```

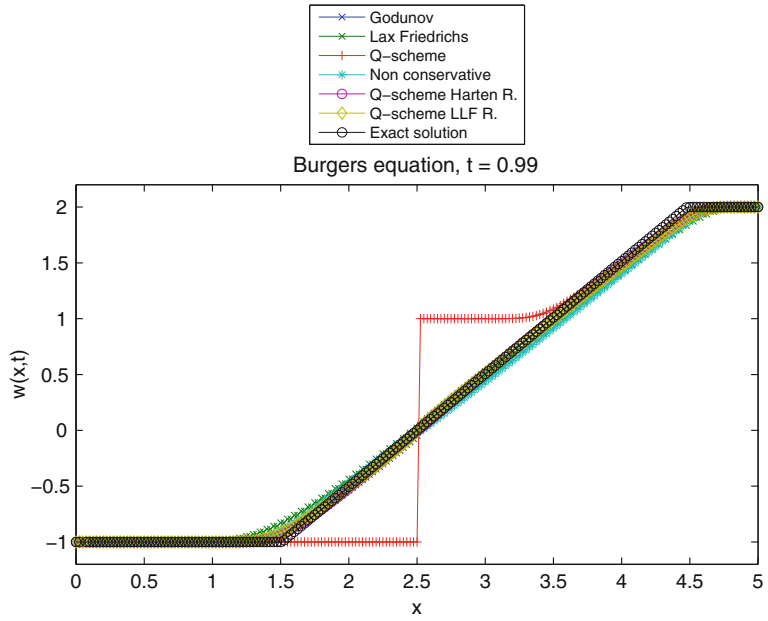


Fig. 9.1 Results for the Riemann problem of the Burgers equation with BEentropy.m

```
w + w w =0
 t     x
Domain: [a,b]
Transmissive boundary conditions
Riemann problem for Burgers equation
w0(x)=wl if x<xb=(b+a)/2; else w0(x)=wr
Exact solution
-------------------------------------------
Lower end of the interval a = 0
Upper end of the interval b = 5
Number of nodes m=200
Initial location of the discontinuity, xb = 2.5
deltax =0.025
Courant number = 0.8
Time step from Courant number = 0.01
Time end = 1
-------------------------------------------
Initial condition:
wl= -1
wr= 2
w0(x)=-1 if x<=2.5
w0(x)=2 if x>2.5
-------------------------------------------
The exact solution is a rarefaction wave
w(x,t)=-1 if (x-2.5)/t<=-1
w(x,t)= (x-2.5)/t if -1<= (x-2.5)/t<=2
w(x,t)=2 if (x-2.5)/t>2
```

```
-------------------------------------------
Numerical methods:
 * Godunov
 * Lax Friedrichs
 * Q-scheme van Leer / Roe
 * Q-scheme with
Harten regularization (HR)
      eps for Harten regularization= 0.9
 * Q-scheme with
Local Lax Friedrichs regularization (LLFR)
-------------------------------------------
-------------------------------------------
Error at time t=0.99
* Godunov max(|we-wngod|)=.........0.091609
* Lax Friedrichs m\'{a}x|we-wnlxf|=.....0.17242
* Non conservative max|we-wnnc|=....0.1393
* Q-scheme max|we-wnlxf|=..........1
* Q-scheme + Harten R. max|we-wnnc|=0.090024
* Q-scheme + LLF R. max|we-wnqllfr|=0.10219
-------------------------------------------
-------------------------------------------
```

9.4 Functions to Solve Riemann Problem for the Burgers Equation with All the Studied Schemes: BERP.m, steger_warming_btbc.m, vijaya_btbc.m, vijaya_hr_btbc.m and vijaya_llfr_btbc.m

This code solves a Riemann problem for the Burgers equation with all the schemes studied in Chap. 6 for Burgers equation:

- Godunov method (see Sect. 6.1.1).
- Lax Friedrichs scheme (see Sect. 6.1.2).
- Q-scheme of van Leer/Roe (see Sect. 6.5):
  - Without regularization.
  - With Harten regularization (see (6.46)).
  - With Local Lax Friedrichs regularization (see (6.47)).
- Non conservative scheme (see Sect. 6.1.3).
- Steger-Warming flux splitting (see Sect. 6.8.1).
- Vijayasundaram flux splitting (see Sect. 6.8.2):
  - Without regularization.
  - With Harten regularization (see (6.46)).
  - With Local Lax Friedrichs regularization (see (6.47)).

```
function BERP
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
%   Finite volume methods for Burgers equation
%
%       w + w w =0
%        t         x
%
%   Domain: [a,b]
%
%   Riemann Problem
%
%   w0(x)=wl if x<xb=(b+a)/2; else w0(x)=wr
%
%   Transmissive boundary conditions
%   Deltat is obtained from CFL number
%
%   Numerical methods:
%     * Godunov method
%     * Lax Friedrichs
%     * Q-scheme van Leer / Roe
%     * Q-scheme
%               + Harten regularization
%               + Local Lax Friedrichs regularization
%     * Non conservative
%     * Steger-Warming flux splitting
%     * Vijayasundaram splitting
%     * Vijayasundaram splitting
%               + Local Lax Friedrichs regularization

%               + Harten regularization
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
 clear all
 clc
 disp('-------------------------------------------')
 disp('Finite volume methods for Burgers equation')
 disp('-------------------------------------------')
 disp('w + w w =0')
 disp(' t     x')
 disp('Domain: [a,b]')
 disp('Transmissive boundary conditions')
 disp('Riemann problem for Burgers equation')
 disp('w0(x)=wp if x=xp0=(b-a)/2; else w0(x)=0')
 disp('Exact solution')
 disp('-------------------------------------------')
 a=0;
 disp(['Lower end of the interval a = ',num2str(a)])
 b=5;
 disp(['Upper end of the interval b = ',num2str(b)])
 m=200;
 disp(['Number of nodes m=', num2str(m)])
```

```
 deltax=(b-a)/(m);
 xb=(b+a)/2;
 disp(['Initial location of the discontinuity, xb = '...
      , num2str(xb)])
 disp(['deltax =', num2str(deltax)])
 x=[a:deltax:b];
 cfl=0.5;
 disp(['Courant number = ', num2str(cfl)])
%
 wl=-1;
 wr=2;
%
% Deltat from Courant number
 deltat=cfl*deltax/max(abs(wl),abs(wr));
%
 disp(['Time step from Courant number = ',...
       num2str(deltat)])
 tmax=1;
 disp(['Time end = ', num2str(tmax)])
 mt=tmax/deltat;
disp('--------------------------------------------')
disp(['Initial condition:'])
disp(['wl= ',num2str(wl)])
disp(['wr= ',num2str(wr)])
disp(['w0(x)=',num2str(wl),' if x<=',num2str(xb)])
disp(['w0(x)=',num2str(wr),' if x>',num2str(xb)])
%
disp('--------------------------------------------')
if wl>wr
        s=(wl+wr)/2;
    disp(['The speed of the shock'])
    disp(['from the Rankine Hugoniot condition s=',...
          num2str(s)])
    disp(['The exact solution is an entropy shock'])
    disp(['w(x,t)=',num2str(wl),...
         ' if (x-',num2str(xb),')/t<=',num2str(s)])
    disp(['w(x,t)=',num2str(wr),...
         ' if (x-',num2str(xb),')/t>',num2str(s)])
elseif wl<wr
        s=(wl+wr)/2;
    disp(['The exact solution is a rarefaction wave'])
   disp(['w(x,t)=',num2str(wl),...
         ' if (x-',num2str(xb),')/t<=',num2str(wl)])
    disp(['w(x,t)= (x-',num2str(xb),')/t',...
         ' if ',num2str(wl),'<= (x-',num2str(xb),...
                              ')/t<=',num2str(wr)])
    disp(['w(x,t)=',num2str(wr),...
        ' if (x-',num2str(xb),')/t>',num2str(wr)])
elseif wl==wr
    disp('The exact solution is')
    disp('a constant solution w(x,t)=wl=wr')
end
```

```
disp('-------------------------------------------')
disp(['Numerical methods:'])
disp([' * Godunov'])
disp([' * Lax Friedrichs'])
disp([' * Q-scheme van Leer / Roe'])
disp([' * Q-scheme with'])
disp(['   Harten regularization (HR)'])
epshr=0.9;
disp(['      eps for Harten regularization= ',...
      num2str(epshr)])
disp([' * Q-scheme with'])
disp(['   Local Lax Friedrichs regularization (LLFR)'])
disp([' * Steger-Warming flux splitting'])
disp([' * Vijayasundaram flux splitting'])
disp([' * Vijayasundaram flux splitting with'])
disp(['   Local Lax Friedrichs regularization (LLFR)'])
disp('-------------------------------------------')
%
% Plot the initial condition at [a,b]
%
 for i=1:m+1
     if x(i)<=xb
        w0(i)=wl;
     else
        w0(i)=wr;
     end
 end
%
 wmin=min(wl,wr)-0.2; %lower limit for the y axis
 wmax=max(wl,wr)+0.2; %upper limit for the y axis
 figure(1)
%
    plot(x,w0,'ok')
    axis([a b wmin wmax])
 xlabel('x'); ylabel('w(x,0)');
title('Initial condition');
hold off
%
% Initialization
%
t=0;
%
we=w0;%Exact solution
wagod=w0;% Godunov method
walxf=w0;% Lax Friedrichs
waq=w0;% Q-scheme (Roe=van leer for Burgers equation)
waqhr=w0;% Q-scheme Harten regularization
waqllfr=w0;% Q-scheme LLF regularization
wanc=w0;% Non conservative scheme
wasw=w0;% Steger-Warming flux splitting
wav=w0;% Vijayasundaram flux splitting
```

```
wavhr=w0;% Vijayasundaram flux splitting Harten regularization
wavllfr=w0;% Vijayasundaram flux splitting LLF regularization
%
dtdx=deltat/deltax;
%
for n=1:mt
% Godunov method
wngod=god_btbc(wagod,dtdx,m);
% Lax Friedrichs scheme
wnlxf=lxf_btbc(walxf,dtdx,m);
% Q-scheme
wnq=qscheme_btbc(waq,dtdx,m);
% Non conservative scheme
wnnc=ncon_btbc(wanc,dtdx,m);
% Q-scheme Harten regularization
wnqhr=qscheme_hr_btbc(waqhr,dtdx,m,epshr);
% Q-scheme LLF regularization
wnqllfr=qscheme_llfr_btbc(waqllfr,dtdx,m);
% Steger-Warming flux splitting
wnsw=steger_warming_btbc(wasw,dtdx,m);
% Vijayasundaram flux splitting
wnv=vijaya_btbc(wav,dtdx,m);
% Vijayasundaram flux splitting Harten regularization
wnvhr=vijaya_hr_btbc(wavhr,dtdx,m,epshr);
% Vijayasundaram flux splitting LLF regularization
wnvllfr=vijaya_llfr_btbc(wavllfr,dtdx,m);
%
% Exact solution
%
t=t+deltat;
we=exact_burgers(x,t,xb,m,wl,wr,s);
%
%
figure(2)
%
     plot(x,wngod,'-x',x,wnlxf,'-x',...
          x,wnnc,'-*',x,we,'-ok')
    axis([a b wmin wmax])
    xlabel('x'); ylabel('w(x,t)');
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Godunov','Lax Friedrichs',...
           'Non conservative','Exact solution',...
           'Location','NorthOutside')
figure(3)
%
     plot(x,wnq,'-+',x,wnqhr,'-o',x,wnqllfr,'-d',...
          x,we,'-ok')
    axis([a b wmin wmax])
    xlabel('x'); ylabel('w(x,t)');
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Q-scheme','Q-scheme Harten R.',...
           'Q-scheme LLF R.','Exact solution',...
           'Location','NorthOutside')
```

```
figure(4)
%
     plot(x,wnsw,'-v',x,wnv,'-s',x,wnvhr,'-<',...
          x,wnvllfr,'-^',x,we,'-ok')
    axis([a b wmin wmax])
    xlabel('x'); ylabel('w(x,t)');
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Steger-Warming',...
           'Vijayasundaram','Vijayasundaram Harten R.',...
           'Vijayasundaram LLF R.',...
           'Exact solution',...
           'Location','NorthOutside')
%
figure(5)
    plot(x,abs(we-wngod),'-x',x,abs(we-wnlxf),'-x',...
         x,abs(we-wnq),'-+',x,abs(we-wnnc),'-*',...
         x,abs(we-wnqhr),'-o',x,abs(we-wnqllfr),'-d',...
         x,abs(we-wnsw),'-v',x,abs(we-wnv),'-s',...
         x,abs(we-wnvhr),'-<',x,abs(we-wnvllfr),'-')
    xlabel('x'); ylabel('|we(x,t)-w(x,t)|');
    axis([a b wmin wmax])
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Error Godunov','Error Lax Friedrichs',...
           'Error Q-scheme','Error Non conservative',...
           'Error Q-scheme Harten R.',...
           'Error Q-scheme LLF R.',...
           'Error Steger-Warming',...
           'Error Vijayasundaram',...
           'Error Vijayasundaram Harten R.',...
           'Error Vijayasundaram LLF R.',...
           'Location','NorthOutside')
% Update
walxf=wnlxf;
wagod=wngod;
waq=wnq;
waqhr=wnqhr;
waqllfr=wnqllfr;
wanc=wnnc;
wasw=wnsw;
wav=wnv;
wavhr=wnvhr;
wavllfr=wnvllfr;
end
disp('-------------------------------------------')
disp(['Error at time t=',num2str(t)])
disp(['* Godunov max(|we-wngod|)=.........',...
                      num2str( max(abs(we-wngod)) )])
disp(['* Lax Friedrichs m\'{a}x|we-wnlxf|=.....',...
                      num2str( max(abs(we-wnlxf)) )])
disp(['* Non conservative max|we-wnnc|=....',...
                      num2str( max(abs(we-wnnc))  )])
```

```
disp(['* Q-scheme max|we-wnlxf|=..........',...
                        num2str( max(abs(we-wnq))   )])
disp(['* Q-scheme + Harten R. max|we-wnnc|=',...
                       num2str( max(abs(we-wnqhr))  )])
disp(['* Q-scheme + LLF R. max|we-wnqllfr|=',...
                       num2str( max(abs(we-wnqllfr)))])
disp(['* Steger-Warming max|we-wnsw|=.....',...
                        num2str( max(abs(we-wnsw)) )])
disp(['* Vijayasundaram max|we-wnv|=.....',...
                        num2str( max(abs(we-wnv)) )])
disp(['* Vijayasundaram LLF R. max|we-wnvlhr|=',...
                      num2str( max(abs(we-wnvhr)) )])
disp(['* Vijayasundaram LLF R. max|we-wnvllfr|=',...
                      num2str( max(abs(we-wnvllfr)) )])
disp('-------------------------------------------')
disp('-------------------------------------------')
%
% Figures all scheme at final time
%
% Godunov
%
figure(6)
%
     plot(x,wngod,'-x',x,we,'k')
    axis([a b wmin wmax])
    xlabel('x'); ylabel('w(x,t)');
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Godunov','Exact solution',...
           'Location','NorthOutside')
%
% Lax Friedrichs
%
figure(7)
%
     plot(x,wnlxf,'-x',x,we,'k')
    axis([a b wmin wmax])
    xlabel('x'); ylabel('w(x,t)');
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Lax Friedrichs','Exact solution',...
           'Location','NorthOutside')
%
% Non conservative
%
figure(8)
%
     plot(x,wnnc,'-*',x,we,'k')
    axis([a b wmin wmax])
    xlabel('x'); ylabel('w(x,t)');
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Non conservative','Exact solution',...
           'Location','NorthOutside')
```

```
%
% Q-scheme
%
figure(9)
%
     plot(x,wnq,'-+',x,we,'k')
    axis([a b wmin wmax])
    xlabel('x'); ylabel('w(x,t)');
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Q-scheme','Exact solution',...
           'Location','NorthOutside')
figure(10)
%
     plot(x,wnqhr,'-+',x,we,'k')
    axis([a b wmin wmax])
    xlabel('x'); ylabel('w(x,t)');
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Q-scheme + Harten R','Exact solution',...
           'Location','NorthOutside')
figure(11)
%
     plot(x,wnqllfr,'-+',x,we,'k')
    axis([a b wmin wmax])
    xlabel('x'); ylabel('w(x,t)');
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Q-scheme + LLF R','Exact solution',...
           'Location','NorthOutside')
%
% Flux splitting
%
%     Steger-Warming
figure(12)
%
     plot(x,wnsw,'-v',x,we,'k')
    axis([a b wmin wmax])
    xlabel('x'); ylabel('w(x,t)');
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Steger-Warming','Exact solution',...
           'Location','NorthOutside')
%     Vijayasundaram
figure(13)
%
     plot(x,wnv,'-v',x,we,'k')
    axis([a b wmin wmax])
    xlabel('x'); ylabel('w(x,t)');
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Vijayasundaram','Exact solution',...
           'Location','NorthOutside')
figure(14)
%
     plot(x,wnvhr,'-v',x,we,'k')
    axis([a b wmin wmax])
    xlabel('x'); ylabel('w(x,t)');
```

```
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Vijayasundaram + Harten R','Exact solution',...
           'Location','NorthOutside')
figure(15)
%
     plot(x,wnvllfr,'-v',x,we,'k')
    axis([a b wmin wmax])
    xlabel('x'); ylabel('w(x,t)');
    title(['Burgers equation, t = ',num2str(t)]);
    legend('Vijayasundaram + LLF R','Exact solution',...
           'Location','NorthOutside')
```

9.4.1 steger_warming_btbc.m

Function with the Steger-Warming flux splitting for the Burgers Equation 6.8.1.

```
function [wn] = steger_warming_btbc(wa,dtdx,m)
%
%  Steger-Warming flux splitting
%    for the Burgers equation
%
wam=zeros(1,m+1);  % w_(j+1)
wan=zeros(1,m+1);  % w_(j-1)
%
am=zeros(1,m+1);   % a+(0.5*w_j)
amn=zeros(1,m+1);  % a+(0.5*w_(j-1))
an=zeros(1,m+1);   % a-(0.5*w_j)
anm=zeros(1,m+1);  % a-(0.5*w_(j+1))
phim=zeros(1,m+1); % phi(w_j,w_(j+1))
phin=zeros(1,m+1); % phi(w_(j-1),w_j)
%
wam(1:m)=wa(2:m+1);
% Transmissive boundary conditions
wam(m+1)=wa(m);
%
wan(2:m+1)=wa(1:m);
% Transmissive boundary conditions
wan(1)=wa(2);
%
am(1:m+1)=max( 0.5*wa , zeros(1,m+1) );
amn(2:m+1)=am(1:m);
amn(1)=am(1);
%
an(1:m+1)=min( 0.5*wa , zeros(1,m+1) );
anm(1:m)=an(2:m+1);
anm(m+1)=an(m+1);
%
% Numerical flux
%
```

```
phim(1:m+1)=am(1:m+1).*wa(1:m+1)+anm(1:m+1).*wam(1:m+1);
phin(1:m+1)=amn(1:m+1).*wan(1:m+1)+an(1:m+1).*wa(1:m+1);
%
wn(1:m+1)=wa(1:m+1)-dtdx*(phim(1:m+1)-phin(1:m+1));
end
```

9.4.2 vijaya_btbc.m

Function with the Vijayasundatam flux splitting for the Burgers Equation 6.8.2.

```
function [wn] = vijaya_btbc(wa,dtdx,m)
%
%  Vijayasundaram flux splitting
%    for the Burgers equation
%
wam=zeros(1,m+1);   % w_(j+1)
wan=zeros(1,m+1);   % w_(j-1)
% a=0.5*wa
am=zeros(1,m+1);    % a+(0.25(w_j+w_(j+1)))
amn=zeros(1,m+1);   % a+(0.25(w_j+w_(j-1)))
an=zeros(1,m+1);    % a-(0.25(w_j+w_(j+1)))
ann=zeros(1,m+1);   % a-(0.25(w_j+w_(j-1)))
phim=zeros(1,m+1); % phi(w_j,w_(j+1))
phin=zeros(1,m+1); % phi(w_(j-1),w_j)
%
wam(1:m)=wa(2:m+1);
% Transmissive boundary conditions
wam(m+1)=wa(m);
%
wan(2:m+1)=wa(1:m);
% Transmissive boundary conditions
wan(1)=wa(2);
%
for i=1:m+1
am(i)=max( 0.25*(wa(i)+wam(i)) , 0 );
an(i)=min( 0.25*(wa(i)+wam(i)) , 0 );
end
%
amn(2:m+1)=am(1:m);
amn(1)=am(1);
%
ann(2:m+1)=an(2:m+1);
anm(m+1)=an(m+1);
%
% Numerical flux
%
phim(1:m+1)=am(1:m+1).*wa(1:m+1)+an(1:m+1).*wam(1:m+1);
phin(1:m+1)=amn(1:m+1).*wan(1:m+1)+ann(1:m+1).*wa(1:m+1);
%
wn(1:m+1)=wa(1:m+1)-dtdx*(phim(1:m+1)-phin(1:m+1));
end
```

9.4.3 vijaya_hr_btbc.m

Function with the Vijayasundatam flux splitting for the Burgers equation with the Harten regularization Sect. 6.8.2 and (6.46).

```
function [wn] = vijaya_hr_btbc(wa,dtdx,m,epshr)
%
%  Vijayasundaram flux splitting
%    for the Burgers equation
%
%  Harten regularization
%
wam=zeros(1,m+1);  % w_(j+1)
wan=zeros(1,m+1);  % w_(j-1)
%
am=zeros(1,m+1);   % a+(0.25(w_j+w_(j+1)))
amn=zeros(1,m+1);  % a+(0.25(w_j+w_(j-1)))
an=zeros(1,m+1);   % a-(0.25(w_j+w_(j+1)))
ann=zeros(1,m+1);  % a-(0.25(w_j+w_(j-1)))
phim=zeros(1,m+1); % phi(w_j,w_(j+1))
phin=zeros(1,m+1); % phi(w_(j-1),w_j)
%
wam(1:m)=wa(2:m+1);
% Transmissive boundary conditions
wam(m+1)=wa(m);
%
wan(2:m+1)=wa(1:m);
% Transmissive boundary conditions
wan(1)=wa(2);
%
% Harten regularization

%
epsr=epshr*ones(1,m+1);
%
ablanm=0.25*abs(wa(1:m+1)+wam(1:m+1));
regm=ablanm+0.5*(1+sign(epsr-ablanm)).*...
            ((ablanm.^2+epsr.^2)./(2*epsr)-ablanm);
ablann=0.25*abs(wa(1:m+1)+wan(1:m+1));
regn=ablann+0.5*(1+sign(epsr-ablann)).*...
            ((ablann.^2+epsr.^2)./(2*epsr)-ablann);
%
am(1:m+1) =0.5*(0.25.*(wa(1:m+1)+wam(1:m+1))+regm(1:m+1) );
amn(1:m+1)=0.5*(0.25.*(wa(1:m+1)+wan(1:m+1))+regn(1:m+1) );
an(1:m+1) =0.5*(0.25.*(wa(1:m+1)+wam(1:m+1))-regm(1:m+1) );
ann(1:m+1)=0.5*(0.25.*(wa(1:m+1)+wan(1:m+1))-regn(1:m+1) );
%
% Numerical flux
%
phim(1:m+1)=am(1:m+1).*wa(1:m+1)+an(1:m+1).*wam(1:m+1);
phin(1:m+1)=amn(1:m+1).*wan(1:m+1)+ann(1:m+1).*wa(1:m+1);
%
wn(1:m+1)=wa(1:m+1)-dtdx*(phim(1:m+1)-phin(1:m+1));
end
```

9.4.4 vijaya_llfr_btbc.m

Function with the Vijayasundatam flux splitting for the Burgers equation with the Local Lax Friedrichs regularization Sect. 6.8.2 and (6.47).

```
function [wn] = vijaya_llfr_btbc(wa,dtdx,m)
%
%  Vijayasundaram flux splitting
%    for the Burgers equation
%
%  Local Lax Friedrichs regularization
%
wam=zeros(1,m+1);  % w_(j+1)
wan=zeros(1,m+1);  % w_(j-1)
% a=0.5*wa
am=zeros(1,m+1);   % a+(0.25(w_j+w_(j+1)))
amn=zeros(1,m+1);  % a+(0.25(w_j+w_(j-1)))
an=zeros(1,m+1);   % a-(0.25(w_j+w_(j+1)))
ann=zeros(1,m+1);  % a-(0.25(w_j+w_(j-1)))
phim=zeros(1,m+1); % phi(w_j,w_(j+1))
phin=zeros(1,m+1); % phi(w_(j-1),w_j)
%
wam(1:m)=wa(2:m+1);
% Transmissive boundary conditions
wam(m+1)=wa(m);
%
wan(2:m+1)=wa(1:m);
% Transmissive boundary conditions
wan(1)=wa(2);
%
%  Local Lax Friedrichs regularization
%
regm(1:m+1)=0.5*max(abs(wa(1:m+1)),abs(wam(1:m+1)));
regn(1:m+1)=0.5*max(abs(wa(1:m+1)),abs(wan(1:m+1)));
%
am(1:m+1) =0.5*(0.25.*(wa(1:m+1)+wam(1:m+1))+regm(1:m+1) );
amn(1:m+1)=0.5*(0.25.*(wa(1:m+1)+wan(1:m+1))+regn(1:m+1) );
an(1:m+1) =0.5*(0.25.*(wa(1:m+1)+wam(1:m+1))-regm(1:m+1) );
ann(1:m+1)=0.5*(0.25.*(wa(1:m+1)+wan(1:m+1))-regn(1:m+1) );
%
% Numerical flux
%
phim(1:m+1)=am(1:m+1).*wa(1:m+1)+an(1:m+1).*wam(1:m+1);
phin(1:m+1)=amn(1:m+1).*wan(1:m+1)+ann(1:m+1).*wa(1:m+1);
%
wn(1:m+1)=wa(1:m+1)-dtdx*(phim(1:m+1)-phin(1:m+1));
end
```

9.4.5 Output of This Code

This code provides the following information at the command prompt:

```
-------------------------------------------
Finite volume methods for Burgers equation
-------------------------------------------
w + w w =0
 t     x
Domain: [a,b]
Transmissive boundary conditions
Riemann problem for Burgers equation
w0(x)=wl if x<xb=(b+a)/2; else w0(x)=wr
Exact solution
-------------------------------------------
Lower end of the interval a = 0
Upper end of the interval b = 5
Number of nodes m=200
Initial location of the discontinuity, xb = 2.5
deltax =0.025
Courant number = 0.5
Time step from Courant number = 0.00625
Time end = 1
-------------------------------------------
Initial condition:
wl= -1
wr= 2
w0(x)=-1 if x<=2.5
w0(x)=2 if x>2.5
-------------------------------------------
The exact solution is a rarefaction wave
w(x,t)=-1 if (x-2.5)/t<=-1
w(x,t)= (x-2.5)/t if -1<= (x-2.5)/t<=2
w(x,t)=2 if (x-2.5)/t>2
-------------------------------------------
Numerical methods:
 * Godunov
 * Lax Friedrichs
 * Q-scheme van Leer / Roe
 * Q-scheme with
   Harten regularization (HR)
      eps for Harten regularization= 0.9
 * Q-scheme with
   Local Lax Friedrichs regularization (LLFR)
 * Steger-Warming flux splitting
 * Vijayasundaram flux splitting
 * Vijayasundaram flux splitting with
   Harten regularization
      eps for Harten regularization= 0.9
 * Vijayasundaram flux splitting with
   Local Lax Friedrichs regularization (LLFR)
-------------------------------------------
-------------------------------------------
Error at time t=1
* Godunov max(|we-wngod|)=.........0.12031
* Lax Friedrichs m\'{a}x|we-wnlxf|=.....0.23174
```

```
* Non conservative max|we-wnnc|=....0.1505
* Q-scheme max|we-wnlxf|=..........1
* Q-scheme + Harten R. max|we-wnnc|=0.11859
* Q-scheme + LLF R. max|we-wnqllfr|=0.1288
* Steger-Warming max|we-wnsw|=.....0.12031
* Vijayasundaram max|we-wnv|=.....0.066673
* Vijayasundaram Harten R. max|we-wnvlhr|=0.072193
* Vijayasundaram LLF R. max|we-wnvllfr|=0.072282
-------------------------------------------
-------------------------------------------
```

Chapter 10
Biographical Summary of Professor Eleuterio Francisco Toro

Abstract Professor Eleuterio Francisco Toro has remarkable contributions in the context of Riemann solvers.

Professor Eleuterio Francisco Toro was born in Chile on July the 16th 1946. He was the second of a family of four sisters and four brothers. In 1967 he graduated from the *Escuela Normal de Victoria* as an elementary school teacher, with specialty in music. From 1969 to 1973 he studied pedagogy of mathematics at the *Universidad Católica de Temuco*. While studying, he taught mathematics at the *Liceo Pablo Neruda* and *Liceo Gabriela Mistral* in Temuco.



Fig. 10.1 Professor Eleuterio Francisco Toro at the University of de Santiago de Compostela (July 2011)

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He was also involved in students affairs, becoming president of the students union of his university in 1971. Three months before completing his studies, on orders from the Pinochet military dictatorship, he was expelled from the university. After a brief period at the *Universidad Católica de Santiago* he re-initiated his university studies at the Mathematics Institute, University of Warwick (UK), from which he obtained a BSc honours in pure mathematics in 1977. In 1978 he obtained a Master in Applied Mathematics from the University of Dundee (UK) and in 1982 he obtained his PhD in Computational Mathematics from Teesside University (UK). He then initiated his academic career in mathematics as a post-doctoral research fellow at the University of Leeds (UK). Subsequent academic appointments include: senior lecturer in computational fluid dynamics, Cranfield University (UK); full professor of Applied Mathematics, Manchester Metropolitan University (UK) and senior research fellow, Isaac Newton Institute for Mathematical Sciences, University of Cambridge (UK). In 2002 he was called by the University of Trento, Italy, as *Professore di Chiara Fama*, to take up a chair in numerical analysis, position held untill the present.

Professor Toro always projects his *teaching profile*, with which he started his academic CV. His books always reflect his profesionalism and his commitment with student training, producing a great influence in the scientific community. Moreover, every new edition is a challenge with new contributions. The book [30] is one of the references listed by Godunov in his recent work [11] in 2014. He is also the author of NUMERICA, a Library of Source Codes for teaching, research and applications, in which he implemented, among others, the methods described in [30]. This library is available on the Web site http://www.numetitek.com. For all these reasons, I consider him the best reference possible for the third part of the book.

His contributions to finite volume methods are not minor, nor is their impact on different fields. He has been working on the construction of non-linear numerical methods of arbitrary order of accuracy in space and time, known as the ADER methods [30]. He is currently focused on the study of the physics of neurodegenerative diseases and their potential link to venous haemodynamics. He also maintains contributions on numerical methods for Euler equations and the shallow water equations [29], for which he has a special motivation, after he survived the Chile Tsunami in May 22 1960, when he was only 14 years old.

Professor Toro has received several honours and distinctions, which include the honorary Order of the British Empire from Queen Elizabeth II (Buckingham Palace, London, 2000); Distinguished Citizen of the City of Carahue (Chile, 2001); Life Fellow, Claire Hall, University of Cambridge (UK, 2003); Fellow of the Indian Society for Shock Wave Research (Bangalore, 2005); Doctor Honoris Causa (Universidad de Santiago de Chile, 2008); William Penney Fellow, University of Cambridge (UK, 2010); Doctor Honoris Causa (*Universidad de la Frontera*, Chile, 2012).

The large record of his contributions, mentions and honors, can be reviewed in the following links:

- Eleuterio F. Toro in MathSciNet:
  http://www.ams.org/mathscinet/search/author.html?mrauthid=205696
- Eleuterio F. Toro in Scholar Google:
  https://scholar.google.es/citations?user=UFc2S58AAAAJ&hl=es
- Eleuterio F. Toro in his personal web site:
  http://www.ing.unitn.it/~toroe/

One of the great gifts of a scientific career is the opportunity to meet wonderful people such as Peter Lax (Fig. 4.1), Sergei Konstantinovich Godunov (Fig. 7.1) and Eleuterio Francisco Toro (Fig. 10.1). Admiration is the natural corollary of their awards and contributions. Building new contributions with the people you admire is a source of satisfaction and joy. Moreover, you have the friendship of your colleagues, which is the most valuable gift. My admiration for Professor Toro started a long time ago resulting from our collaboration in several works. But during the organization of the Congress *Numerical Methods for Hyperbolic Equations: Theory and Applications* [33], held in the Faculty of Mathematics, University of Santiago de Compostela, in July 2011 (Fig. 10.1) to honor Toro in his 65th birthday, I realised that I was not the only one. The admiration and steem for him were shared by a great number of personalities and students of the field, who wanted to pay tribute to this vital scientist. Again we have the same feeling: *every one is here!*. Moreover, the *Short Course on Numerical Methods for Hyperbolic Equations* held from the 2nd to the 4th of July 2011 [36], before the international conference, was organized in recognition of his contribution to education and training. *Thank you Tito for this gift!*.

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